

# KNESER'S TETRATION CONTINUED AROUND THE CUSP $e^{1/e}$ : SEPARATION FROM REGULAR ITERATION AND THE INVERSE HORN MAP OF $e^u - 1$

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ABSTRACT. For  $1 < b < \eta = e^{1/e}$  the equation  $F(z+1) = b^{F(z)}$ ,  $F(0) = 1$ , has the regular solution  $R_b$  at the attracting fixed point. We prove that Kneser's real solution for  $b > \eta$ , continued analytically in the base around the cusp  $\eta$ , crosses the Shell–Thron boundary (jointly holomorphically in base and height, whatever the arithmetic of the neutral multiplier), extends single-valued to the whole upper half of the Shell–Thron region, and has boundary values  $K_b$  on  $(1, \eta)$  along every path there. These are a solution  $K_b^W$  obtained by sewing the two regular charts, characterised, for  $b$  near  $\eta$ , by the position of its seam. Moreover, with a computer-assisted certificate, Kneser's family continues to a single-valued holomorphic family on the whole upper half  $b$ -plane: Kneser's own straight initial region survives on the entire exterior, and the family crosses every point of the open upper Shell–Thron boundary arc. So the continuation is path independent, it coincides with Paulsen's base-continued family, and it is holomorphic across every real base in  $(0, \eta) \setminus \{1\}$  (in particular at the neutral endpoint  $e^{-e}$ ). Writing  $K_b = R_b \circ (\text{id} + \sum_{n \geq 0} c_n e^{2\pi i n z})$  and  $\Lambda = \exp(4\pi^2 / \log \lambda)$ , we show  $c_n / \Lambda^n \rightarrow B_n e^{2\pi i n a}$  as  $b \rightarrow \eta^-$  for every  $n$ , where  $B_n$  are the Fourier coefficients of the inverse Écalle–Voronin horn map of  $e^u - 1$  and  $a$  is the Fatou coordinate of the base point;  $B_1 \neq 0$ . The proof combines a confluence theorem for the transition map between the two regular charts, a contraction in a weighted Wiener algebra, and a family of tilted initial regions whose quotients are uniformly quasiconformal to a flat cylinder. The corrections have an asymptotic expansion to all orders in the unfolding parameter with coefficients given by convergent parabolic series. Numerically, the sewn solution reproduces Paulsen's 120-digit value at  $b = \sqrt{2}$  in all quoted digits.

## 1. INTRODUCTION

The tetration equation

$$(1) \quad F(z+1) = b^{F(z)}, \quad F(0) = 1,$$

has, for real  $b > \eta := e^{1/e}$ , a distinguished real-analytic solution constructed by Kneser [10]. It tends to the two complex-conjugate fixed points of  $w \mapsto b^w$  as  $\text{Im } z \rightarrow \pm\infty$ . Its conformal uniformisation is uniquely characterised by the initial-region criterion of [23]. For  $1 < b < \eta$  the fixed points are real:  $L = e^\lambda$  is attracting, with  $0 < \lambda < 1$ , and  $L_2 = e^{\lambda_2}$  is repelling, with  $\lambda_2 > 1$ . The Koenigs construction at  $L$  yields the regular solution  $R_b$ . This is the solution usually meant in that range [12]. Paulsen [18] proposed continuing Kneser's two-fixed-point construction through the upper half of the Shell–Thron region. If this base continuation  $\kappa$  has a boundary value, denote it by

$$(2) \quad K_b = \lim_{y \downarrow 0} \kappa_{b+iy}.$$

High-precision algorithms for the relevant pointwise constructions are developed in [11, 12, 22]. We construct a two-sided sewing solution  $K_b^W$  independently and determine its difference from  $R_b$ . We then prove that the continuation of Kneser's solution through the upper half-disc about

$\eta$ , and then near the real axis, has boundary values  $K_b = K_b^W$  (Theorem D), so every result about  $K_b^W$  holds for these boundary values.

**Notation.** Let  $h = 2\pi/|\log \lambda|$  be the imaginary period of  $R_b$ , and put

$$\Lambda = e^{-2\pi h} = \exp(4\pi^2/\log \lambda).$$

Both  $R$  and  $K^W$  are normalised by  $F(0) = 1$ . The map  $P = R^{-1} \circ K^W - \text{id}$  is 1-periodic with  $P(0) = 0$ . In the relevant region it has only non-negative modes,  $P = \sum_{n \geq 0} c_n e^{2\pi i n z}$ , and the separation is

$$K_b^W(z) - R_b(z) = c_1 R'_b(z)(e^{2\pi i z} - 1) + O(c_2) + O(c_1^2).$$

At  $b = \eta$  the map becomes the parabolic germ  $f(u) = e^u - 1$  in  $w = e(1+u)$ . Let  $\Phi_{\text{att}}, \Phi_{\text{rep}}$  be its Fatou coordinates, both normalised by the formal Abel function  $\alpha(u) = -2/u + \frac{1}{3} \log(-u) + O(u)$ . Let

$$\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}(z) = z + \sum_{n \geq 1} B_n e^{2\pi i n z}$$

be the inverse horn map on the upper gate [6, 24], and  $a = \Phi_{\text{att}}(1/e - 1)$  the attracting Fatou coordinate of the base point  $w = 1$ . Numerically

$$|B_1| = 0.0890584364122133\dots, \quad \frac{B_2}{B_1^2} = -0.0253204931 + 1.5742190652i,$$

$$a = 3.0292972144180\dots$$

(Section 2). The transition map between the two regular solutions is  $T_b = R_b^{-1} \circ S_b$ , where  $S_b$  is the regular solution at the repelling point. Write  $T_b(z) - z = \sum t_n e^{2\pi i n z}$  and  $\tau_n = t_n e^{-2\pi i n t_0}$  (Section 3).

*Symbols with more than one use.* A few letters are unavoidably overloaded; their meaning is fixed within each section.

- $\varepsilon$ :  $1 - \lambda$  in Proposition 2.2 and, as a complex parameter, in Section 9;  $|\log \lambda|$  in Sections 6–8. The two agree to first order.
- $\tau$ : the repelling Koenigs map (Section 3); the invariants  $\tau_n$ ; and  $T - \text{id} - t_0$  in Theorem 6.3.
- $a$ : the Fatou coordinate  $\Phi_{\text{att}}(1/e - 1)$  above;  $a = \log b$  only inside Remark 1.2 and Propositions 9.11 and 9.50.
- $h$ : the period  $2\pi/|\log \lambda|$ ; the horn map and its inverse are written  $h^{\pm 1}$  only in Theorem 3.6, Section 4 and Section 5.
- $s$ :  $1 - e \log b$  in Section 5; a shift  $s(b)$  in Theorems 3.6 and 4.4; an interpolation parameter in Section 9.
- $\kappa$ : the base-continued Kneser family  $\kappa_b$ ;  $\kappa_n = c_n/c_1^n$  and the first-order coefficients  $\kappa^{(n)}$  carry indices.

## Main results.

**Theorem A** (Confluence). *For every  $n \geq 1$ ,  $\tau_n(b) \rightarrow B_n e^{2\pi i n a}$  as  $b \rightarrow \eta^-$ . Moreover,  $B_1 \neq 0$  and  $t_n/t_1^n \rightarrow B_n/B_1^n$  for every  $n \geq 1$ .*

**Theorem B** (The sewing solution). *There is  $b_1 < \eta$  such that for  $b \in (b_1, \eta)$  the following holds. There is exactly one solution  $K_b^W$  of (1) with the following three properties:*

- it has a representation  $R_b \circ (\text{id} + P)$  on an upper half-plane;
- it has a representation  $S_b \circ (\text{id} + Q)$  on a lower half-plane;
- $P$  and  $Q$  are bounded and 1-periodic, and the seam lies half a period below the real axis.

The precise statement is Definition 8.1. Its coefficients satisfy

$$\left| \frac{c_n}{\Lambda^n} - \tau_n(b) \right| \leq C_n \Lambda$$

with explicit constants  $C_n$  independent of  $b$ .

**Corollary 1.1.** For  $K_b^W$ ,  $\hat{c}_n := c_n/\Lambda^n \rightarrow B_n e^{2\pi i n a}$  for every  $n \geq 1$ . In particular

$$\lim_{b \rightarrow \eta^-} \frac{|c_1|}{\Lambda} = |B_1|.$$

The limiting phase is  $\arg B_1 + 2\pi a$ , and  $c_n/c_1^n \rightarrow B_n/B_1^n$  for every  $n \geq 1$ .

**Theorem C** (First-order term). Let  $s = 1 - e \log b$  and  $p = |\log \lambda| \log \lambda_2 = 2s + O(s^2)$ . For every  $n \geq 1$  with  $B_n \neq 0$  (in particular  $n = 1$  by Proposition 2.4),

$$\log \frac{\tau_n(b)}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^2) \quad (b \rightarrow \eta^-),$$

where  $\kappa^{(n)}$  is given by an absolutely convergent series over orbits of the parabolic germ  $e^u - 1$  (Theorem 5.6; the remainder  $O(p^2)$  is Theorem 5.8 with  $m = 1$ ). More generally,  $\log(\tau_n/B_n e^{2\pi i n a})$  has an asymptotic expansion  $\sum_{j \geq 1} \kappa_j^{(n)} p^j$  to all orders, and every coefficient is built from convergent parabolic series of the same kind (Theorem 5.8).

The sewn solution is defined for  $b < \eta$  without reference to Kneser's construction. The next theorem shows that it is exactly what Kneser's solution becomes when continued in the base around the cusp.

**Theorem D** (Continuation around the cusp). Kneser's real-normalised solution for real  $b > \eta$  continues analytically in the base through the upper half-disc about  $\eta$ , across the Shell–Thron boundary arc there (whatever the arithmetic of the neutral multiplier), and single-valued along every path in the upper half of the Shell–Thron region. Its boundary values on  $(1, \eta)$  are  $K_b = K_b^W$ , so Corollary 1.1 holds for  $K_b$ :  $c_n/\Lambda^n \rightarrow B_n e^{2\pi i n a}$  for every  $n \geq 1$ , and  $|c_1|/\Lambda \rightarrow |B_1| > 0$ . Continuation through the lower half-disc gives the complex-conjugate branch.

**Theorem E** (The whole upper half-plane; computer-assisted). The continuation of Theorem D is a single-valued holomorphic family on the whole upper half  $b$ -plane. It crosses every point of the open upper Shell–Thron boundary arc holomorphically, it is holomorphic at the neutral endpoint  $b = e^{-e}$ , and it extends holomorphically across every point of  $(0, \eta) \setminus \{1\}$ ;  $b = 0$  and  $b = 1$  are genuine singularities. In particular the continuation is path independent, and it is the base-continued family of [18].

The precise statements are Theorems 9.14 and 9.22 for Theorem D, and Theorems 9.33, 9.34, 9.40, 9.43, 9.45 and Proposition 9.46 for Theorem E.

- *Around the cusp* (Section 9), Kneser's initial region (the chord between the fixed points and its image) is tilted continuously along the path. In the coordinate  $\log \frac{u-u_1}{u-u_2}$  it is a thin strip whose quotient is uniformly quasiconformal to a flat cylinder, so the neutral boundary is no obstacle. At real  $b < \eta$  it is a fundamental domain of the sewn surface, and the seam condition of Definition 8.1 identifies the limit with  $K_b^W$ . A quasiconformal transport of the dynamics extends the family over the whole upper Shell–Thron region (Section 9.9).
- *Away from the cusp* (Sections 9.10–9.13), Kneser's own straight chord remains a crescent with a canonical marked point on the whole exterior part of the upper half-plane and across the boundary arc. This is verified by interval arithmetic, in a parametrisation by the multiplier of the continued fixed point.

- The family of [18] is defined by analytic continuation in the base, but there the crossing of the Shell–Thron boundary is argued only numerically, and the uniqueness statement needs an additional root-selection condition (Remark 1.2). Theorem E settles the first point. The nine-digit agreement at  $b = \sqrt{2}$  is a numerical check.

By Lemma 2.1 and Proposition 2.2,  $\Lambda$  has the saddle-node form

$$\Lambda = \exp(-C_\eta(\eta - b)^{-1/2}(1 + o(1))), \quad C_\eta = \frac{4\pi^2}{\sqrt{2}e^{1-1/e}} = 20.3508 \dots$$

Thus the coefficients of the separation have the stated exponentially small upper bounds. Proposition 2.4 makes the first-mode asymptotic sharp with a strictly positive constant, and Proposition 2.6 gives a certified enclosure of it.

**Numerical results.** Theorems A and B are confirmed and extended numerically in Section 10.

- (1) Ladder computations of  $K_b$  reproduce  $\tau_1$  and  $t_2/t_1^2$  to their full precision, arguments included; for example, eight digits at  $b = 1.1$ .
- (2) At  $b = \sqrt{2}$ , far from  $\eta$ , the sewing equation gives  $\text{Im } K_{\sqrt{2}}^W(\frac{1}{2}) = -1.18899697185 \cdot 10^{-48}$ , which agrees with Paulsen’s 120-digit value  $-1.18899697 \cdot 10^{-48}$  for his base-continued family [18, §6] in all nine quoted digits, sign included, as Theorem E predicts.
- (3)  $\tau_1$ ,  $t_2/t_1^2$  and  $t_3/t_1^3$  have been followed up to  $\eta - b = 3 \cdot 10^{-5}$ .
- (4) The parabolic series of Theorem C gives

$$\log \frac{\tau_1}{B_1 e^{2\pi i a}} = (-0.010199006345897402441 - 0.013250956182506712250 i) p + O(p^2),$$

in agreement with fits from the side  $b < \eta$  to their eleven digits. The second coefficient,  $\kappa_2^{(1)} = -7.88992074991511334 \cdot 10^{-5} + 3.6554727325799459 \cdot 10^{-4} i$ , agrees with the fits to within one unit in their sixth digit.

**Method.** Four observations drive the proofs.

- (1) The separation is a property of the transition map  $T_b$  between two classical Koenigs coordinates. A welding identity expresses  $c_n$  through the Fourier modes of  $T_b$  (Theorem 3.2). The scale  $\Lambda$  appears because the seam sits half a period below the real axis, and the gate carrying the positive modes a further half period below it.
- (2) The limit of  $T_b$  is a confluence problem for the saddle-node unfolding in the direction where both fixed points are real (the “Glutsyuk” direction [9, 15, 5]). In the complementary Lavaurs direction, the corresponding statements are the classical results of parabolic implosion [13, 21, 16]. For vector fields, see [20]. We prove the needed statement directly (Section 4), using the model time  $A_1 \log(u - u_1) + A_2 \log(u - u_2)$ , whose one-step error vanishes at both fixed points.
- (3) The Kneser-type solution is the conformal sewing of two half-cylinders along  $T_b$ . A contraction in a weighted Wiener algebra gives it with explicit bounds (Theorem 6.3). A proper-map argument proves uniqueness in the class of Definition 8.1 (Theorem 8.2).
- (4) Kneser’s initial region can be tilted continuously around the cusp. In the coordinate  $\log \frac{u-u_1}{u-u_2}$  it is a thin strip on which the map is a translation up to relative error  $O(|\varepsilon|)$ , so its quotient is uniformly quasiconformal to a flat cylinder, whatever the multiplier. The Ahlfors–Bers theorem gives holomorphic dependence on the base, and at  $b < \eta$  the tilted region is a fundamental domain of the sewn surface (Section 9).

**Organisation.** Section 2 collects the parametrisation and the parabolic data. Section 3 introduces  $T_b$ , the welding identity and the sewing construction. Section 4 proves Theorem A, and Section 5 proves Theorem C. Sections 6–8 prove Theorem B (existence with Wiener bounds, uniqueness) and its base dependence, and Section 9 proves Theorem D, including the crossing away from the cusp (Section 9.10). Section 10 reports the numerics, and Section 11 lists open problems.

**Remark 1.2** (On the definition and uniqueness in [18]). In [18],  $\kappa_b$  is defined by analytic continuation in  $b$  of Kneser's solution  $\kappa_b(z_0)$ ,  $\operatorname{Re} z_0 > -2$ , from  $(\eta, \infty)$  (Prop. 3 there). That proposition asserts real-analytic dependence of Kneser's conformal uniformisation on the real base without a separate parameter-dependence proof; Proposition 9.11 supplies such a proof locally along the real exterior ray. Near the cusp, Theorem D carries the family through the Shell–Thron boundary by a different route. Away from the cusp, Theorems 9.33 and 9.34 cross the boundary arcs with Kneser's untilted chord. Off that boundary, Lemma 1 of that paper asserts two regular representations  $\psi_1^{-1}(\rho_1)$  and  $\psi_2^{-1}(\rho_2)$  from the limits at  $\pm i\infty$ . We impose their degree-one form explicitly in Definition 8.1. A fixed-point limit alone leaves an integer end label open: near an end one only gets  $\sigma(F(z)) = e^{\ell z} q^m h(q)$  with  $q = e^{\pm 2\pi i z}$ ,  $h(0) \neq 0$  and some integer  $m \geq 0$ , and only  $m = 0$  is a time change  $\operatorname{id} + (\text{bounded periodic})$ . The continuation across the Shell–Thron boundary is argued in [18, §5] by the continuity of a numerical scheme; it is not proved there.

The uniqueness statement [18, Prop. 2] requires only analyticity on  $\operatorname{Re} z > -2$ ,  $F(0) = 1$  and the two limits. As stated it is false for every real  $b > \eta$ . Let  $K = K_b$  be Kneser's solution,  $a = \log b > 1/e$ , and  $u_n = K(n) \rightarrow \infty$ .

- *Large image discs.* Choose  $N$  with  $au_{N+1} > 1$ , and  $\delta > 0$  with  $B(u_N, \delta) \subset K(B(N, \frac{1}{2}))$  and  $u_{N+1}(1 - e^{-a\delta}) > \delta$ . The inclusion  $E_b(B(t, \delta)) \supset B(e^{at}, e^{at}(1 - e^{-a\delta}))$  (apply the principal logarithm and  $|\log(1+v)| \leq -\log(1-|v|)$ ) propagates these discs. So  $K(B(n+1, \frac{1}{2}))$  contains a disc about  $u_{n+1}$  of radius  $u_{n+1}(1 - e^{-a\delta}) \rightarrow \infty$ .
- *A shifted solution.* For large  $n$  and  $k = \lceil a \exp(au_{n+1})/(2\pi) \rceil$ , the point  $w_k = a^{-1}(\log(2\pi k/a) + i\pi/2)$  lies in that disc, and  $E_b(w_k) = 2\pi i k/a$ ,  $E_b^2(w_k) = 1$ . Choose  $z_k \in B(n+1, \frac{1}{2})$  with  $K(z_k) = w_k$  and put  $F_k(z) = K(z + z_k + 2)$ . Then  $F_k$  is analytic on  $\operatorname{Re} z > -2$ , has  $F_k(0) = 1$  and the same vertical limits as  $K$ , but  $F_k(-1) = 2\pi i k/a \neq 0 = K(-1)$ .

Since  $k$  is unbounded, this gives infinitely many solutions. The gap in [18, Prop. 2] is the last step of its proof, which infers  $C = 0$  from  $F_1(C) = 1$ . For real  $b > \eta$  the uniqueness theorem of [17] is not affected: it also assumes analyticity on  $\mathbb{C} \setminus (-\infty, -2]$  and  $F(\bar{z}) = \overline{F(z)}$ , both of which  $F_k$  violates. Indeed  $z_k$  is nonreal, because  $K$  is real on the real axis and  $w_k$  is not. And  $K$  cannot extend holomorphically through  $-2$ , since  $K(-1) = 0$  whereas an exponential never vanishes. Hence  $F_k$  is singular at  $-4 - z_k$ , off the standard cut. For complex bases, some root-selection condition must be added to [18, Prop. 2]. The seam-height condition of Definition 8.1 is such a condition, and Theorem 8.2 shows that it suffices near  $(1, \eta)$ .

## 2. SETUP AND PARABOLIC DATA

Throughout,  $1 < b < \eta$  and  $E(w) = b^w$ .

**Lemma 2.1** (Parametrisation). *Let  $L$  be a fixed point of  $E$  with multiplier  $\lambda = L \log b$ . Then  $L = e^\lambda$  and  $\log b = \lambda e^{-\lambda}$ . The map  $\lambda \mapsto \lambda e^{-\lambda}$  increases on  $(0, 1)$  and decreases on  $(1, \infty)$ , with maximum  $e^{-1} = \log \eta$ . Hence  $\lambda \in (0, 1)$  and  $\lambda_2 \in (1, \infty)$  parametrise  $(1, \eta)$  through the attracting and the repelling fixed point, and the two branches meet at  $b = \eta$ .*

*Proof.*  $L = b^L$  gives  $\log L = L \log b = \lambda$ . □

**Proposition 2.2.** *With  $\lambda = 1 - \varepsilon$  one has  $\varepsilon = \sqrt{2e(\log \eta - \log b)}(1 + O(\varepsilon))$ , and as  $b \rightarrow \eta^-$*

$$\Lambda = \exp\left(-\frac{C_\eta}{\sqrt{\eta - b}}(1 + o(1))\right), \quad C_\eta = \frac{4\pi^2}{\sqrt{2e^{1-1/e}}}.$$

*Proof.* Expand  $\lambda e^{-\lambda} = e^{-1}(1 - \varepsilon^2/2 - \varepsilon^3/3 + O(\varepsilon^4))$  and use  $|\log \lambda| = \varepsilon + O(\varepsilon^2)$  and  $\log \eta - \log b \sim (\eta - b)/\eta$ .  $\square$

**Remark 2.3** (Translation invariance). A simultaneous shift  $K(\cdot + \delta)$ ,  $R(\cdot + \delta)$  sends  $c_n \mapsto c_n e^{2\pi i n \delta}$ . Thus  $\kappa_n = c_n/c_1^n$ , when defined, is invariant under complex translations, while  $|c_1|$  is invariant only under real ones. The normalisation  $F(0) = 1$  fixes the imaginary origin. This is why normalised first-mode limits require a specified base point, and why the constant  $a$  enters Corollary 1.1.

**2.1. The parabolic germ.** At  $b = \eta$ , put  $w = e(1 + u)$ . Then  $E$  becomes  $f(u) = e^u - 1$ , with attracting petal on  $\operatorname{Re} u < 0$  and formal Abel function

$$(3) \quad \alpha(u) = -\frac{2}{u} + \frac{\log(-u)}{3} + \sum_{k \geq 1} c_k^\alpha u^k, \quad \alpha(f(u)) = \alpha(u) + 1.$$

The Fatou coordinates are  $\Phi_{\text{att}}(u) = \lim[\alpha(f^n u) - n]$  and  $\Phi_{\text{rep}}(u) = \lim[\alpha(f^{-n} u) + n]$ , with  $\log(-u)$  continued along orbits. On the upper gate, the horn map  $\Phi_{\text{rep}} \circ \Phi_{\text{att}}^{-1}(z) = z + \sum_{n \geq 1} A_n e^{2\pi i n z}$  and its inverse  $\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}(z) = z + \sum_{n \geq 1} B_n e^{2\pi i n z}$  have no constant mode (Section 4, item (d)). Formal inversion gives

$$(4) \quad B_1 = -A_1, \quad B_2 = -A_2 + 2\pi i A_1^2.$$

**Proposition 2.4.** *The first coefficient of the inverse upper horn map satisfies  $B_1 \neq 0$ .*

*Proof.* Let  $A$  be the immediate parabolic basin of  $f(u) = e^u - 1$ . The negative real ray is contained in  $A$ , since  $u < f(u) < 0$  for  $u < 0$ . The entire map  $f$  has no critical points and exactly one finite singular value,  $-1$ , its asymptotic value along that ray. By [4, Lemma 87], the singular values of  $f|_A : A \rightarrow A$  are contained in  $A \cap \{-1\}$ ; the negative ray shows that  $-1$  does occur. Hence  $f$  belongs to the one-petal unisingular class  $S_\infty$  of [4, Definition 8]. The parabolic-renormalisation theorem [4, Theorem 9 and Corollary 11] gives a renormalisation  $R[f]$  with exactly one attracting petal at its parabolic fixed point. The upper renormalisation is the exponential projection of  $\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}$ , up to nonzero linear scalings of source and target [4, Appendix A]. In our normalisation, writing  $\zeta = e^{2\pi i z}$ , this projection is  $H(\zeta) = \zeta + 2\pi i B_1 \zeta^2 + O(\zeta^3)$ . Since  $H'(0) = 1$ , nonzero linear scalings that again give multiplier one preserve whether the quadratic coefficient vanishes. One attracting petal means that this coefficient is nonzero. Therefore  $B_1 \neq 0$ .  $\square$

**Remark 2.5** (The mechanism). The underlying reason is the classical one. The map  $f$  is of finite type in the sense of Epstein: it has no critical points and a single finite asymptotic value. Its horn maps are again of finite type, with singular values in  $\mathbb{C}^*$  given by the projections of the single singular orbit [7]. So the upper horn map has exactly one singular value in  $\mathbb{C}^*$ . By the Fatou-type theorem for finite-type maps [7], the immediate basin of each attracting petal of a parabolic fixed point contains a singular value other than the fixed point. If  $B_1 = 0$ , then  $H(\zeta) - \zeta = O(\zeta^3)$ , and  $H$  would have at least two attracting petals at 0, each needing its own singular value.

Computing both maps directly on horizontal lines gives

$$(5) \quad \begin{aligned} |B_1| &= 0.089058436412213331563, & |B_2| &= 0.012487384111564700811, \\ \kappa_2^\infty &:= \frac{B_2}{B_1^2} = -0.02532049309975755123 + 1.5742190652319516267i, \\ \frac{B_3}{B_1^3} &= -2.254799124523803111 - 0.024408714296883171i. \end{aligned}$$

Also  $a = \Phi_{\text{att}}(1/e - 1) = 3.0292972144180360989\dots$ , so the limiting phase in Corollary 1.1 is  $\arg B_1 + 2\pi a = 1.5865330757\dots \pmod{2\pi}$ . These values are stable to all digits shown under changes of working precision (40–60 digits), transform size ( $N = 16, 24, 32$ ) and sampling height ( $\text{Im } z = 1, 1.5, 2$ ). Digits beyond the intervals below remain numerical. The first three horn coefficients have independent certified enclosures of Proposition 2.6; nonvanishing of  $B_1$  also follows from Proposition 2.4. The base-point coordinate  $a$  has the certified enclosure

$$3.0292972144180360989249938 < a < 3.0292972144180360989249940.$$

(Appendix A.5).

**Proposition 2.6** (Certified first horn coefficients). *The Fourier coefficients of the inverse upper horn map satisfy*

$$0.0890584364122133315631 < |B_1| < 0.0890584364122133315633, \quad 0.01248738411156469 < |B_2| < 0.01248738$$

*The ratios used in the numerical comparisons also satisfy*

$$\begin{aligned} -0.025320493100 < \text{Re}(B_2/B_1^2) < -0.025320493099, & 1.574219065231 < \text{Im}(B_2/B_1^2) < 1.574219065233, \\ -2.25479914 < \text{Re}(B_3/B_1^3) < -2.25479910, & -0.02440873 < \text{Im}(B_3/B_1^3) < -0.02440869. \end{aligned}$$

*Proof.* Appendix A.5. □

**Remark 2.7** (Formal invariants). Put  $\zeta = e^{2\pi iz}$ . The inverse horn map becomes the cylinder germ  $F(\zeta) = \zeta \exp(2\pi i \sum B_n \zeta^n) = \zeta + a_2 \zeta^2 + a_3 \zeta^3 + \dots$ . Since  $B_1 \neq 0$ , its formal invariants under tangent-to-identity conjugacy include  $a_2 = 2\pi i B_1$  and the iterative residue  $\rho = 1 - a_3/a_2^2$  [3]. One has

$$(6) \quad \kappa_2^\infty = 2\pi i \left(\frac{1}{2} - \rho\right), \quad \rho = 0.249455254258832 - 0.004029881638351i.$$

When defined, the higher ratios  $B_n/B_1^n$  are not formal invariants of  $F$ . They are, however, invariants under the translations that preserve the normalisation of Fatou coordinates, and Theorem A compares them for every  $n$ .

### 3. THE SEPARATION IS A TRANSITION-MAP INVARIANT

The separation problem compares a Kneser-type solution  $K_b$  with  $R_b$ . This section removes  $K_b$  from the problem; Section 9 identifies it with the sewn solution constructed below. The separation coefficients turn out to be determined, up to a relative error of order  $\Lambda$ , by the transition map between the two *regular* solutions at the two real fixed points — an object built from two convergent Schröder series in a fraction of a second, at any  $b < \eta$ .

**3.1. The transition map.** Let  $\tau$  be the Koenigs coordinate at the repelling point  $L_2$ ,  $\tau(E(w)) = \lambda_2 \tau(w)$ ,  $\tau'(L_2) = 1$ , and put

$$S(z) = \tau^{-1}(-\lambda_2^z),$$

continued to an entire function by  $S(z+1) = E(S(z))$ . It maps  $\mathbb{R}$  decreasingly onto the segment  $(L, L_2)$  and has period  $2\pi i / \log \lambda_2$ . Let  $R^{-1}(w) = \log(\sigma(w)/\sigma(1)) / \log \lambda$  be the attracting Abel

coordinate,  $R^{-1}(1) = 0$ , which is multivalued with period  $ih = 2\pi i/|\log \lambda|$ . On the horizontal strip where  $S$  takes values in the immediate basin of  $L$  and a continuous branch is chosen, put

$$T(z) := R^{-1}(S(z)), \quad T(z+1) = T(z) + 1, \quad T(z) - z = \sum_{n \in \mathbb{Z}} t_n e^{2\pi i n z}.$$

**Lemma 3.1** (Real structure). *For real  $b \in (1, \eta)$ ,  $\operatorname{Im} T(x) \equiv h/2 \pmod{h}$  for every real  $x$ . Consequently, on the branch with  $\operatorname{Im} t_0 = h/2$ ,  $T(x) - x - ih/2$  is real,  $t_{-n} = \overline{t_n}$ , and the line  $\operatorname{Im} w = -h/2$  of attracting time is the image of  $\mathbb{R}$  under the branch  $T - ih$ .*

*Proof.*  $S(x) \in (L, L_2)$ , where  $\sigma > 0$ , while  $1 < L$  gives  $\sigma(1) < 0$ . Hence  $\sigma(S(x))/\sigma(1)$  is negative and  $\operatorname{Im} \log(\sigma(S(x))/\sigma(1)) \in \pi + 2\pi\mathbb{Z}$ ; divide by  $\log \lambda = -2\pi/h$ .  $\square$

Translations of  $S$  ( $S \mapsto S(\cdot + \delta)$ ,  $\delta \in \mathbb{C}$ ) send  $t_0 \mapsto t_0 + \delta$ ,  $t_n \mapsto t_n e^{2\pi i n \delta}$ , so

$$(7) \quad \tau_n(b) := t_n e^{-2\pi i n t_0} \quad (\operatorname{Im} t_0 = h/2)$$

does not depend on the normalisation of  $S$ ; it depends on that of  $R$  only through the real translation fixed by  $R(0) = 1$ . The condition  $\operatorname{Im} t_0 = h/2$  refers to the real normalisation of  $S$  and fixes the branch of  $R^{-1}$ ; a branch shift by  $ikh$  would multiply  $\tau_n$  by  $\Lambda^{-nk}$ . The branch is fixed once in the real normalisation and then carried along.

### 3.2. The welding identity.

**Theorem 3.2.** *Let  $K$  be a solution of (1), holomorphic on the sets below where it is represented, and suppose there are 1-periodic holomorphic maps  $\varphi_{\pm} = \operatorname{id} + (\text{periodic})$  with*

- (1)  $K = R \circ \varphi_+$  on an upper half-plane  $U_+$ , where  $P = \varphi_+ - \operatorname{id} = \sum_{n \geq 0} c_n e^{2\pi i n z}$  converges on  $U_+$ ;
- (2)  $K = S \circ \varphi_-$  on a lower half-plane  $U_-$ , where  $\varphi_-$  is univalent,  $G := \varphi_-^{-1} - \operatorname{id}$  is bounded on  $V_- = \varphi_-(U_-)$ , which contains a lower half-plane, and  $G(w) \rightarrow g_0$  as  $\operatorname{Im} w \rightarrow -\infty$ ;
- (3)  $U_+ \cap U_-$  contains a horizontal line  $\ell$  with  $\varphi_-(\ell)$  inside the strip of  $T$ .

Then, with the branch of  $T$  fixed by continuity from  $\varphi_+ \circ \varphi_-^{-1}$ , for every  $n \geq 1$

$$(8) \quad t_n = \sum_{m \geq n} c_m [e^{2\pi i m G}]_{n-m}, \quad t_0 = g_0 + \sum_{m \geq 0} c_m [e^{2\pi i m G}]_{-m},$$

where  $[F]_k$  is the  $k$ -th Fourier coefficient; in particular  $[e^{2\pi i m G}]_0 = e^{2\pi i m g_0}$  and  $[e^{2\pi i m G}]_k = 0$  for  $k > 0$ .

*Proof.* On  $\ell$ ,  $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$ , so  $\varphi_+ - T \circ \varphi_-$  takes values in the discrete period lattice of  $R$  and is constant; absorb it into the branch of  $T$ . Hence  $T = \varphi_+ \circ \varphi_-^{-1}$  on the curve  $\varphi_-(\ell)$ , i.e.  $T(w) - w = G(w) + P(w + G(w))$  there. The Fourier coefficients are computed by Cauchy's theorem on this curve. The region below it lies in  $V_-$ , and  $\sum c_m e^{2\pi i m \varphi_-^{-1}(w)}$  converges uniformly on the curve  $\varphi_-(\ell)$  itself, because  $\ell \subset U_+$ . Since  $G$  is bounded and holomorphic on  $V_-$ , with limit  $g_0$  at  $-i\infty$ , so is  $e^{2\pi i m G}$ , whose Fourier series therefore contains only modes  $k \leq 0$  and has constant term  $e^{2\pi i m g_0}$ . Expanding  $P(w + G) = \sum_m c_m e^{2\pi i m w} e^{2\pi i m G(w)}$  and collecting modes gives (8); the Fourier coefficients of  $T - \operatorname{id}$  do not depend on the line inside its strip of holomorphy.  $\square$

**Corollary 3.3.** *Under the hypotheses of Theorem 3.2, for every  $n \geq 1$ ,*

$$t_n e^{-2\pi i n g_0} = c_n + e^{-2\pi i n g_0} \sum_{m > n} c_m [e^{2\pi i m G}]_{n-m}.$$

If  $c_n \neq 0$ , this may be written as

$$t_n e^{-2\pi i n g_0} = c_n (1 + \rho'_n), \quad |\rho'_n| \leq \sum_{m>n} \frac{|c_m|}{|c_n|} |[e^{2\pi i m G}]_{n-m}| |e^{-2\pi i n g_0}|,$$

and  $g_0 = t_0 - c_0 - \sum_{m \geq 1} c_m [e^{2\pi i m G}]_{-m}$ . Heuristically, if the  $c_m$  decay like  $\Lambda^m$  and  $G$  is bounded on a line at height comparable to  $h$  above the gluing line, both corrections are  $O(\Lambda)$ . The precise statement for the sewing solution is Theorem 6.3(4). On the branch of Lemma 3.1,  $g_0 = t_0 - ih + O(\Lambda)$ , so that for the sewing solution, by Theorem 6.3(4),

$$(9) \quad \hat{c}_n := \frac{c_n}{\Lambda^n} = \tau_n(b) + O(\Lambda).$$

Since  $B_1 \neq 0$ , also  $c_1 \neq 0$  for  $b$  sufficiently close to  $\eta$  and  $\kappa_n := c_n/c_1^n = t_n/t_1^n + O(\Lambda)$  for each fixed  $n$ .

The identity  $e^{-2\pi i n(t_0 - ih)} = \Lambda^n e^{-2\pi i n t_0}$  is where the scale  $\Lambda$  enters: the  $\Lambda$  law is the statement that the gluing line sits at attracting time  $-h/2$ , half a period below the real axis, while the gate carrying the positive modes of  $T$  sits a further half period below it. Nothing in the argument uses Kneser's construction.

### 3.3. Welding construction.

**Proposition 3.4.** *For  $b \in (1, \eta)$  glue the half-cylinders  $C_- = \{\text{Im } w' \leq 0\}/\mathbb{Z}$  (repelling time) and  $C_+ = \{\text{Im } w \geq -h/2\}/\mathbb{Z}$  (attracting time) along their boundaries by  $w = T(w') - ih$ . The result is conformally  $\mathbb{C}^*$ ; its universal-cover coordinate  $z$  defines a solution*

$$K^W(z) = R(w(z)) \text{ on } C_+, \quad K^W(z) = S(w'(z)) \text{ on } C_-,$$

unique up to a translation in  $z$ , which satisfies the hypotheses of Theorem 3.2. It is holomorphic except where  $w(z)$  meets the singular set  $\bigcup_k ((-\infty, -2] + ikh)$  of  $R$ .

Unlike  $S$ , which is entire (the Poincaré function of a repelling point),  $R$  is not: its inverse Koenigs map extends only through inverse branches of  $E$ . So  $K^W$  carries, besides the cut near  $(-\infty, -2]$ , the copies near  $(-\infty, -2] + ikh$ ,  $k \geq 1$ , lying in its  $R$ -represented region. The same holds for the Kneser-type solutions at complex bases inside the Shell–Thron region, whose upper representation is through the attracting regular superfunction.

*Proof.* By Lemma 3.1 the gluing map is a real-analytic diffeomorphism between the boundary circles that extends holomorphically to a neighbourhood, so the glued surface  $X$  is a Riemann surface; each end is a punctured disc, hence  $X \cup \{\text{two points}\}$  is a simply connected compact Riemann surface and  $X \cong \mathbb{C}^*$ . The definitions agree on the seam because  $R \circ (T - ih) = S$ . The generator of  $\pi_1(X)$  lifts to  $z \mapsto z + 1$  and acts as  $w \mapsto w + 1$ ,  $w' \mapsto w' + 1$ , whence  $K^W(z + 1) = E(K^W(z))$ . Near each puncture the change of coordinate is a holomorphic map of punctured discs fixing the puncture, which is the statement that  $w - z$  (resp.  $w' - z$ ) has only non-negative (resp. non-positive) Fourier modes. Uniqueness is the uniqueness of the uniformisation.  $\square$

**Proposition 3.5** (Local parameter dependence of the sewn family). *For every  $b_0 \in (1, \eta)$ , the normalised geometric sewing of Proposition 3.4 extends to a holomorphic family over a complex disc  $\Delta_{b_0}$  about  $b_0$ . On every common regular domain of the universal-cover coordinate,  $K_b^W(z)$  is jointly holomorphic in  $(b, z)$ . These local families agree on overlaps and therefore define the inner sewn family on an open complex neighbourhood  $N_W$  of the entire real interval  $(1, \eta)$ , the union of the discs  $\Delta_{b_0}$ . No uniform width of that neighbourhood is asserted.*

*Proof.* Fix  $b_0$ . Both fixed points and their Koenigs coordinates vary holomorphically near  $b_0$ . The compact set  $S_{b_0}([0, 1]) \subset (L, L_2)$  lies in the attracting basin and avoids its fixed point. Thus the chosen branch of  $T_b = R_b^{-1} \circ S_b$  is jointly holomorphic on a parameter disc times a narrow strip about  $[0, 1]$ ; the relation  $T_b(w' + 1) = T_b(w') + 1$  extends it to the whole strip. The real circle map is orientation preserving:  $\tau'_b(S_b(x)) > 0$  by inverse iteration to  $L_2$ ,  $\sigma'_b(S_b(x)) > 0$  by forward iteration to  $L$ , and hence  $S'_b(x) < 0$  and  $(R_b^{-1})'(S_b(x)) = \sigma'_b(S_b(x))/(\sigma_b(S_b(x)) \log \lambda_b) < 0$ . Consequently  $T'_{b_0}(x) > 0$ . Compactness of the circle and local injectivity give a smaller strip on which  $T_{b_0}$  is injective modulo 1; shrinking the parameter disc preserves this property and makes the image of the real circle an analytic Jordan curve in the cylinder.

Put  $U_b = T_b + \varpi_b$ , where  $\varpi_b = 2\pi i / \log \lambda_b$ ; at  $b_0$  this is  $T_{b_0} - ih$ . Use  $U_b$  to identify a thin open annular collar in the repelling cylinder with its image in the attracting cylinder. More explicitly, for a sufficiently small  $\delta > 0$  take the repelling chart  $\{\text{Im } w' < \delta/2\}/\mathbb{Z}$  and the attracting chart above the curve  $U_b(\{\text{Im } w' = -\delta/2\})$ ; identify them on  $|\text{Im } w'| < \delta/2$ . After shrinking the parameter disc, these charts and the transition  $(b, w') \mapsto (b, U_b(w'))$  form a Hausdorff holomorphic family of cylinders. At  $b = b_0$  it is the analytic-collar version of the boundary welding in Proposition 3.4. Adjoin the two ends using the parameter independent local coordinates  $e^{2\pi i w}$  and  $e^{-2\pi i w'}$ . This yields a proper holomorphic submersion whose fibres are compact Riemann surfaces of genus zero. The two added points and the upper-chart point  $w = 0$  are three disjoint holomorphic sections (the last remains above the seam after shrinking the disc).

Local holomorphic triviality of a proper family with mutually biholomorphic compact fibres [8], followed by the unique fibrewise Möbius map sending these three sections to  $0, \infty, 1$ , gives a jointly holomorphic coordinate  $q_b$  on the compactified fibres. Choose the local lift  $z = (2\pi i)^{-1} \log q_b$  for which the marked upper point  $w = 0$  has  $z = 0$ . On the lift of the overlap,  $R_b(U_b(w')) = S_b(w')$  because  $\varpi_b$  is a period of  $R_b$ . The two regular superfunctions therefore define  $K_b^W$ , jointly holomorphic wherever their represented branches are regular, and  $K_b^W(0) = R_b(0) = 1$ . On the real parameter slice this is exactly the geometric welding. Its marked coordinate makes the local families unique, so the identity theorem patches them on overlapping discs.  $\square$

For non-real  $b$  sufficiently close to any fixed real  $b_0 \in (1, \eta)$ , the sewn solution thus depends holomorphically on  $b$ . Theorem 7.1 gives the stronger quantitative Wiener bounds near  $\eta$ . For  $\text{Im } b > 0$  near  $(b_1, \eta)$  it tends (Theorem 7.1(2)) to  $L$  and  $L_2$  as  $\text{Im } z \rightarrow \pm\infty$ , respectively. Those limits alone do not characterise it: Proposition 2 of [18] has the counterexample in Remark 1.2. The base-continued Kneser family is identified with  $K^W$  in Theorem D; the seam condition of Definition 8.1 is what selects the right root there. Table 1 gives numerical evidence:  $\tau_1$  and  $t_2/t_1^2$  reproduce the ladder values of  $\hat{c}_1$  and  $\kappa_2$  — moduli *and* arguments — to the precision of the ladder extrapolation, e.g. eight digits at  $b = 1.10$ .

TABLE 1. Kneser ladders (Sections 10.1, 10.2, extrapolated to  $\text{Im } b = 0$ ) against the Kneser-free invariants  $\tau_1 = t_1 e^{-2\pi i t_0}$  and  $t_2/t_1^2$  (script `transition_map.py`).

| $b$  | $\hat{c}_1$ (ladder)         | $\tau_1$                     | $\kappa_2$ (ladder)      | $t_2/t_1^2$              |
|------|------------------------------|------------------------------|--------------------------|--------------------------|
| 1.05 | $0.0850235 e^{1.534136i}$    | $0.0850226 e^{1.534165i}$    | $1.374049 e^{1.413420i}$ | $1.374336 e^{1.413645i}$ |
| 1.10 | $0.08642720 e^{1.55109214i}$ | $0.08642720 e^{1.55109214i}$ | $1.440851 e^{1.477690i}$ | $1.440851 e^{1.477690i}$ |
| 1.15 | $0.0872053 e^{1.561012i}$    | $0.0872052 e^{1.561013i}$    | $1.479111 e^{1.511255i}$ | $1.479107 e^{1.511258i}$ |
| 1.30 | 0.088434                     | $0.0884321 e^{1.577580i}$    | $1.541516 e^{1.561918i}$ | $1.541515 e^{1.561919i}$ |

**3.4. Reduction to confluence.** Write  $a := \Phi_{\text{att}}(1/e - 1) = 3.0292972144180360989 \dots$  for the  $\alpha$ -normalised attracting Fatou coordinate of the base point  $w = 1$  at  $b = \eta$  (real, since the orbit of  $u = 1/e - 1$  stays on the negative axis).

**Theorem 3.6.** *Assume the following confluence statement (C): there are constants  $s(b) \in \mathbb{C}$  such that, as  $b \rightarrow \eta^-$ ,*

$$R_b^{-1}(e(1+u)) \rightarrow \Phi_{\text{att}}(u) - a, \quad S_b^{-1}(e(1+u)) + s(b) \rightarrow \Phi_{\text{rep}}(u),$$

*uniformly on compact subsets of the upper gate, with the branches continued as in Section 2.1. Then for every  $n \geq 1$*

$$\tau_n(b) \longrightarrow B_n e^{2\pi i n a}.$$

*Consequently, also  $t_n/t_1^n \rightarrow B_n/B_1^n$  for every  $n \geq 1$ . Together with Corollary 3.3 and the identification of  $K_b$  with  $K_b^W$  in Theorem 9.14, this gives  $\hat{c}_n \rightarrow B_n e^{2\pi i n a}$  for all  $n$ . The limiting phase  $\arg \hat{c}_1 \rightarrow \arg B_1 + 2\pi a = 1.5865330757 \dots \pmod{2\pi}$ .*

*Proof.* Replace  $S_b$  by  $S_b(\cdot - s(b))$ , which leaves  $\tau_n$  unchanged. By (C),  $T_b \rightarrow \Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1} - a = h^{-1} - a$  uniformly on  $[0, 1] + iY_g$  for fixed  $Y_g$  in the gate, so the Fourier coefficients on that line converge:  $t_n \rightarrow B_n$  ( $n \geq 1$ ) and  $t_0 \rightarrow -a$ , the latter because  $B_0 = 0$  in the  $\alpha$  normalisation. Hence  $\tau_n = t_n e^{-2\pi i n t_0} \rightarrow B_n e^{2\pi i n a}$ .  $\square$

Statement (C) is a confluence theorem for the saddle-node unfolding  $u \mapsto e^{\lambda e^{-\lambda} e(1+u)}/e - 1$  in the direction where the two fixed points are real (the “Glutsyuk” direction): the Koenigs coordinates at the two hyperbolic points converge, on the gate, to the sectorial Fatou coordinates of the parabolic germ. Results of this type are due to Glutsyuk [9] and, for the full unfolded modulus, Mardešić–Roussarie–Rousseau [15] and Christopher–Rousseau [5]. Rather than matching their normalisations, we prove (C) directly for this family in Section 4, so Theorem 3.6 holds unconditionally.

#### 4. PROOF OF THEOREM A: CONFLUENCE

We prove (C) directly for the family at hand; no normalisation from the literature is needed. In  $u$ -coordinates ( $w = e(1+u)$ ,  $\mu = e \log b$ )

$$f_b(u) = e^{\mu-1+\mu u} - 1, \quad f_\eta(u) = e^u - 1.$$

For  $b < \eta$  close to  $\eta$  the map has, in a fixed disc  $D_r$ , exactly the two fixed points  $u_1 = e^{\lambda-1} - 1 < 0 < u_2 = e^{\lambda_2-1} - 1$  (Rouché), and

$$(10) \quad f_b(u) - u = (u - u_1)(u - u_2) k_b(u), \quad k_b \rightarrow k_\eta, \quad k_\eta(u) = \frac{1}{2} + \frac{u}{6} + \dots,$$

uniformly on  $D_r$ , with  $k_b$  zero-free there. Put  $A_j = 1/\log \lambda_j$ , so  $A_1 < 0 < A_2$ .

**Lemma 4.1** (Residues). *As  $b \rightarrow \eta^-$ :  $A_1(u_1 - u_2) \rightarrow 2$ ,  $A_1 + A_2 \rightarrow \nu := 1 - 4k'_\eta(0) = \frac{1}{3}$ , hence  $A_1 u_1 + A_2 u_2 \rightarrow 2$  and  $A_j u_j^2 \rightarrow 0$ .*

*Proof.*  $\lambda_1 = 1 + (u_1 - u_2)k_b(u_1)$  and  $\lambda_2 = 1 + (u_2 - u_1)k_b(u_2)$ . With  $d = u_1 - u_2 \rightarrow 0$  and  $1/\log(1+x) = 1/x + \frac{1}{2} + O(x)$ ,  $A_1 + A_2 = \frac{k_b(u_2) - k_b(u_1)}{d k_b(u_1) k_b(u_2)} + 1 + O(d) \rightarrow 1 - 4k'_\eta(0)$ , and  $A_1 d \rightarrow 1/k_\eta(0) = 2$ .  $\square$

Numerically  $A_1 + A_2 = 0.3333331469$  and  $A_1(u_1 - u_2) = 1.99676$  at  $\lambda = 0.99$ . Define the *model time*

$$\Psi_b(u) = A_1 \log(u - u_1) + A_2 \log(u - u_2),$$

which is exactly the time of the vector field  $(u - u_1)(u - u_2)/(\dots)$  whose time-one map has multipliers  $\lambda_1, \lambda_2$ , and let  $\alpha_2(u) = -2/u + \frac{1}{3} \log(-u)$ . By Lemma 4.1, for  $u, p$  in a simply connected region at positive distance from 0, with logarithms continued along the same path,

$$(11) \quad \Psi_b(u) - \Psi_b(p) \longrightarrow \alpha_2(u) - \alpha_2(p).$$

**Lemma 4.2** (Uniform one-step error).  *$F_b := \Psi_b \circ f_b - \Psi_b - 1$  is single-valued and holomorphic on a fixed disc  $D_{r'}$ ,  $F_b \rightarrow F_\eta := \alpha_2 \circ f_\eta - \alpha_2 - 1$  uniformly there, and*

$$|F_b(u)| \leq M |u - u_1| |u - u_2| \quad (u \in D_{r'/2})$$

with  $M$  independent of  $b$ .

*Proof.* By (10),  $(f(u) - u_1)/(u - u_1) = 1 + (u - u_2)k(u)$  and  $(f(u) - u_2)/(u - u_2) = 1 + (u - u_1)k(u)$ , so

$$F = A_1 \log \frac{1 + (u - u_2)k}{1 + (u - u_1)k} + (A_1 + A_2) \log(1 + (u - u_1)k) - 1,$$

holomorphic for  $|u| < r'$  small. The first logarithm is  $O(|u_1 - u_2|)$  uniformly, and  $A_1(u_1 - u_2)$ ,  $A_1 + A_2$  are bounded (Lemma 4.1); so  $|F| \leq M_0$  on  $|u| = r'$  uniformly, and the displayed formula converges. At  $u = u_1$  the first term is  $A_1 \log \lambda_1 = 1$  and the second vanishes, so  $F(u_1) = 0$ ; likewise  $F(u_2) = 0$ . The maximum principle applied to  $F/((u - u_1)(u - u_2))$  on  $D_{r'}$  gives the bound, and letting  $b \rightarrow \eta$  gives  $F_\eta = O(u^2)$ .  $\square$

**Lemma 4.3** (Uniform petals and summability). *Put  $Z_b(u) = A_1 \log \frac{u - u_1}{u - u_2}$  and  $\Omega_{R,b} = \{u : \operatorname{Re} Z_b(u) > R\}$  (a disc of Apollonius about  $u_1$ ). There is  $R_0$  such that for  $R \geq R_0$  and all  $b$  near  $\eta$ :  $\Omega_{R,b} \subset D_{r'/2}$ ,  $f_b(\Omega_{R,b}) \subset \Omega_{R,b}$ ,  $\operatorname{Re} Z_b(f_b^k u) \geq \operatorname{Re} Z_b(u) + \frac{3}{4}k$ , and*

$$\sum_{k \geq N} |f_b^k u - u_1| |f_b^k u - u_2| \leq \frac{C}{R + \frac{3}{4}N} \quad (u \in \Omega_{R,b}),$$

with  $C$  independent of  $b$  and  $u$ . As  $b \rightarrow \eta$ ,  $\Omega_{R,b}$  converges on compacta to the petal  $\{\operatorname{Re}(-2/u) > R\}$ .

*Proof.*  $Z(f(u)) - Z(u) = A_1 \log \frac{1 + (u - u_2)k}{1 + (u - u_1)k} = 1 + O(|u| + |u_1| + |u_2|)$  uniformly, by the expansion in the proof of Lemma 4.2; on  $\Omega_{R,b} \subset D_{C/R + |u_1|}$  this is within  $\frac{1}{4}$  of 1 once  $R$  is large, which gives invariance and the growth of  $\operatorname{Re} Z$ . With  $q = (u - u_1)/(u - u_2)$ ,  $|q| = e^{-\sigma}$ ,  $\sigma = |\log \lambda_1| \operatorname{Re} Z$ , one has  $u - u_2 = (u_2 - u_1)/(q - 1)$ , whence

$$|u - u_1| |u - u_2| = \frac{|q| |u_2 - u_1|^2}{|1 - q|^2} \leq \left( \frac{|u_2 - u_1|}{|\log \lambda_1|} \right)^2 \frac{(1 + \sigma)^2 e^{-\sigma}}{(\operatorname{Re} Z)^2},$$

using  $1 - e^{-\sigma} \geq \sigma/(1 + \sigma)$ . The first factor is bounded (Lemma 4.1) and  $(1 + \sigma)^2 e^{-\sigma} \leq 4$ ; summing  $(R + \frac{3}{4}k)^{-2}$  over  $k \geq N$  gives the bound. The last assertion follows from  $\operatorname{Re} Z_b(u) \rightarrow \operatorname{Re}(-2/u)$ , i.e.  $A_1(u_2 - u_1) \rightarrow -2$ .  $\square$

**Theorem 4.4** (Confluence). *Statement (C) holds. More precisely, with branches continued along paths in the respective parabolic basins:*

- (1)  $R_b^{-1}(e(1 + u)) \rightarrow \Phi_{\text{att}}(u) - a$  uniformly on compact subsets of the parabolic basin of  $f_\eta$ ;
- (2) for a fixed  $p'$  in the gate (below), putting  $s(b) = \Phi_{\text{rep}}(p') - \mathcal{S}_b(p')$ ,  $\mathcal{S}_b(u) + s(b) \rightarrow \Phi_{\text{rep}}(u)$  uniformly on compact subsets of the repelling basin (backward orbits tending to 0). Here  $\mathcal{S}_b$  is the real-normalised repelling time, i.e. the branch of  $S_b^{-1}$  that is real on the segment  $(u_1, u_2)$  and is continued through the band of Lemma 6.1, then along backward-invariant paths.

The gate of height  $Y_*$  is  $\mathcal{G}(Y_*) = \Phi_{\text{rep}}^{-1}(\{Y_* \leq \text{Im } z \leq Y_* + 1, -1 \leq \text{Re } z \leq 2\})$ . For  $Y_*$  large it is a compact subset of both parabolic basins of  $f_\eta$ . For  $b$  near  $\eta$  it lies in the band  $\mathcal{B}_b(Y_* - 2)$ , because  $Z_b \rightarrow -2/u$  uniformly on it (Lemma 4.1) and  $\text{Im}(-2/u) = \text{Im } \alpha(u) + O(1) = \text{Im } \Phi_{\text{rep}}(u) + O(1)$  there.

*Proof.* (1) On  $\Omega_{R,b}$  put  $\mathcal{A}_b(u) = \lim_n [\Psi_b(f_b^n u) - n] = \Psi_b(u) + \sum_{k \geq 0} F_b(f_b^k u)$ ; the series converges absolutely by Lemmas 4.2 and 4.3. Since  $f^n u \rightarrow u_1$  and the Koenigs map satisfies  $\sigma(f^n u) = \lambda_1^n \sigma(u)$ ,  $\Psi_b(f^n u) - n = A_1 \log \sigma(u) + A_2 \log(u_1 - u_2) + o(1)$ , because  $A_1 \log \lambda_1 = 1$ . Thus  $\mathcal{A}_b$  is the Koenigs Abel coordinate up to an additive constant, and  $R_b^{-1}(e(1+u)) = \mathcal{A}_b(u) - \mathcal{A}_b(u^*)$ ,  $u^* = 1/e - 1$ . Let  $K$  be a compact subset of the petal  $P_R = \{\text{Re}(-2/u) > R\}$  and  $U \supset K \cup \{f_\eta^{N_0} u^*\}$  a simply connected region in  $P_R$  at positive distance from 0. For  $b$  near  $\eta$ ,  $U \subset \Omega_{R,b}$  and  $u_1, u_2 \notin U$ . For  $u, p \in U$ ,

$$\mathcal{A}_b(u) - \mathcal{A}_b(p) = \Psi_b(u) - \Psi_b(p) + \sum_{k \geq 0} [F_b(f_b^k u) - F_b(f_b^k p)].$$

The first term converges by (11). In the sum, each term converges because  $f_b^k \rightarrow f_\eta^k$  and  $F_b \rightarrow F_\eta$  uniformly. The tails beyond  $N$  are  $\leq 2MC/(R + \frac{3}{4}N)$  uniformly in  $b$  (Lemma 4.3). Hence the difference converges uniformly to  $\alpha_2(u) - \alpha_2(p) + \sum_k [F_\eta(f_\eta^k u) - F_\eta(f_\eta^k p)] = \Phi_{\text{att}}(u) - \Phi_{\text{att}}(p)$ . The last identity holds because  $\Phi_{\text{att}} = \lim[\alpha(f^n u) - n]$  and  $\alpha - \alpha_2 = O(u) \rightarrow 0$  along orbits. A general compact  $K$  of the basin satisfies  $f_\eta^N(K) \Subset P_R$  for some  $N$ . Then  $f_b^N(K) \Subset P_R$  for  $b$  near  $\eta$ , and  $\mathcal{A}_b = \mathcal{A}_b \circ f_b^N - N$ . Taking  $p = f_b^{N_0} u^*$ , which tends to  $f_\eta^{N_0} u^* \in U$ , and using uniform convergence on a neighbourhood gives (1). (On  $\Omega_{R,b}$  the function  $\mathcal{A}_b$  is multivalued, since this punctured Apollonius disc surrounds  $u_1$ . It is single-valued on  $U$ , which avoids  $u_1$ . On  $K$ ,  $\sigma_b \neq 0$  because  $f_b^N(K)$  avoids  $u_1$ . On non-immediate components of the basin the branch is defined by  $\mathcal{A}_b \circ f_b^N - N$ . The Koenigs map is normalised by  $\sigma'_b(u_1) = 1$ .)

(2) Apply (1)'s argument to  $G_b(v) = -f_b^{-1}(-v)$ , the principal inverse branch reflected. Then  $G_\eta(v) = -\log(1-v) = v + \frac{v^2}{2} + \frac{v^3}{3} + \dots$ , which is of the same prepared form with  $k'_\eta(0) = \frac{1}{3}$  and residue  $-\frac{1}{3}$ . The attracting fixed point of  $G_b$  is  $-u_2$ , with multiplier  $1/\lambda_2$ . Its Koenigs Abel coordinate is  $-\mathcal{S}_b^{-1}(e(1-v))$  up to a constant, and its parabolic Fatou coordinate is  $-\Phi_{\text{rep}}(-v)$  up to a constant (since  $\alpha_f(-v) = -\alpha_G(v) + \text{const}$ ). The free constant is fixed by  $p'$ .  $\square$

With Theorem 4.4, Theorem 3.6 becomes unconditional. Four points need checking; each is settled here. The proof of Lemma 6.1 uses only Lemmas 4.1–4.3, so there is no circularity.

(a) *Inversion on the gate line.* Fix  $Y_* \geq Y_B + 6$  and the line  $\ell = [0, 1] + iY_*$ . By Rouché, the solution  $v_b(z)$  of  $\mathcal{S}_b(v) + s(b) = z$  near  $\Phi_{\text{rep}}^{-1}(z)$  exists and converges uniformly on  $\ell$ . Moreover  $\mathcal{S}_b(z - s(b)) = e(1 + v_b(z))$ , because  $\mathcal{S}_b$  is a branch of  $\mathcal{S}_b^{-1}$ .

(b) *The gate line lies in the strip of holomorphy of  $T$ .* The points  $v_b(z)$ ,  $z \in \ell$ , lie within  $o(1)$  of the gate, hence in the band  $\mathcal{B}_b(Y_B + 3)$ . By Lemma 6.1(ii) their  $S$ -times  $z - s(b)$  satisfy  $|\text{Im}(z - s(b))| \leq h/2 - Y_B - 3 + \pi/3 + C/Y_B + C\varepsilon < h/2 - Y_B - 2$ . So the shifted line  $\ell - s(b)$  lies in the strip  $\Sigma$  of Lemma 6.1(iii), where  $T$  is holomorphic. Therefore the Fourier coefficients of  $\tilde{T} = T(\cdot - s)$  on  $\ell$  are  $t_n e^{-2\pi i n s}$  and  $\tilde{t}_0 = t_0 - s$ , with the  $t_n$  those of  $T$  on  $\mathbb{R}$ . No prior knowledge of  $\text{Im } s$  is used.

(c) *The branch of  $R_b^{-1}$ .* Continue  $R_b^{-1}$  from  $u^*$  along the negative axis to  $f_\eta^{N_0} u^*$ , where it is real. Then go inside  $P_R \setminus D_\delta(0)$  through the upper half-plane to the gate, and then inside the band down to the segment  $(u_1, u_2)$ . This path passes above  $u_1$  and does not wind around  $u_2$ . Along it  $R_b^{-1} = \Psi_b + \sum_k F_b \circ f_b^k + c_R$ , so the change of  $\text{Im } R_b^{-1}$  is  $A_1 \Delta \arg(u - u_1) + A_2 \Delta \arg(u - u_2) + O(C/Y + \varepsilon) = h/2 + 0 + \text{small}$ . Lemma 3.1 forces the exact value, so on the segment  $\text{Im } R_b^{-1} = +h/2$ . The branch in Theorem 4.4(1) is therefore the one with  $\text{Im } t_0 = h/2$  used in (7).

(d)  $B_0 = 0$ . At  $b = \eta$ , on the gate,  $\Phi_{\text{att}} - \alpha_2 = \sum_{k \geq 0} F_\eta(f^k u)$  and  $\Phi_{\text{rep}} - \alpha_2 = -\sum_{k \geq 1} F_\eta(f^{-k} u)$ , with the same principal  $\log(-u)$  in  $\alpha_2$ . That logarithm stays continuous along both orbits, since  $-f^{\pm k} u$  remains in the lower half-plane. By Lemma 4.2 and  $f^{\pm k} u \approx -2/(x \pm k + iY)$ , both sums are  $O(1/Y)$ . Hence  $\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}(z) - z \rightarrow 0$  as  $\text{Im } z \rightarrow +\infty$ , so its constant Fourier mode vanishes:  $B_0 = 0$ . Numerically  $|B_0| < 10^{-59}$ .

By (a)–(d) and Theorem 4.4,  $\tilde{T} \rightarrow h^{-1} - a$  uniformly on  $\ell$ , where  $h^{-1}$  denotes the inverse horn map (not the period  $h$ ). So  $\tilde{t}_n \rightarrow B_n$  and  $\tilde{t}_0 \rightarrow -a$ , and  $\tau_n = \tilde{t}_n e^{-2\pi i n \tilde{t}_0}$  gives:

**Corollary 4.5.** *For every  $n \geq 1$ ,  $\tau_n(b) \rightarrow B_n e^{2\pi i n a}$  as  $b \rightarrow \eta^-$ . Since  $B_1 \neq 0$ , also  $t_n/t_1^n \rightarrow B_n/B_1^n$  for every  $n \geq 1$ .*

A direct pointwise check of Theorem 4.4 on two gate points (`confluence_check.py`) gives errors 0.086, 0.021, 0.0050, 0.00079 (attracting side) at  $\lambda = 0.8, 0.9, 0.95, 0.98$ , decreasing like  $\varepsilon^2$ .

## 5. THE FIRST-ORDER TERM: A PARABOLIC SERIES

Theorem A gives the limit of  $\tau_n$ . Here we prove that  $\tau_n$  is differentiable at  $b = \eta$  from the side  $b < \eta$ . The derivative is given by an absolutely convergent series computed entirely at the parabolic germ. Put

$$s = 1 - e \log b = 1 - \mu, \quad f_s(u) = e^{-s+(1-s)u} - 1,$$

so  $s \in (0, s_0]$  corresponds to  $b < \eta$  near  $\eta$  and  $f_0(u) = e^u - 1$ . The map  $(s, u) \mapsto f_s(u)$  is entire.

**5.1. Why termwise differentiation fails, and the remedy.** With the model time  $\Psi_b$  of Section 4, the one-step error is only  $O(|u - u_1||u - u_2|)$  (Lemma 4.2). Along a parabolic orbit  $u_k \approx -2/k$  one has  $\partial_s u_k \asymp k$ . Hence both  $\partial_s F(u_k)$  and  $F'(u_k) \partial_s u_k$  are of order one, and the differentiated series diverges. The remedy is the idea behind the prepared form of Mardešić–Roussarie–Rousseau [15]. We add to the model time a polynomial correction, holomorphic in  $s$ , that makes the one-step error vanish to order  $N$  at both fixed points simultaneously.

By the Weierstrass preparation theorem,

$$(12) \quad f_s(u) - u = q_s(u) k_s(u), \quad q_s(u) = u^2 - \sigma_1(s)u + \sigma_2(s),$$

on a polydisc  $|s| < s_0$ ,  $|u| < r$ . Here  $\sigma_1, \sigma_2$  are holomorphic,  $k_s$  is holomorphic and zero-free, and  $k_0(0) = \frac{1}{2}$ . From  $f_s(0) = -s + O(s^2)$  and  $f'_s(0) - 1 = -2s + O(s^2)$  we get  $\sigma_2 = -2s + O(s^2)$ ,  $\sigma_1 = O(s)$ , and the discriminant

$$D(s) = \sigma_1^2 - 4\sigma_2 = 8s + O(s^2)$$

is biholomorphic near 0. For real  $s > 0$  the roots  $u_1 < 0 < u_2$  are the fixed points of Section 4. Put  $t = \sqrt{D(s)}$ . Then  $u_{1,2} = \frac{1}{2}(\sigma_1 \mp t)$  are holomorphic functions of  $t$ , interchanged by  $t \mapsto -t$ . Since  $f'_s(u_1) = 1 + (u_1 - u_2)k_s(u_1)$ , the residues are  $A_1 = a_s(u_1, u_2)$  and  $A_2 = a_s(u_2, u_1)$ , where

$$a_s(x, y) = \frac{1}{\log(1 + (x - y)k_s(x))} = \frac{1}{(x - y)k_s(x)} + \frac{1}{2} + (x - y)h_s(x, y)$$

with  $h_s$  holomorphic.

**Lemma 5.1** (Uniform model). *Fix  $N \geq 1$ . After shrinking  $s_0$  and  $r$ , there are holomorphic functions  $e_1, \dots, e_{2N-2}$  on  $\{|s| < s_0\}$  with the following property. Put  $P_s(u) = \sum_i e_i(s)u^i$  and, for  $s \neq 0$ ,*

$$\Psi_s(u) = A_1 \log(u - u_1) + A_2 \log(u - u_2) + P_s(u).$$

*Then:*

- (1)  $\nu(s) = A_1 + A_2$  extends holomorphically to  $|s| < s_0$ . For every  $\delta \in (0, r)$  there is  $s_\delta > 0$  such that  $\Psi_s(u) - \nu(s) \log u$  extends to a holomorphic function of  $(s, u) \in \{|s| < s_\delta\} \times \{\delta < |u| < r\}$ . Here, for  $|u| > \delta$ , the branches are fixed by  $\log(u - u_j) := \log u + \text{Log}(1 - u_j/u)$  with one common branch of  $\log u$ . Individually  $A_j \sim \pm 1/\sqrt{2}s$ , so any other convention would introduce  $2\pi i A_j$ -jumps that are not holomorphic in  $s$ .
- (2) The one-step error

$$F_s(u) = A_1 \log \frac{f_s(u) - u_1}{u - u_1} + A_2 \log \frac{f_s(u) - u_2}{u - u_2} + P_s(f_s(u)) - P_s(u) - 1$$

(principal logarithms) extends to a holomorphic function on  $\{|s| < s_0\} \times \{|u| < r\}$ , and  $F_s = q_s^N \Gamma_s$  with  $\Gamma$  holomorphic and  $|\Gamma| \leq M$  there.

*Proof. Removability.* Let  $\Theta(s, x, y, u) = a_s(x, y)\ell(y) + a_s(y, x)\ell(x)$ , where  $\ell$  is holomorphic in  $(s, x, y, u)$ . By the expansion of  $a_s$ , its singular part is

$$\frac{\ell(y)/k_s(x) - \ell(x)/k_s(y)}{x - y}.$$

The numerator vanishes on  $x = y$ , so  $\Theta$  extends holomorphically across the diagonal. Evaluate  $\Theta$  at  $(x, y) = (u_1, u_2)$ . The result is a holomorphic function of  $t$ , even in  $t$ , hence a holomorphic function of  $t^2 = D(s)$ , hence of  $s$ . The same holds for the other pairing  $\Theta'(s, x, y, u) = a_s(x, y)\ell(x) + a_s(y, x)\ell(y)$ , whose singular part  $[\ell(x)/k_s(x) - \ell(y)/k_s(y)]/(x - y)$  is again removable and symmetric. These arguments apply to the following choices of  $\ell$ :

- $\ell \equiv 1$  in  $\Theta'$ , which gives  $\nu$ ;
- $\ell(x) = \log(1 - x/u)$  in  $\Theta'$ , which gives  $\Psi_s - \nu \log u - P_s$  for  $|u| > \delta$  once  $|u_j| < \delta/2$ ;
- $\ell(x) = \log(1 + (u - x)k_s(u))$  in  $\Theta$ , which gives  $F_s^{\log} := A_1 \log \frac{f_s - u_1}{u - u_1} + A_2 \log \frac{f_s - u_2}{u - u_2} - 1$ , because  $(f_s(u) - u_1)/(u - u_1) = 1 + (u - u_2)k_s(u)$ .

This proves (1) and the holomorphy of  $F^{\log}$ .

*Divisibility.* For  $s \neq 0$ ,  $F_s^{\log}(u_j) = A_j \log f'_s(u_j) - 1 = 0$  (as in Lemma 4.2). Weierstrass division by  $q_s$  has a remainder  $r_0(s) + r_1(s)u$  that is holomorphic in  $s$  and vanishes at two distinct points for  $s \neq 0$ . So the remainder is 0, and  $F^{\log} = q_s \varphi$  with  $\varphi$  holomorphic. Likewise  $D_i := (f_s(u))^i - u^i = q_s d_i$  (powers, not iterates). At  $s = 0$ ,  $D_i = u^i((1 + \frac{u}{2} + \dots)^i - 1)$ , so  $d_i = \frac{i}{2}u^{i-1} + O(u^i)$ .

*The linear system.* Divide  $\varphi + \sum e_i d_i$  by the Weierstrass polynomial  $q_s^{N-1}$  of degree  $2N - 2$ . The  $2N - 2$  coefficients of the remainder depend holomorphically on  $s$  and affinely on  $e$ , with matrix  $M(s)$ . At  $s = 0$  the remainder is the Taylor polynomial modulo  $u^{2N-2}$ , so  $M(0)$  is lower triangular with diagonal entries  $i/2 \neq 0$ . Hence  $e(s) = -M(s)^{-1} \text{rem}(\varphi)$  is holomorphic near 0. Then  $F_s = F_s^{\log} + \sum e_i D_i = q_s(\varphi + \sum e_i d_i)$  is divisible by  $q_s \cdot q_s^{N-1}$ . The bound on  $\Gamma$  follows after shrinking the polydisc.  $\square$

Only the one-step error enters the estimates below, and it is single-valued. The logarithms in  $\Psi_s$  are continued along orbits exactly as in the proof of Theorem 4.4. Since  $P_s$  is single-valued, replacing  $\Psi_b$  by  $\Psi_s$  changes no branch. The path of item (c) in Section 4 runs outside  $D_\delta(0)$  and does not pass between  $u_1$  and  $u_2$ . So along it the convention of Lemma 5.1(1) agrees with the continuation used there.

**5.2. Orbits for complex parameters.** Let  $P_R = \{\text{Re}(-2/u) > R\}$  and put  $\zeta = -2/u$ .

**Lemma 5.2** (Finite orbits). *Let  $K \subset P_R$  be compact, with  $R$  large, and let  $Z = \max_K |2/u|$ . There is  $c > 0$  with the following property. For every integer  $J \geq 1$ , every  $\sigma \in \mathbb{C}$  with  $|\sigma| \leq c/J^2$ , every  $u \in K$  and every  $0 \leq k \leq J$ , the iterate  $u_k = f_\sigma^k(u)$  is defined and  $\zeta_k = -2/u_k$  satisfies*

$$\text{Re } \zeta_k \geq \text{Re } \zeta_0 + k/2, \quad |\zeta_k| \leq |\zeta_0| + 2k.$$

In particular  $|u_k| \leq 2/(R + k/2)$ . For  $\sigma = 0$  this holds for all  $k$ .

*Proof.* For  $|u| \leq r$  we have  $-2/f_0(u) = -2/u + 1 + \epsilon(u)$  with  $|\epsilon(u)| \leq C_0|u|$ , and  $|f_0(u)| \geq |u|/2$ . Write  $f_\sigma = f_0 + \sigma g$  with  $|g| \leq G$ . Suppose the claim holds up to  $k < J$ . Then  $|u_k| \leq 2/R \leq r$  and  $w := \sigma g(u_k)/f_0(u_k)$  satisfies

$$|w| \leq G|\sigma||\zeta_k| \leq Gc(Z + 2) \leq \frac{1}{2}.$$

Hence

$$\zeta_{k+1} = \frac{-2}{f_0(u_k)}(1 + w)^{-1} = \zeta_k + 1 + e_k, \quad |e_k| \leq \frac{2C_0}{R} + 4G|\sigma||\zeta_k|^2 \leq \frac{2C_0}{R} + 4Gc(Z + 2)^2.$$

This is at most  $\frac{1}{2}$  for  $R \geq 8C_0$  and  $c$  small. So  $\operatorname{Re} \zeta_{k+1} \geq \operatorname{Re} \zeta_k + \frac{1}{2}$  and  $|\zeta_{k+1}| \leq |\zeta_k| + \frac{3}{2}$ .  $\square$

**Lemma 5.3.** *Let  $u \in K$ ,  $u_k = f_0^k(u)$  and  $\dot{u}_k = \partial_\sigma f_\sigma^k(u)|_{\sigma=0}$ . Then*

$$|\dot{u}_k| \leq C(k + 1), \quad |\partial_\sigma F_\sigma(v)|_{\sigma=0} \leq M'|v|^{2N-2}, \quad |F'_0(v)| \leq M'|v|^{2N-1} \quad (|v| \leq r/2).$$

*Proof.* For the first bound, apply Cauchy's estimate on  $|\sigma| = c/(k+1)^2$  together with Lemma 5.2 (with  $J = k+1$ ). This gives  $|\dot{u}_k| \leq \frac{2}{R+k/2} \cdot \frac{(k+1)^2}{c}$ . For the other two, write  $F_\sigma = q_\sigma^N \Gamma_\sigma$  with  $q_0(v) = v^2$  and  $\partial_\sigma q_\sigma$  bounded, and bound  $\partial_\sigma \Gamma$  and  $\Gamma'_0$  by Cauchy's estimates.  $\square$

**5.3. One-sided differentiability.** For  $u$  in a compact  $K \subset P_R$  and real  $s \in [0, s_0]$  put

$$\mathcal{A}_s(u) = \Psi_s(u) + \sum_{k \geq 0} F_s(f_s^k u),$$

with a fixed branch of  $\log u$  on a simply connected neighbourhood  $U \subset P_R$  of  $K$ . For  $s > 0$  this is the attracting Koenigs Abel coordinate up to an additive constant, by the argument of Theorem 4.4(1). For  $s = 0$ ,  $\mathcal{A}_0 = \Phi_{\text{att}} + \text{const}$ , because  $\Psi_0 - \alpha = \text{const} + O(u)$ .

**Proposition 5.4.** *Let  $N \geq 3$ . Uniformly for  $u \in K$ , as  $s \rightarrow 0^+$ ,*

$$\mathcal{A}_s(u) = \mathcal{A}_0(u) + s\dot{\mathcal{A}}(u) + O(s^{10/9}),$$

where

$$(13) \quad \dot{\mathcal{A}}(u) = \partial_s \Psi_s(u)|_{s=0} + \sum_{k \geq 0} \left[ \partial_\sigma F_\sigma(u_k)|_{\sigma=0} + F'_0(u_k) \dot{u}_k \right], \quad u_k = f_0^k(u).$$

The series converges absolutely and uniformly on  $K$ , with terms  $O(k^{2-2N})$ . Hence  $\dot{\mathcal{A}}$  is holomorphic on the interior of  $K$ .

*Proof.* The  $\Psi$ -part is holomorphic in  $s$  by Lemma 5.1(1), so it contributes  $O(s^2)$ . Fix  $J \geq 1$  and split the sum at  $J$ .

*Head.*  $H_J(\sigma) = \sum_{k < J} F_\sigma(f_\sigma^k u)$  is holomorphic on  $|\sigma| < c/J^2$ . There  $|u_j(\sigma)| \leq C\sqrt{|\sigma|} \leq C\sqrt{c}/J$ . Since  $k < J$  gives  $1/J \leq (R+1)/(R+k/2)$ , Lemma 5.2 yields  $|q_\sigma(u_k)| \leq C_3(R+k/2)^{-2}$ . Therefore

$$|H_J(\sigma)| \leq MC_3^N \sum_k (R+k/2)^{-2N} =: M_1,$$

uniformly in  $J$ . Cauchy's estimate gives  $|H_J''| \leq 8M_1 J^4/c^2$  on  $[0, c/(2J^2)]$ , so

$$|H_J(s) - H_J(0) - sH_J'(0)| \leq 4M_1 c^{-2} s^2 J^4.$$

*Tails.* We may assume  $K \subset P_{R+1}$ . Since  $Z_s \rightarrow -2/u$  uniformly on  $K$  (Lemma 4.1),  $K \subset \Omega_{R,s}$  for all small  $s > 0$ . If  $r$  was shrunk in Lemma 5.1, increase  $R$  so that  $\Omega_{R,s} \subset D_{r/2}$ , and reach a general  $K$  by  $f^m$  as below. For real  $s \in (0, s_0]$  the proof of Lemma 4.3 gives  $|q_s(f_s^k u)| \leq C_1(R + \frac{3}{4}k)^{-2}$ . For  $s = 0$ , Lemma 5.2 gives the analogous bound  $|u_k|^2 \leq 4(R+k/2)^{-2}$ .

Hence  $|\sum_{k \geq J} F_s(f_s^k u)| \leq C_4 J^{1-2N}$  for all  $s \in [0, s_0]$ . By Lemma 5.3, the terms of (13) are  $O((R + k/2)^{2-2N})$ . So  $H'_J(0)$  differs from the series by at most  $C_5 J^{3-2N}$ .

*Balance.* For  $s \leq c/(2J^2)$ ,

$$|\mathcal{A}_s - \mathcal{A}_0 - s\dot{\mathcal{A}}| \leq Cs^2 + 4M_1 c^{-2} s^2 J^4 + 2C_4 J^{1-2N} + C_5 s J^{3-2N}.$$

Take  $J = \lceil s^{-2/9} \rceil$ , which is admissible for small  $s$ . With  $N \geq 3$  every term is  $O(s^{10/9})$ .  $\square$

**Remark 5.5.** Taking  $J = s^{-2/(2N+3)}$  instead gives the error  $O(s^{2-8/(2N+3)})$ , so  $O(s^{2-\epsilon})$  for any  $\epsilon > 0$  with  $N$  large. The proof needs  $N \geq 3$  only because of the crude bound on the head ( $N = 2$  would give  $s^{6/7}$ ). The series converges for  $N \geq 2$ .

The derivative does not depend on  $N$ . The condition  $F_0 = O(u^{2N})$  forces  $\Psi_0$  to agree with the formal Abel function  $\alpha$  up to a constant and  $O(u^{2N-1})$ , so the  $e_i(0)$  are the coefficients of  $\alpha$  and do not depend on  $N$ . Hence  $\mathcal{A}_s^{(N)} - \mathcal{A}_s^{(N')}$ , which equals the difference of the two polynomials evaluated at  $u_1(s) = O(\sqrt{s})$ , is a constant plus  $O(s^{3/2})$ . In any case additive constants cancel in  $\dot{\mathcal{A}}(v) - \dot{\mathcal{A}}(u^*)$ . Numerically,  $N = 3, 4, 6, 8$  give the same  $\kappa^{(1)}$  to 16–20 digits.

**5.4. The derivative of  $\tau_n$ .** Compact subsets of the basin are handled by  $\mathcal{A}_s = \mathcal{A}_s \circ f_s^m - m$ , where  $m$  is chosen so that  $f_0^m(K) \subseteq P_R$ . The iterate  $f_s^m$  is jointly holomorphic, so the expansion of Proposition 5.4 persists, with  $\dot{\mathcal{A}} = \dot{\mathcal{A}} \circ f_0^m + \mathcal{A}'_0 \circ f_0^m \cdot \partial_s f_s^m$ .

For the repelling side, apply everything to the reflected family  $G_s(v) = -f_s^{-1}(-v) = -(\log(1-v) + s)/(1-s)$ , exactly as in Theorem 4.4(2). This family is again of the form (12), with fixed points  $-u_2, -u_1$ . Let  $\mathcal{S}_s(u) = -\mathcal{A}_s^G(-u)$ . By Remark 2.3 and (7),  $\tau_n$  does not depend on the additive normalisation of the repelling coordinate. We use  $\mathcal{S}_s^\# = \mathcal{S}_s - \mathcal{S}_s(p') + \Phi_{\text{rep}}(p')$ . This differs from the real normalisation by the constant called  $s(b)$  in Theorem 4.4(2), which we denote  $s_*(b)$  here to avoid a clash with the parameter  $s$ , and it is right-differentiable with derivative  $\dot{\mathcal{S}}^\#$ . On the attracting side,  $R_b^{-1}(e(1+u)) = \mathcal{A}_s(u) - \mathcal{A}_s(u^*)$ , with the branch fixed in item (c) of Section 4.

**Theorem 5.6.** Let  $\ell = [0, 1] + iY_*$  be the gate line of Section 4, and put  $v(z) = \Phi_{\text{rep}}^{-1}(z)$  and  $\tilde{T}_0 = h^{-1} - a$ . Define

$$\delta T(z) = \dot{\mathcal{A}}(v(z)) - \dot{\mathcal{A}}(u^*) - \tilde{T}'_0(z) \dot{\mathcal{S}}^\#(v(z)), \quad \delta t_m = \int_\ell \delta T(z) e^{-2\pi i m z} dz.$$

Then for every  $n \geq 1$ , as  $s \rightarrow 0^+$ ,

$$\tau_n = B_n e^{2\pi i n a} + s \dot{\tau}_n + O(s^{10/9}), \quad \dot{\tau}_n = e^{2\pi i n a} (\delta t_n - 2\pi i n B_n \delta t_0).$$

Moreover  $p = -\log \lambda_1 \log \lambda_2 = 2s + \frac{10}{9}s^2 + O(s^3)$  is holomorphic in  $s$ . Consequently, if  $B_n \neq 0$ ,

$$\log \frac{\tau_n}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^{10/9}), \quad \kappa^{(n)} = \frac{\delta t_n}{2B_n} - \pi i n \delta t_0.$$

*Proof.* By Proposition 5.4 and Cauchy's estimates, both  $\mathcal{A}_s$  and  $\mathcal{S}_s^\#$  have expansions  $X_0 + s\dot{X} + O(s^{10/9})$  in the  $C^1$  topology on a compact neighbourhood of  $v(\ell)$ . By Rouché, the inverse  $v_s = (\mathcal{S}_s^\#)^{-1}$  exists near  $v(\ell)$  and satisfies

$$v_s = v - s \dot{\mathcal{S}}^\#(v) / \Phi'_{\text{rep}}(v) + O(s^{10/9}).$$

Hence  $\tilde{T}_s := R_b^{-1} \circ e(1 + v_s)$  satisfies  $\tilde{T}_s = \tilde{T}_0 + s \delta T + O(s^{10/9})$  uniformly on  $\ell$ , where we used  $\mathcal{A}'_0(v) / \Phi'_{\text{rep}}(v) = \tilde{T}'_0$ . By items (a)–(c) of Section 4, the Fourier coefficients  $\tilde{t}_m(s)$  of  $\tilde{T}_s - \text{id}$  on  $\ell$  satisfy  $\tau_n = \tilde{t}_n e^{-2\pi i n \tilde{t}_0}$ . Differentiate, using  $\tilde{t}_n(0) = B_n$  and  $\tilde{t}_0(0) = -a$ . The expansion of  $p$  follows from the removability argument of Lemma 5.1 applied to  $\log \lambda_1 \log \lambda_2$ , together with a direct Taylor expansion. Finally,  $\log(\tau_n / \tau_n(0)) = s \dot{\tau}_n / \tau_n(0) + O(s^{10/9})$  and  $s = p/2 + O(p^2)$ .  $\square$

**5.5. Higher orders.** The argument of Proposition 5.4 extends to every order, at the price of a larger  $N$ .

**Proposition 5.7.** *Let  $m \geq 1$  and  $N \geq m^2 + m + 1$ . For  $u \in K$  and  $j \geq 0$  put*

$$(14) \quad \mathcal{A}^{(j)}(u) = \partial_s^j \Psi_s(u)|_{s=0} + \sum_{k \geq 0} \partial_\sigma^j [F_\sigma(f_\sigma^k u)]_{\sigma=0}.$$

*For  $j \leq m$  the series converges absolutely and uniformly on  $K$ , and its terms are  $O(k^{2j-2N})$ . Uniformly for  $u \in K$ , as  $s \rightarrow 0^+$ ,*

$$\mathcal{A}_s(u) = \sum_{j=0}^m \frac{s^j}{j!} \mathcal{A}^{(j)}(u) + O(s^{m+\gamma}), \quad \gamma = 1 - \frac{2(m+1)^2}{2N+2m+1} > 0.$$

*Proof. Term bounds.* Fix  $k \geq 0$  and apply Lemma 5.2 with  $J = k+1$ . For  $|\sigma| \leq c/(k+1)^2$  the head estimate in the proof of Proposition 5.4 gives  $|q_\sigma(f_\sigma^k u)| \leq C_3(R+k/2)^{-2}$ , hence  $|F_\sigma(f_\sigma^k u)| \leq MC_3^N(R+k/2)^{-2N}$ . By Cauchy's estimate on this disc,

$$|\partial_\sigma^j [F_\sigma(f_\sigma^k u)]_{\sigma=0}| \leq j! MC_3^N c^{-j} (k+1)^{2j} (R+k/2)^{-2N} = O(k^{2j-2N}).$$

This is summable for  $j \leq m < N - \frac{1}{2}$ . For  $m = 1$  this recovers the bounds of Lemma 5.3, without computing  $\dot{u}_k$ .

*Head.*  $H_J(\sigma) = \sum_{k < J} F_\sigma(f_\sigma^k u)$  is holomorphic and bounded by  $M_1$  on  $|\sigma| < \rho := c/J^2$ . For real  $0 \leq s \leq \rho/2$ , the Taylor remainder of order  $m$  satisfies

$$\left| H_J(s) - \sum_{j \leq m} \frac{s^j}{j!} H_J^{(j)}(0) \right| \leq M_1 \sum_{j > m} (s/\rho)^j \leq 2M_1 c^{-m-1} s^{m+1} J^{2m+2}.$$

*Tails.* As before,  $|\sum_{k \geq J} F_s(f_s^k u)| \leq C_4 J^{1-2N}$  for  $s \in [0, s_0]$ . By the term bounds, the tails of the  $j$ -th derivative series from  $J$  on are  $O(J^{2j+1-2N})$ . Since  $sJ^2 \leq c/2$ , the tails contribute  $\sum_{j \leq m} s^j J^{2j+1-2N} = O(J^{1-2N})$ .

*Balance.* Write  $\mathcal{T}_J^{(j)}$  for the tail from  $k = J$  on of the  $j$ -th derivative series. The identity

$$\mathcal{A}_s - \sum_{j \leq m} \frac{s^j}{j!} \mathcal{A}^{(j)} = \left[ \Psi_s - \sum_{j \leq m} \frac{s^j}{j!} \partial_s^j \Psi \right] + \left[ H_J(s) - \sum_{j \leq m} \frac{s^j}{j!} H_J^{(j)}(0) \right] + \sum_{k \geq J} F_s(f_s^k u) - \sum_{j \leq m} \frac{s^j}{j!} \mathcal{T}_J^{(j)}$$

holds exactly. The  $\Psi$ -part is holomorphic in  $s$ . Altogether the error is  $O(s^{m+1} J^{2m+2} + J^{1-2N})$ . Take  $J = \lceil s^{-\beta} \rceil$  with  $\beta = (m+1)/(2N+2m+1) < \frac{1}{2}$ , which is admissible. Both terms are then  $O(s^{m+1-2(m+1)\beta}) = O(s^{m+\gamma})$ , and  $\gamma > 0$  exactly when  $N \geq m^2 + m + 1$ .  $\square$

For  $m = 1$  this is Proposition 5.4; the error there is  $O(s^{10/9})$  for  $N = 3$ .

**Theorem 5.8.** *For every  $n \geq 1$  with  $B_n \neq 0$ , the function  $\log(\tau_n/\tau_n(0))$  has an asymptotic expansion to all orders in  $p$  as  $b \rightarrow \eta^-$ :*

$$\log \frac{\tau_n}{B_n e^{2\pi i n a}} \sim \sum_{j \geq 1} \kappa_j^{(n)} p^j, \quad \kappa_1^{(n)} = \kappa^{(n)}.$$

*For each order  $m$ , the remainder after the  $p^m$  term is  $O(p^{m+1})$ . Each coefficient  $\kappa_j^{(n)}$  is a polynomial in the following data:*

- $B_n^{-1}$  and  $a$ ;
- the Taylor coefficients of  $s(p)$ ;
- finitely many Fourier coefficients on  $\ell$  of explicit polynomial expressions in  $\mathcal{A}^{(i)} \circ v$ ,  $\mathcal{S}^{\#(i)} \circ v$  ( $i \leq j$ ), their  $u$ -derivatives, and  $1/\Phi'_{\text{rep}} \circ v$ .

Here  $\mathcal{A}^{(i)}$  and  $\mathcal{S}^{\sharp(i)}$  are given by the convergent series (14) over parabolic orbits of  $f_0$  and  $G_0$ . The coefficients do not depend on  $N$ .

*Proof.* Fix  $m$  and  $N \geq m^2 + m + 1$ . By Proposition 5.7 and the reduction  $\mathcal{A}_s = \mathcal{A}_s \circ f_s^M - M$  (with  $f_s^M$  jointly holomorphic),  $\mathcal{A}_s$  and  $\mathcal{S}_s^\sharp$  have expansions  $\sum_{j \leq m} s^j X_j / j! + O(s^{m+\gamma})$ . These hold uniformly on a compact neighbourhood of  $v(\ell) \cup \{u^*\}$  and are holomorphic in  $u$ , so by Cauchy's estimates they also hold in the  $C^r$  topology on a smaller neighbourhood, for every  $r$ .

*Inverse.* Formal inversion of  $\sum_j s^j \mathcal{S}_j^\sharp / j!$  gives a polynomial approximant  $\tilde{v}_s = v + \sum_{1 \leq j \leq m} s^j V_j$  with  $\mathcal{S}_s^\sharp(\tilde{v}_s) = z + O(s^{m+\gamma})$ . Since  $\mathcal{S}_s^\sharp \rightarrow \Phi_{\text{rep}}$  in  $C^1$  and  $\Phi'_{\text{rep}} \neq 0$  near  $v(\ell)$ , the maps  $\mathcal{S}_s^\sharp$  are uniformly bi-Lipschitz there, and the true inverse satisfies  $v_s = \tilde{v}_s + O(s^{m+\gamma})$ .

*Composition.* Taylor-expand  $\mathcal{A}_s$  about  $v$ , using the bounded  $u$ -derivatives. This gives  $\tilde{T}_s = \sum_{j \leq m} s^j T_j / j! + O(s^{m+\gamma})$  uniformly on  $\ell$ . Hence the Fourier coefficients  $\tilde{t}_i(s)$  and  $\tau_n = \tilde{t}_n e^{-2\pi i n \tilde{t}_0}$  have expansions to order  $m$ . Since  $\tau_n(0) \neq 0$ , so does  $\log(\tau_n / \tau_n(0))$ .

*Change of parameter.* Finally,  $p$  is holomorphic in  $s$  with  $p'(0) = 2$ , so  $s$  is a holomorphic function of  $p$ .

*Independence of  $N$ .* An asymptotic expansion of a fixed function is unique, so the coefficients do not depend on the choice of  $N$ .

*Remainder.* Applying the argument at order  $m + 1$  and absorbing the  $p^{m+1}$  term gives the remainder  $O(p^{m+1})$ .  $\square$

**Remark 5.9.** In particular, Theorem 5.8 with  $m = 2$  improves the remainder in Theorem 5.6 from  $O(p^{10/9})$  to  $O(p^2)$ . Theorem 5.8 gives explicit asymptotic coefficients in the normalisation of  $\tau_n$ , but says nothing about summability. The  $\frac{1}{2}$ -summability of [5], in their normalisation, suggests Gevrey-2 growth  $|\kappa_j^{(n)}| \lesssim C^j (j!)^2$  in  $p \propto \epsilon$  (Gevrey-1 in  $\sqrt{p}$ ). This matches the scale  $\Lambda = \exp(-C/\sqrt{p})$  of the beyond-all-orders terms. The constants in our proof grow faster, because  $N$  must grow like  $m^2$ .

**Remark 5.10** (Relation to the literature). Christopher and Rousseau show that the unfolded modulus, in a particular normalisation, is  $\frac{1}{2}$ -summable in the canonical parameter  $\epsilon = (A_2 - A_1)^{-2}$ , with the Glutsyuk direction as the singular direction [5, Cor. 4.9]. One checks the exact identity  $p = 4\epsilon / (1 - \nu^2 \epsilon)$ . Their result gives the existence of an asymptotic expansion. Mardešić–Roussarie–Rousseau record that at  $\epsilon = 0$  only continuity was known [15, §11]. Theorems 5.6 and 5.8 give the coefficients, in the normalisation of  $\tau_n$ , as explicit convergent series over parabolic orbits. We know of no closed form. An integer-relation search on  $\kappa^{(1)}$  against  $\pi, \pi^2, \log 2, \gamma$  returns only spurious relations.

**5.6. Numerics.** The series (13) is evaluated with  $N = 3, \dots, 8$  on 24 points of the line  $\text{Im } z = 1$  or 1.5, with orbits of length 300–3000. The  $s$ -derivative of each (finite, holomorphic) truncation is taken by a central difference at  $s = \pm 10^{-20}$  to  $\pm 10^{-30}$  in 60–90 digit arithmetic (`mrr_derivative.py`). The results are stable to all printed digits:

$$\begin{aligned} \kappa^{(1)} &= -0.010199006345897402441 - 0.013250956182506712250i, \\ \kappa^{(2)} &= -0.05105655620132933 - 0.06231578053849410i, \\ \kappa^{(3)} &= -0.140044348129 - 0.169224463926i. \end{aligned}$$

These agree with the fits (18) from the side  $b < \eta$  to all of their eleven digits. At  $s = 0$  the same computation reproduces  $\tau_n(0) = B_n e^{2\pi i n a}$  to the precision of the stored  $B_n$ . The second coefficients  $\kappa_2^{(n)}$  of Theorem 5.8 are computed by a five-point stencil ( $s = \pm 10^{-12}, \pm 2 \cdot 10^{-12}$ , 90 digits) applied to the same finite truncations (`mrr_second.py`), with  $N = 7$  ( $\text{Im } z = 1$ , orbits of length 2000) and  $N = 10$  ( $\text{Im } z = 1.5$ , length 3000). The series converge for  $N \geq 3$ ; the

condition  $N \geq m^2 + m + 1 = 7$  is needed only for the proof. The two computations agree to 18, 16 and 12 digits respectively:

$$\begin{aligned}\kappa_2^{(1)} &= -7.88992074991511334 \cdot 10^{-5} + 3.6554727325799459 \cdot 10^{-4} i, \\ \kappa_2^{(2)} &= -1.895481905818775 \cdot 10^{-4} + 1.106655243691997 \cdot 10^{-4} i, \\ \kappa_2^{(3)} &= 4.07137984831 \cdot 10^{-4} - 8.49506527544 \cdot 10^{-3} i.\end{aligned}$$

The fits from the side  $b < \eta$  (`deficit_modes.py`) give  $-7.88991 \cdot 10^{-5} + 3.65547 \cdot 10^{-4} i$ ,  $-1.89548 \cdot 10^{-4} + 1.10666 \cdot 10^{-4} i$  and  $4.07137 \cdot 10^{-4} - 8.49507 \cdot 10^{-3} i$ , in agreement to within one unit in the sixth digit.

## 6. THE SEWING SOLUTION: DECAY ESTIMATE

Throughout,  $b < \eta$  is real and close to  $\eta$ ,  $\varepsilon = |\log \lambda|$ , and  $S$  has the real normalisation of Section 3, so  $T(z) - z = \sum t_n e^{2\pi i n z}$  with  $\text{Im } t_0 = h/2$  and  $t_{-n} = \bar{t}_n$  (Lemma 3.1).

**Lemma 6.1** (The band). *Let  $Z_b(u) = A_1 \log \frac{u-u_1}{u-u_2}$  with the branch  $\text{Im } Z_b \in (0, h)$  on  $\mathbb{C} \setminus ((-\infty, u_1] \cup [u_2, \infty))$ . This branch has  $\text{Im } Z_b = h/2$  on  $(u_1, u_2)$  and  $\text{Im } Z_b < h/2$  on  $\text{Im } u > 0$ . For  $Y > 0$  let*

$$\mathcal{B}_b(Y) = \{u : Y < \text{Im } Z_b(u) < h - Y\},$$

*a lens bounded by two circular arcs through  $u_1, u_2$ . There is an absolute  $Y_B$  such that for  $Y \geq Y_B$  and all  $b$  near  $\eta$ :*

- (i) *every  $u \in \mathcal{B}_b(Y)$  has its forward orbit in  $D_{r'}$  tending to  $u_1$  and its principal backward orbit in  $D_{r'}$  tending to  $u_2$ , with  $\sum_{k \in \mathbb{Z}} |F_b(f_b^k u)| \leq C/Y + C\varepsilon$ ;*
- (ii) *the real-normalised times  $\mathcal{R}_b = \Psi_b + \sum_{k \geq 0} F_b \circ f_b^k + c_R$  and  $\mathcal{S}_b = \Psi_b - \sum_{k \geq 1} F_b \circ f_b^{-k} + c_S$  are holomorphic and injective on  $\mathcal{B}_b(Y)$ . They are branches of  $R_b^{-1}$  and  $S_b^{-1}$ , with  $\mathcal{S}_b$  real on  $(u_1, u_2)$  and  $\mathcal{R}_b$  as in item (c) above, and*

$$|\text{Im } \mathcal{S}_b(u) - (\text{Im } Z_b(u) - h/2)| \leq \frac{\pi}{3} + C/Y + C\varepsilon;$$

- (iii)  *$\mathcal{S}_b$  maps the strip  $\Sigma = \{|\text{Im } w'| < h/2 - Y - 2\}$  into  $\mathcal{B}_b(Y)$  with  $\mathcal{S}_b \circ \mathcal{S}_b = \text{id}$ . Consequently  $T = \mathcal{R}_b \circ \mathcal{S}_b$  is holomorphic on  $\Sigma$ , and*

$$T(w') - w' = (c_R - c_S) + \sum_{k \in \mathbb{Z}} F_b(f_b^k(\mathcal{S}_b(w'))).$$

*Proof.* (i) Put  $q = e^{Z/A_1} = (u - u_1)/(u - u_2)$  and  $\zeta = Z/A_1 = -\varepsilon Z$ , so  $\text{Im } \zeta \in (-2\pi + \varepsilon Y, -\varepsilon Y)$  on the band. On  $-2\pi \leq \text{Im } \zeta \leq 0$  one has  $|e^\zeta - 1| \geq c \min(1, \delta(\zeta)) \max(1, |e^\zeta|)$  with  $\delta(\zeta) = \min(|\zeta|, |\zeta + 2\pi i|)$  and an absolute  $c > 0$ ; the only zeros are 0 and  $-2\pi i$ . On the band  $\delta(\zeta) = \varepsilon \min(|Z|, |Z - ih|) > \varepsilon Y$ . Since  $u - u_2 = (u_2 - u_1)/(q - 1)$  and  $|u_2 - u_1|/\varepsilon \rightarrow 2$  (Lemma 4.1), this gives

$$|u - u_2| \leq C(\varepsilon + Y^{-1}), \quad |u - u_1||u - u_2| = \frac{|q||u_2 - u_1|^2}{|q - 1|^2} \leq \begin{cases} C \min(|Z|, |Z - ih|)^{-2}, & \delta(\zeta) \leq 1, \\ C\varepsilon^2 e^{-\varepsilon|\text{Re } Z|}, & \delta(\zeta) > 1, \end{cases}$$

using  $|q|/\max(1, |q|)^2 = e^{-|\text{Re } \zeta|}$ . So  $\mathcal{B}_b(Y) \subset D_{C/Y + |u_1| + |u_2|} \subset D_{r'/2}$  for  $Y$  large. By the proof of Lemma 4.3,  $Z \circ f - Z = 1 + O(|u| + \varepsilon)$ , so  $\text{Re } Z$  increases by at least  $\frac{3}{4}$  per step while the orbit is in  $D_{r'/2}$ .

The imaginary part is controlled through  $\Psi_b = Z_b + (A_1 + A_2) \log(u - u_2)$ , with step error  $|F_b| \leq M|u - u_1||u - u_2|$  (Lemma 4.2). Summing (15) along  $\text{Re } Z_k \geq \text{Re } Z_0 + \frac{3}{4}k$  bounds  $\sum_k |F_b(u_k)|$  by  $C'/Y + C'\varepsilon$ . Points with  $\delta(\zeta_k) \leq 1$  contribute  $\sum_k (x_k^2 + Y^2)^{-1} \leq C/Y$ , and the

others  $\varepsilon^2 \sum e^{-\varepsilon|x_k|} \leq C\varepsilon$ . Now  $\operatorname{Im} Z - \operatorname{Im} \Psi = -(A_1 + A_2) \arg(u - u_2)$  with  $\arg(u - u_2) \in (0, 2\pi)$  on the lens. So  $\operatorname{Im} Z$  drifts by at most  $C'/Y + C'\varepsilon + \frac{2\pi}{3} + o(1)$  along the orbit, which is  $< Y/2$  for  $Y \geq Y_B \geq 8$ . By induction the orbit stays in  $\mathcal{B}_b(Y/2) \subset D_{r'/2}$ , and  $\operatorname{Re} Z_k \rightarrow +\infty$ , i.e.  $u_k \rightarrow u_1$ . The continued values of  $Z$  along the orbit have  $\operatorname{Im} \in (Y/2, h - Y/2) \subset (0, h)$ , so they agree with the fixed slit-plane branch. The backward orbit is the forward orbit of the reflected inverse family  $G_b(v) = -f_b^{-1}(-v)$ , to which the same argument applies (its prepared form is checked in the proof of Theorem 4.4(2)). Its period is  $h_2 = h + 2\pi/3 + o(1)$ , so  $\mathcal{B}_b(Y)$  is contained in the corresponding band of  $G_b$  at height  $Y(1 + O(\varepsilon)) - 1$ . This is harmless for  $Y \geq Y_B$ .

(ii) As in the proof of Theorem 4.4(1),  $\Psi_b(f^n u) - n \rightarrow A_1 \log \sigma_b(u) + \text{const}$  along forward orbits. So  $\mathcal{R}_b$  is a branch of the attracting Abel coordinate, and  $\mathcal{S}_b$  likewise of the repelling one. Choose the constants  $c_R$  (by item (c) above) and  $c_S$  so that  $\mathcal{S}_b$  is real on the segment. Both  $F_b$  and  $f_b$  are real, so the sums are real there. On the segment  $\operatorname{Im} Z = h/2$  and  $\arg(u - u_2) = \pi$ , which fixes  $\operatorname{Im} c_S$ . The displayed estimate then follows from  $\operatorname{Im} \Psi = \operatorname{Im} Z + (A_1 + A_2) \arg(u - u_2)$ ,  $A_1 + A_2 \rightarrow \frac{1}{3}$  and (i). Injectivity: if  $\mathcal{S}_b(u) = \mathcal{S}_b(u')$  then the principal backward orbits satisfy  $\tau_b(u_{-n}) = \tau_b(u'_{-n})$  near  $u_2$ , where  $\tau_b$  is injective, so  $u_{-n} = u'_{-n}$ . Moreover  $f_b$  is injective on  $D_{r'}$  (its non-injectivity has period  $2\pi i / \log b$  in  $w$ , far larger than the disc), so  $u = u'$ . The same argument with  $\sigma_b$  and forward orbits proves injectivity of  $\mathcal{R}_b$ .

(iii) Fix  $x \in \mathbb{R}$ .  $\mathcal{S}_b(x) \in (u_1, u_2) \subset \mathcal{B}_b(Y)$  and  $\mathcal{S}_b(\mathcal{S}_b(x)) = x$ , since both sides are the real-normalised repelling time. Move  $y'$  from 0 towards  $\pm(h/2 - Y - 2)$ . As long as  $\mathcal{S}_b(x + iy')$  stays in the lens, continuation gives  $\mathcal{S}_b(\mathcal{S}_b(x + iy')) = x + iy'$ . It cannot leave through the corners  $u_1, u_2$ , where  $|\operatorname{Re} \mathcal{S}_b| \rightarrow \infty$ . It cannot leave through the arcs either: there  $\operatorname{Im} Z \in \{Y, h - Y\}$ , so by (ii)  $|\operatorname{Im} \mathcal{S}_b| \geq h/2 - Y - \pi/3 - o(1) > |y'|$ . Hence  $\mathcal{S}_b(\Sigma) \subset \mathcal{B}_b(Y)$ ,  $T = \mathcal{R}_b \circ \mathcal{S}_b$  on  $\Sigma$ , and  $T(w') - w' = \mathcal{R}_b(u) - \mathcal{S}_b(u)$  with  $u = \mathcal{S}_b(w')$  is the displayed bi-infinite sum.  $\square$

**Lemma 6.2** (Uniform mode bound). *Put  $Y_0 := Y_B + 3$ . There is  $M$ , which can be taken  $\leq 1$ , and  $b_1 < \eta$  such that for  $b \in (b_1, \eta)$  and all  $n \neq 0$ ,*

$$|t_n| \leq M e^{-2\pi|n|d}, \quad d := \frac{h}{2} - Y_0 - 1.$$

*Proof.* Take  $Y = Y_B + 1$  in Lemma 6.1(iii), so that the estimate (ii) is available on the closure of the lens.  $\Sigma$  has half-width  $h/2 - Y_B - 3 = d + 1$ , so the lines  $\operatorname{Im} w' = \pm d$  lie in  $\Sigma$ . On them, by (iii) and (i),  $T - \operatorname{id} - t_0 = \sum_k F_b(f_b^k \mathcal{S}_b) - \text{mean}$  has modulus at most  $2(C/Y_B + C\varepsilon)$ . Let  $M$  be the supremum of this over  $b \in (b_1, \eta)$ . Cauchy's estimate on these lines gives the bound for  $n > 0$  and  $n < 0$ . This does not use Theorem 4.4.  $\square$

For  $H > 0$  let  $\mathfrak{A}_H$  be the space of 1-periodic  $g = \sum g_n e^{2\pi i n z}$  with  $\|g\|_H = \sum |g_n| e^{2\pi|n|H} < \infty$ . It is a Banach algebra (the weight is submultiplicative), its elements are holomorphic and bounded by  $\|g\|_H$  on  $|\operatorname{Im} z| \leq H$ , and the projections  $\Pi_{<0}, \Pi_{\geq 0}$  onto negative and non-negative modes have norm 1. Also  $\|e^g\|_H \leq e^{\|g\|_H}$  and  $\|e^{g_1} - e^{g_2}\|_H \leq \|g_1 - g_2\|_H e^{\max \|g_j\|_H}$ .

**Theorem 6.3** (W2). *Let  $M, d$  be as in Lemma 6.2,  $\rho = \frac{1}{4}$ , and choose  $D \geq 1$  with  $4\pi M e^{-2\pi D} \leq \rho(1 - e^{-2\pi D})^2$ . Put  $H = d - D - \rho$  and  $\tau = T - \operatorname{id} - t_0$ . Assume  $H > 1$ , which holds for  $b$  near  $\eta$  since  $d \sim \pi/\varepsilon$ , and that  $b$  is so close to  $\eta$  that  $C_1 \Lambda e^{4\pi(Y_0 + D + 1)} < \frac{1}{2}$ , where  $C_1 = 4M + 2\rho$ . Then:*

- (1) *The map  $\mathcal{T}(Q) = -\Pi_{<0}[\tau \circ (\operatorname{id} + Q)]$  is a contraction, with constant  $\leq \frac{1}{2}$ , of the ball  $\{Q \in \Pi_{<0}\mathfrak{A}_H : \|Q\|_H \leq \rho\}$  into itself. Its fixed point  $Q^*$  and  $P' := \Pi_{\geq 0}[\tau \circ (\operatorname{id} + Q^*)]$  satisfy*

$$T \circ (\operatorname{id} + Q^*) = \operatorname{id} + t_0 + P' \quad \text{on } |\operatorname{Im} z| < H.$$

- (2)  *$\varphi_- = \operatorname{id} + Q^*$  is univalent on  $\{\operatorname{Im} z < H - 1\}$  and  $\varphi_+ = \operatorname{id} + t_0 + P'$  on  $\{\operatorname{Im} z > -(H - 1)\}$ . The function equal to  $R \circ (\varphi_+ - ih)$  above and  $S \circ \varphi_-$  below is a solution of (1); it is the sewing solution  $K_b^W$  of Proposition 3.4.*

- (3) The equation  $\varphi_+(z_0) = ih$  has a unique solution with  $|z_0 - (ih - t_0)| < 1$ , and  $|z_0 - (ih - t_0)| \leq C_1 \Lambda e^{4\pi(Y_0+D+1)}$ . After the translation  $z \mapsto z + z_0$  one has  $K_b^W(0) = 1$  and  $P = R^{-1} \circ K_b^W - \text{id}$  with  $P(0) = 0$ .
- (4) For every  $n \geq 1$  the Fourier coefficients of this  $P$  satisfy

$$\left| \frac{c_n}{\Lambda^n} - \tau_n(b) \right| \leq C_n \Lambda, \quad C_n = C M(M+1) e^{2\pi(n+2)(Y_0+D+2)},$$

with an absolute constant  $C$ .

*Proof.* (1) Write  $\tau \circ (\text{id} + Q) = \sum_{m \neq 0} t_m e^{2\pi i m z} e^{2\pi i m Q}$ . Then

$$\|\tau \circ (\text{id} + Q)\|_H \leq \sum_{m \neq 0} |t_m| e^{2\pi|m|(H+\rho)} \leq 2M \sum_{m \geq 1} e^{-2\pi m D} \leq \rho$$

by Lemma 6.2 and the choice of  $D$ . Similarly

$$\|\tau \circ (\text{id} + Q_1) - \tau \circ (\text{id} + Q_2)\|_H \leq \sum_{m \neq 0} |t_m| e^{2\pi|m|(H+\rho)} 2\pi|m| \|Q_1 - Q_2\|_H \leq \frac{4\pi M e^{-2\pi D}}{(1 - e^{-2\pi D})^2} \|Q_1 - Q_2\|_H,$$

which is at most  $\frac{1}{2} \|Q_1 - Q_2\|_H$ . Since  $|\text{Im}(z + Q)| \leq H + \rho < d$  on the strip, the Fourier series of  $T$  converges at  $z + Q(z)$  and equals  $T$  there (Lemma 6.1). The identity follows by splitting  $\tau \circ (\text{id} + Q^*) = P' + \Pi_{<0}[\dots] = P' - Q^*$ .

(2) The series of  $Q^*$  has only negative modes and converges on  $\text{Im } z \leq H$ , so it converges on the whole lower half-plane  $\text{Im } z < H$ . There  $|Q^{*'}| \leq 2\pi e^{-2\pi} \rho / (1 - e^{-2\pi})^2 < 1$  on  $\text{Im } z \leq H - 1$ , so  $\text{Re } \varphi'_- > 0$  and  $\varphi_-$  is univalent (Noshiro–Warschawski). The same holds for  $\varphi_+$  above. On the strip,  $R \circ (\varphi_+ - ih) = R \circ (T \circ \varphi_- - ih) = R \circ T \circ \varphi_- = S \circ \varphi_-$ , because  $ih$  is a period of  $R$  and  $R \circ T = S$ . The functional equation holds because  $\varphi_{\pm}$  commute with  $z \mapsto z + 1$ . The resulting function realises the sewing of Proposition 3.4: the images of  $\varphi_{\pm}$  are the two half-cylinders and the seam in the  $z$ -plane is  $\Gamma = \varphi_-^{-1}(\mathbb{R})$ , a curve in  $|\text{Im } z| \leq \rho$ . Since  $\varphi_{\pm} - \text{id}$  is bounded, the argument principle shows that the restrictions of  $\varphi_{\pm}$  to the two sides of  $\Gamma$  map onto the two half-cylinders. By uniqueness of the uniformisation it is  $K_b^W$  up to translation. (The identity  $R \circ T = S$  is used on the band, where orbits stay in  $D_{r'}$  and  $\mathcal{R}_b$  inverts  $R$ ; Lemma 6.1.)

(3) For  $\text{Im } z \geq h/2 - 1$  we have  $|P'(z) - p'_0| \leq \sum_{n \geq 1} |p'_n| e^{-2\pi n(h/2-1)}$  and  $|p'_n| \leq \rho e^{-2\pi n H}$ . Moreover  $p'_0 = \sum_{m \geq 1} t_m [e^{2\pi i m Q^*}]_{-m}$ , and  $|[e^{2\pi i m Q^*}]_{-m}| \leq e^{2\pi m \rho} e^{-2\pi m H}$ , so  $|p'_0| \leq 2M e^{-2\pi(d+H-\rho)} \leq 2M \Lambda e^{2\pi(2Y_0+D+2+2\rho)}$ , and  $\sum_{n \geq 1} |p'_n| e^{-2\pi n(h/2-1)} \leq 2\rho \Lambda e^{2\pi(Y_0+D+2+\rho)}$ . Both are bounded by

$$\frac{1}{2} C_1 \Lambda e^{4\pi(Y_0+D+1)}.$$

Since  $\text{Im}(ih - t_0) = h/2$  and  $|P''| \leq \rho$  there, Rouché's theorem on the disc of radius 1 gives the unique root and the bound, using the smallness hypothesis. Then  $K_b^W(z_0) = R(\varphi_+(z_0) - ih) = R(0) = 1$ , and after translation  $P(z) = \varphi_+(z + z_0) - ih - z$ , so  $P(0) = 0$ .

(4) For  $n \geq 1$  the  $n$ -th mode of  $P$  is  $c_n = p'_n e^{2\pi i n z_0}$ . Because  $Q^*$  has no constant or positive modes,  $[e^{2\pi i m Q^*}]_0 = 1$  and  $[e^{2\pi i m Q^*}]_k = 0$  for  $k > 0$ . Hence  $p'_n = t_n + \sum_{m > n} t_m [e^{2\pi i m Q^*}]_{n-m}$ , and

$$|p'_n - t_n| \leq \sum_{m > n} M e^{-2\pi m d} e^{2\pi m \rho} e^{-2\pi(m-n)H} \leq 2M e^{-2\pi n(d-\rho)} e^{-2\pi(d+H-\rho)}.$$

With  $z_0 = ih - t_0 + \zeta$ ,  $|\zeta| \leq C_1 \Lambda e^{4\pi(Y_0+D+1)}$  by (3), we have  $e^{2\pi i n z_0} = \Lambda^n e^{-2\pi i n t_0} e^{2\pi i n \zeta}$  and  $|t_n e^{-2\pi i n t_0}| = |\tau_n|$ , so  $|t_n| = |\tau_n| e^{-\pi n h}$ . Dividing,

$$\frac{c_n}{\Lambda^n} - \tau_n = \tau_n (e^{2\pi i n \zeta} - 1) + (p'_n - t_n) e^{-2\pi i n t_0} e^{2\pi i n \zeta}.$$

In the first term  $|\tau_n| = |t_n| e^{\pi n h} \leq M e^{2\pi n(Y_0+1)}$  and  $|e^{2\pi i n \zeta} - 1| \leq 4\pi n |\zeta|$  when  $2\pi n |\zeta| \leq 1$ ; in the second,  $|e^{2\pi i n \zeta}| \leq e$  in that case. If instead  $2\pi n |\zeta| > 1$ , then  $\Lambda \geq (2\pi n C_1)^{-1} e^{-4\pi(Y_0+D+1)}$ , and

the trivial bound  $|c_n|\Lambda^{-n} + |\tau_n| \leq C'Me^{2\pi n(Y_0+D+2)}$  is already  $\leq C_n\Lambda$  (this uses  $D \geq 1$ ). In the second term,  $|(p'_n - t_n)e^{-2\pi i n t_0}| \leq 2Me^{2\pi n(Y_0+1+\rho)}\Lambda e^{2\pi(2Y_0+2+D+2\rho)}$ , by  $d = h/2 - Y_0 - 1$  and  $H = d - D - \rho$ . Adding the two gives the stated bound, since  $n(Y_0 + 1 + \rho) + 2Y_0 + 2D + 2.5 \leq (n+2)(Y_0 + D + 2)$ .  $\square$

**Corollary 6.4.** *For the sewing solution,  $\hat{c}_n(K_b^W) \rightarrow B_n e^{2\pi i n a}$  for every  $n \geq 1$  as  $b \rightarrow \eta^-$ . Moreover,  $\kappa_n(K_b^W) \rightarrow B_n/B_1^n$  and  $\lim \arg \hat{c}_1 = \arg B_1 + 2\pi a$ .*

*Proof.* Theorem 6.3(4) and Corollary 4.5:  $C_n$  does not depend on  $b$  and  $\Lambda \rightarrow 0$ . Proposition 2.4 makes  $c_1$  nonzero for  $b$  near  $\eta$ , justifying division and the limiting phase.  $\square$

**Corollary 6.5** (Local parabolic limit of the sewn solution). *Put  $u^* = 1/e - 1$  and  $a = \Phi_{\text{att}}(u^*)$ . On some fixed disk about  $z = 0$ , the function*

$$F_\eta^{\text{att}}(z) = e(1 + \Phi_{\text{att}}^{-1}(z + a))$$

*is holomorphic, satisfies  $F_\eta^{\text{att}}(0) = 1$ , and has  $(F_\eta^{\text{att}})'(0) \neq 0$ . As  $b \rightarrow \eta^-$  through real bases,  $K_b^W \rightarrow F_\eta^{\text{att}}$  locally uniformly on that disk, with all  $z$ -derivatives. Consequently there are  $r > 0$  and  $b_2 < \eta$  such that, for  $b_2 < b < \eta$ , the only preimage of 1 under  $K_b^W$  in  $B(0, r)$  is 0; both  $K_b^W$  and  $1/K_b^W$  are uniformly bounded there.*

*Proof.* Theorem 4.4(1) gives locally uniform convergence of  $A_b(u) := R_b^{-1}(e(1+u))$  to  $A_\eta(u) := \Phi_{\text{att}}(u) - a$  near  $u^*$ , with  $A_b(u^*) = 0$ . The derivative  $A'_\eta(u^*)$  is nonzero: the series in the proof of Theorem 4.4, Lemmas 4.2 and 4.3, and Cauchy's estimate give  $\Phi'_{\text{att}}(u) = 2/u^2 + O(1/u)$  on a narrower attracting petal. The Abel identity and  $f'_\eta(u) \neq 0$  propagate nonvanishing back along the orbit of  $u^*$ . Uniform convergence of these local inverses therefore gives  $R_b \rightarrow F_\eta^{\text{att}}$  on a fixed disk about 0.

In Theorem 6.3, write  $P_b(z) = \zeta_b + P'_b(z + z_0(b))$ , where  $\zeta_b = z_0(b) + t_0(b) - ih(b) \rightarrow 0$ . The proof of that theorem gives  $p'_0(b) \rightarrow 0$  and  $|p'_n(b)| \leq \rho e^{-2\pi n H(b)}$  for  $n \geq 1$ , while  $H(b) \rightarrow \infty$  and  $\text{Im } z_0(b) = h(b)/2 + o(1)$ . Hence  $P_b \rightarrow 0$  uniformly on every fixed bounded  $z$ -disk on which the upper representation is defined. That representation contains a fixed disk about 0 for all sufficiently large  $b$ , and  $K_b^W(z) = R_b(z + P_b(z))$  there. This proves the asserted convergence; Cauchy's formula gives convergence of derivatives. Choose a small closed disk on which  $F_\eta^{\text{att}}$  is nonzero and  $F_\eta^{\text{att}} - 1$  has only its simple zero at 0. Rouché's theorem and  $K_b^W(0) = 1$  give the final assertions.  $\square$

The construction is directly computable (`weld_solve.py`). At  $b = 1.10$  the iteration contracts by  $\sim 10^{-4}$  per step and gives  $\hat{c}_1 = 0.0864272029 e^{1.5510921403i}$ , against  $\tau_1 = 0.0864272022 e^{1.5510921408i}$ . The relative difference is  $8.8 \cdot 10^{-9} = 0.38\Lambda$ , and at  $b = 1.30$  it is  $3.9 \cdot 10^{-19} = 0.40\Lambda$ . At these bases  $h$  is too small for the hypotheses of Theorem 6.3, so the agreement is not an instance of the theorem. It does match the first-order expansion of the correction,  $c_1/(\Lambda\tau_1) - 1 \approx \Lambda[-4\pi^2|\tau_1|^2 - 2\pi i\tau_1 - 4\pi i\tau_2\bar{\tau}_1/\tau_1]$ , whose modulus is  $0.3834\Lambda$  and  $0.3984\Lambda$  respectively.

## 7. DEPENDENCE ON THE BASE

This section proves that  $K_b^W$  depends holomorphically on the base near  $(b_1, \eta)$ , with quantitative bounds; the identification with the boundary value  $K_b$  of (2) is completed in Section 9.

**Theorem 7.1** (W1, local). *Let  $b_0 \in (b_1, \eta)$ . There is a disc  $\Delta \ni b_0$  in the base plane such that:*

- (1) *For  $b' \in \Delta$ , with  $L(b'), L_2(b')$  the continued fixed points,  $R_{b'}, S_{b'}$  the regular superfunctions,  $\varpi(b') = 2\pi i / \log \lambda(b')$  the period of  $R_{b'}$ , and  $\hat{T}_{b'} = R_{b'}^{-1} \circ S_{b'} + \varpi/2$ , the contraction of Theorem 6.3 works with  $\rho$  and  $D$  unchanged,  $2M$  in place of  $M$ , and  $H - \frac{1}{2}$  in place of  $H$ . The solution is  $K_{b'}^W = R_{b'} \circ (\varphi_+ + \varpi)$  above, where  $\varphi_+ = T_{b'} \circ \varphi_-$ , and  $S_{b'} \circ \varphi_-$*

below. It is normalised by the root of  $\varphi_+(z_0) = -\varpi$  in the disc of radius 1 about  $-\varpi - t_0$ . (For real  $b'$ ,  $\varpi = -ih$ .) Its fixed point, the resulting solution  $K_{b'}^W$ , and the normalising root  $z_0(b')$  are holomorphic in  $b'$ .

- (2) For  $b' \in \Delta$  with  $\text{Im } b' > 0$  one has  $\arg \lambda > 0$ ,  $\arg \lambda_2 < 0$ , and  $K_{b'}^W(z) \rightarrow L(b')$  as  $\text{Im } z \rightarrow +\infty$ ,  $K_{b'}^W(z) \rightarrow L_2(b')$  as  $\text{Im } z \rightarrow -\infty$ , uniformly for  $\text{Re } z$  in compacts. In the upper and lower half-planes (of the translated coordinate) it has the representations  $R_{b'} \circ (\text{id} + \theta_{\text{up}})$  and  $S_{b'} \circ (\text{id} + \theta_{\text{dn}})$ . These are the defining representations of the complex-base construction in `_cbuild.py`.
- (3) (Local uniqueness.) If a solution  $F$  with  $F(0) = 1$  admits such representations and, after the translation that puts its seam representation in the form  $T \circ (\text{id} + Q) = \text{id} + t_0 + P'$ , has  $\|Q\|_H \leq \rho$  and its normalising root in that disc, then  $F = K_{b'}^W$ . (Without the root condition the statement is false: the translates of  $K_{b'}^W$  by the other roots  $c_j$ ,  $j \geq 1$ , have the same  $Q$ .)

*Proof.* (1) By Lemma 6.1(iii),  $T$  is holomorphic on the strip of half-width  $d + 1$ , so there is slack 1 in  $d$ ; we use half of it. The Koenigs maps at  $L(b')$ ,  $L_2(b')$  and the principal logarithms defining  $R^{-1}$  depend holomorphically on  $b'$ . To choose one parameter disc for the unbounded strip, first work on the compact fundamental rectangle  $0 \leq \text{Re } z \leq 1$ ,  $|\text{Im } z| \leq d(b_0) + \frac{1}{2}$ . Its image under  $S_{b_0}$  is compactly contained in the common domain of the two chosen Abel charts; their local parameter continuations therefore give a single disc  $\Delta$  on this rectangle. The identity  $T_{b'}(z + 1) = T_{b'}(z) + 1$  extends the result to the whole strip. Thus  $\hat{T}_{b'} - \text{id}$  is jointly holomorphic on  $\Delta \times \{|\text{Im } z| < d(b_0) + \frac{1}{2}\}$ . Shrinking  $\Delta$  again, its modes satisfy  $|t_n(b')| \leq 2Me^{-2\pi|n|(d(b_0)-1/2)}$  (Cauchy), and the estimates of Theorem 6.3(1) hold with  $2M$  in place of  $M$ : the original choice of  $D$  gives a Lipschitz constant at most  $2\rho = \frac{1}{2}$  and a ball-image norm at most  $\rho(1 - e^{-2\pi D})/\pi < \rho$ . The iterates  $\mathcal{T}^k(0)$  are holomorphic in  $b'$  and converge uniformly, so  $Q^*$  is holomorphic in  $b'$ . So are  $P'$  and, by the implicit function theorem (Rouché),  $z_0$ .

(2) From  $\lambda e^{-\lambda} = \log b'$ ,  $\partial\lambda/\partial b = e^\lambda/(b(1 - \lambda))$  is positive for  $\lambda < 1$  and negative for  $\lambda_2 > 1$ . Hence  $\text{Im } b' > 0$  gives  $\text{Im } \lambda > 0 > \text{Im } \lambda_2$  for  $\Delta$  small. Above,  $K^W = R(z + c + o(1))$ , and  $R(w) = \sigma^{-1}(c'\lambda^w) \rightarrow L$  when  $\text{Re}(w \log \lambda) \rightarrow -\infty$ . This is the case as  $\text{Im } w \rightarrow +\infty$ , since  $\text{Re}(w \log \lambda) = \text{Re } w \log |\lambda| - \text{Im } w \arg \lambda$ . Below, the same argument with  $\arg \lambda_2 < 0$  applies. The representations are those of Theorem 6.3(2), transported by  $z \mapsto z + z_0$ .

(3) The fixed point of a contraction is unique in its ball.  $P'$  and  $z_0$  are then determined, the latter by  $F(0) = 1$  together with the root condition.  $\square$

Theorem 7.1 gives quantitative control of the sewn family on complex discs about real bases near  $\eta$ . The base continuation from  $b > \eta$  passes the Shell–Thron boundary, where  $|\Lambda| = 1$  and the contraction does not apply; Section 9 reaches the sewn family by a different route, a tilted initial region that is insensitive to  $|\Lambda|$ .

## 8. CHARACTERISATION OF THE SEWN SOLUTION

The sewn solution is characterised by its two regular representations and the position of its seam. This is the uniqueness statement in Theorem B; it is also what fixes the root in the identification of Section 9.

**Definition 8.1.** For real  $b \in (1, \eta)$  and  $Y > 0$  let  $\mathcal{C}_b(Y)$  be the set of solutions  $F$  of (1) with  $F(0) = 1$  such that (“solution” means holomorphic off a closed singular set and satisfying the equation where defined)

- (a)  $F = R \circ (\text{id} + P)$  on  $\{\text{Im } z > -2Y\}$ , with  $P$  holomorphic, 1-periodic and bounded there, and  $P(0) = 0$  (the branch  $R^{-1}(1) = 0$ );

- (b)  $F = S \circ (\text{id} + Q)$  on  $\{\text{Im } z < -Y\}$ , with  $Q$  holomorphic, 1-periodic and bounded there, and  $|\text{Im } Q - h/2| \leq 1$  on the strip  $O = \{-2Y < \text{Im } z < -Y\}$ .

Condition (b) says that the real line of repelling time, i.e. the segment  $(L, L_2)$ , sits half a period of attracting time below the real axis of  $F$ . Heuristically, some such condition is necessary. For the sewing solution the roots of  $\varphi_+^W(c) \in i\hbar\mathbb{Z}$  near the axis are  $c_j = c_0 + ij\hbar + O(\Lambda)$ . The translates  $K^W(\cdot + c_j)$  keep an  $R$ -representation above and an  $S$ -representation on some lower half-plane, and give  $|c_1| \asymp \Lambda^{1+j}$ . Condition (b) excludes  $j \neq 0$ , as the proof shows.

**Theorem 8.2** (Characterisation). *There are  $b_1 < \eta$  and  $Y_B$  such that for  $b \in (b_1, \eta)$  and  $Y_0 + D + 3 < Y < (h - Y_0 - D - 3)/2$  (with  $Y_0, D$  as in Theorem 6.3),*

$$\mathcal{C}_b(Y) = \{K_b^W\},$$

with  $K_b^W$  normalised as in Theorem 6.3(3). In particular  $\hat{c}_n \rightarrow B_n e^{2\pi i n a}$  for every  $n$  holds for the unique element of  $\mathcal{C}_b(Y)$ .

*Proof.*  $K_b^W \in \mathcal{C}_b(Y)$ . In the normalised coordinate,  $\varphi_-(z) = z + z_0 + Q^*(z + z_0)$  is defined for  $\text{Im}(z + z_0) < H$ , i.e. for  $\text{Im } z < -(Y_0 + 1 + D + \rho) + O(\Lambda)$ , which contains  $\{\text{Im } z < -Y\}$ . There  $|Q - z_0| \leq \rho$ , so  $|\text{Im } Q - h/2| \leq \rho + O(\Lambda) < 1$ . The  $R$ -representation holds for  $\text{Im}(z + z_0) > -(H - 1)$ , i.e. above  $-(h - Y_0 - D - 2 - \rho)$ , which contains  $\{\text{Im } z > -2Y\}$ .  $P$  and  $Q$  are bounded there.

*Uniqueness.* Let  $F \in \mathcal{C}_b(Y)$ ,  $\varphi_+ = \text{id} + P$  and  $\varphi_- = \text{id} + Q$ . Work with the strip  $\Sigma_q = \{|\text{Im } w'| < d - D\}$  of  $S$ -time. By Lemma 6.2 and the choice of  $D$ , on  $\overline{\Sigma_q}$

$$|T - \text{id} - t_0| \leq \frac{2Me^{-2\pi D}}{1 - e^{-2\pi D}} \leq \frac{\rho}{2\pi}, \quad |T' - 1| \leq \rho < 1,$$

so  $T$  is univalent on the convex set  $\Sigma_q$  (Noshiro–Warschawski), continuous up to the boundary, and within  $\rho/2\pi$  of a translation.

(i) *The overlap.* Let  $O = \{-2Y < \text{Im } z < -Y\}$ . For  $z \in O$ ,  $\text{Im } \varphi_-(z) \in (h/2 - 2Y - 1, h/2 - Y + 1) \subset (-(d - D), d - D)$ , because  $2Y < h - Y_0 - D - 2$  and  $Y > Y_0 + D + 2$ . Since  $\sigma \circ R(w) = \sigma(1)\lambda^w$  wherever  $R$  is defined,  $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$  gives  $\lambda^{\varphi_+} = \lambda^{T \circ \varphi_-}$ . Hence  $\varphi_+ = T \circ \varphi_- - i\hbar + kih$  on  $O$  for a constant  $k \in \mathbb{Z}$ . No injectivity is needed for this step.

(ii) *The lifted sewn surface.* Let  $V_- = \{\text{Im } w' < d - D\}$  (repelling time) and let  $V_+$  be the region above the curve  $T(\{\text{Im } w' = -(d - D)\}) - i\hbar$  (attracting time). Let  $\tilde{X}$  be  $V_- \sqcup V_+$  modulo  $w = T(w') - i\hbar$  for  $w' \in \Sigma_q$ . Since  $T$  is a homeomorphism of  $\overline{\Sigma_q}$  onto its image, the identification graph is closed and  $\tilde{X}$  is a Hausdorff Riemann surface. Translation by 1 acts freely on it, and  $X = \tilde{X}/\mathbb{Z} \cong \mathbb{C}^*$ , both ends being punctured discs. The function  $\tilde{K}$ , equal to  $R$  on  $V_+$  and  $S$  on  $V_-$ , is well defined, because  $R(T(w') - i\hbar) = S(w')$  on the band. The sewing solution is  $\tilde{K} \circ \tilde{\Phi}^W$ , where  $\tilde{\Phi}^W$  is the lift of its uniformisation; sewing along the real line or along any other essential curve in  $\Sigma_q$  gives the same  $\tilde{X}$ .

Define  $\tilde{\Phi} : \mathbb{C} \rightarrow \tilde{X}$  by  $\varphi_+ - kih$  on  $\{\text{Im } z > -2Y\}$  and  $\varphi_-$  on  $\{\text{Im } z < -Y\}$ . These agree on  $O$  by (i). They land in  $V_{\mp}$  as follows.  $\text{Im } Q$  is bounded and harmonic on a half-cylinder, hence extends harmonically over the puncture. So on  $\{\text{Im } z \leq -Y - \delta\}$  it is bounded by its values on the line  $\text{Im } z = -Y - \delta \subset O$ . This gives  $\text{Im } \varphi_- \leq -Y + h/2 + 1 < d - D$ . Likewise, on a line in  $O$  near  $\text{Im } z = -2Y$ , (i) gives  $\text{Im}(\varphi_+ - kih) \geq \text{Im } z - 1 - \rho/2\pi$ . The minimum principle then gives  $\text{Im}(\varphi_+ - kih) \geq -2Y - 1 - \rho/2\pi$  above that line. This exceeds the height  $-h/2 - d + D + \rho/2\pi$  of the lower boundary of  $V_+$ , because  $2Y < h - Y_0 - D - 3$ .

(iii)  $\tilde{\Phi}$  is a translate of  $\tilde{\Phi}^W$ .  $\tilde{\Phi}$  commutes with  $z \mapsto z + 1$  and descends to a holomorphic  $\Phi : \mathbb{C}/\mathbb{Z} \rightarrow X \cong \mathbb{C}^*$ .  $\Phi$  is proper, since  $\varphi_{\pm} - \text{id}$  are bounded, and has local degree one at each end, since  $P \rightarrow p_0$  and  $Q \rightarrow q_0$ . A proper holomorphic map of  $\mathbb{C}^*$  that fixes the ends with local

degree one is linear. Hence  $\tilde{\Phi} = \tilde{\Phi}^W(\cdot + c)$  for some  $c \in \mathbb{C}$ , and  $F = \tilde{K} \circ \tilde{\Phi} = K_b^W(\cdot + c)$  in the coordinate of Theorem 6.3(1)–(2). Univalence of  $\varphi_{\pm}$  is a conclusion, not an assumption.

(iv) *The root.* Take a line  $\ell \subset O$  with  $\operatorname{Im} z = -Y''$  and  $Y''$  close to  $2Y$ . There  $\operatorname{Im} \varphi_- \leq -2Y + h/2 + 1 + o(1) < H - \rho$ , so  $\varphi_-(\ell)$  lies in  $\varphi_-^W(\{\operatorname{Im} < H\})$ . The inclusion  $\varphi_-^W(\{\operatorname{Im} < H\}) \supset \{\operatorname{Im} < H - \rho\}$  holds because  $|Q^*| \leq \rho$ : for  $w$  in the smaller set,  $\varphi_-^W - w$  has winding number 1 around 0 on the boundary of a large fundamental rectangle. By injectivity of  $\tilde{\Phi}^W$ , the points  $z + c$  with  $z \in \ell$  lie in  $\{\operatorname{Im} < H\}$ . There  $Q(z) = c + Q^*(z + c)$ , so  $|\operatorname{Im} c - h/2| \leq 1 + \rho$ . On the upper chart,  $\varphi_+(z) - kih = \varphi_+^W(z + c) - ih$ , and  $P(0) = 0$  gives  $\varphi_+^W(c) = ih(1 - k)$ . Since  $|P'| \leq \rho$  where  $\varphi_+^W$  is defined,  $|c - (ih(1 - k) - t_0)| \leq \rho$ . Comparing imaginary parts with the bound on  $\operatorname{Im} c$  gives  $|k|h \leq 1 + 2\rho$ , so  $k = 0$ . Then  $c$  lies in the Rouché disc of Theorem 6.3(3), so  $c = z_0$  and  $F = K_b^W$ . The last sentence of the theorem is Corollary 6.4.  $\square$

## 9. CONTINUATION OF KNESER'S SOLUTION AROUND THE CUSP

This section proves Theorem D, for the continuation that passes near  $\eta$ . The continuation leaves the real exterior ray, crosses the upper Shell–Thron boundary arc close to the cusp, and returns to the real axis below  $\eta$  from the upper half-plane. Near the cusp the two fixed points are close, and Kneser's initial region (the chord between the fixed points together with its image) can be tilted continuously into an initial region for the attracting–repelling configuration below  $\eta$ . The multiplier of the upper fixed point crosses the unit circle on the way, but this plays no role. In the logarithmic coordinate of the pair of fixed points the tilted region is a thin strip on which the map is a translation up to a relative error  $O(|\varepsilon|)$ , so its quotient is quasiconformally a flat cylinder, whether or not the fixed points are linearisable. At the end of the path the tilted region is a fundamental domain for the sewn surface of Proposition 3.4, and a proper-map argument identifies the continued solution with  $K_b^W$ .

*Conventions for this section.* Here  $\varepsilon$  is the complex parameter  $1 - \lambda_1$  defined below; for real  $b < \eta$  it differs from the quantity  $|\log \lambda|$  called  $\varepsilon$  in Sections 6–8 by  $O(\varepsilon^2)$ , and we use results of those sections only in the form stated there. The letters  $\alpha$  (a tilt angle),  $\varsigma$  (a normal coordinate),  $s \in [0, 1]$  (an interpolation parameter),  $H, X, M, q$  and  $f$  have local meanings (in the subsection “Consequences at the cusp”  $f = F_{\eta}^{\text{att}}$ ).  $L_1$  denotes the fixed point with multiplier  $1 - \varepsilon$ : it is the upper fixed point  $L_+$  for real  $b > \eta$  and the attracting fixed point  $L$  for real  $b < \eta$ . Constants  $C, c$  do not depend on  $\varepsilon$ ;  $r_0$  is decreased finitely many times.

**9.1. The cusp parameter.** For small complex  $\varepsilon$  put

$$\lambda_1 = 1 - \varepsilon, \quad \log b(\varepsilon) = \lambda_1 e^{-\lambda_1} = (1 - \varepsilon)e^{\varepsilon-1},$$

and let  $\lambda_2(\varepsilon)$  be the other root of  $\lambda e^{-\lambda} = \log b(\varepsilon)$  near 1. It is holomorphic,  $\lambda_2 = 1 + \varepsilon + \frac{2}{3}\varepsilon^2 + O(\varepsilon^3)$ . By Lemma 2.1,  $L_j = e^{\lambda_j}$  are fixed points of  $E_b$  with multipliers  $\lambda_j$ . In the coordinate  $w = e(1 + u)$ ,

$$f_{\varepsilon}(u) = e^{\mu(1+u)-1} - 1, \quad \mu = e \log b = (1 - \varepsilon)e^{\varepsilon},$$

with fixed points  $u_j = e^{\lambda_j-1} - 1$ , so that  $u_1 - u_2 = -2\varepsilon(1 + O(\varepsilon))$ . For  $0 < r \leq \frac{1}{10}$  let

$$\mathcal{S}_r^c = \left\{ \varepsilon : 0 < |\varepsilon| < r, -\frac{7\pi}{12} < \arg \varepsilon < \frac{\pi}{12} \right\}, \quad \mathcal{S}_r^{c'} = \left\{ \varepsilon \in \mathcal{S}_r^c : -\frac{7\pi}{12} + \frac{1}{10} < \arg \varepsilon < \frac{\pi}{12} - \frac{1}{10} \right\}.$$

The lemmas are stated on  $\mathcal{S}_r^c$ ; the family is built on  $\mathcal{S}_r^{c'}$ , so that the parameter discs of Proposition 9.9 stay in  $\mathcal{S}_r^c$ . Put  $u^* = e^{-1} - 1$ , the base point  $w = 1$ , and  $u_n^* = f_{\varepsilon}^n(u^*)$ .

**Lemma 9.1** (Cusp parameter). *There are  $r_0, r_1 > 0$  with the following properties.*

- (1)  $\varepsilon \mapsto b(\varepsilon)$  is injective on  $\mathcal{S}_{r_0}^c$ . For every  $r \leq r_0$  there is  $r' > 0$  such that  $b(\mathcal{S}_r^{c'})$  contains  $\{b : |b - \eta| < r', \operatorname{Im} b > 0\}$  and the segments  $(\eta - r', \eta)$  and  $(\eta, \eta + r')$ . Real  $b \in (\eta - r_1, \eta)$  corresponds to real  $\varepsilon \in (0, r_0)$ , and then  $L_1 = L$  is the attracting and  $L_2$  the repelling real

fixed point. Real  $b \in (\eta, \eta + r_1)$  corresponds to  $\varepsilon(\theta) = 1 - \theta \cot \theta - i\theta = \frac{1}{3}\theta^2 - i\theta + O(\theta^4)$ ,  $0 < \theta < \theta_1$ ; then  $L_1 = L_+$  is the fixed point in the upper half-plane used in Proposition 9.11, and  $\lambda_2(\varepsilon(\theta)) = \overline{\lambda_1}$ ,  $u_2 = \overline{u_1}$ .

- (2) On  $D_{\rho_0}$ ,  $\rho_0 = \frac{1}{2}$ ,  $f_\varepsilon(u) - u = (u - u_1)(u - u_2)k_\varepsilon(u)$  with  $k_\varepsilon$  jointly holomorphic in  $(\varepsilon, u)$  and  $|k_\varepsilon - \frac{1}{2}| \leq \frac{1}{4}$ . Moreover  $f'_\varepsilon(u_j) = \lambda_j$ .
- (3)  $g_\varepsilon(v) = (\text{Log}(1+v)+1)/\mu-1$  is jointly holomorphic on  $\{\text{Re}(1+v) > 0\}$ , and  $g_\varepsilon(f_\varepsilon(u)) = u$  whenever  $|\text{Im}(\mu(1+u))| < \pi$ , in particular on  $D_{\rho_0}$  and near the real segment  $[u^*, 0]$ .

*Proof.* (1)  $(1 - \varepsilon)e^\varepsilon - 1 = -\frac{1}{2}\varepsilon^2(1 + O(\varepsilon))$ , so  $\log b - \log \eta = -(2e)^{-1}\psi_0(\varepsilon)^2$  with  $\psi_0$  holomorphic and  $\psi_0(\varepsilon) = \varepsilon(1 + O(\varepsilon))$ . For small  $r_0$ ,  $\psi_0$  is injective and maps  $\mathcal{S}_{r_0}^c$  into a sector of opening  $2\pi/3 + o(1) < \pi$ , on which squaring is injective. The image of  $\mathcal{S}_{r_0}^{c'}$  under  $-\psi_0^2$  contains, near 0, all directions in  $(-\frac{\pi}{6} + 0.2 + o(1), \frac{7\pi}{6} - 0.2 - o(1)) \supset [0, \pi]$ . Since  $b \mapsto \log b$  is biholomorphic near  $\eta$ , (1) follows except for the identification of the real segments. For real  $\varepsilon \in (0, 1)$ ,  $\lambda_1 \in (0, 1)$  and Lemma 2.1 applies. For real  $b > \eta$  the upper fixed point has multiplier  $L_+ \log b = \theta \cot \theta + i\theta$  with  $\theta = \text{Im}(L_+ \log b) \in (0, \pi)$  (the computation in the proof of Proposition 9.11), and  $\lambda_1 = 1 - \varepsilon$  gives  $\varepsilon(\theta)$ , whose argument lies in  $(-\pi/2, -\pi/2 + \theta)$ . The conjugate fixed point  $L_- = \overline{L_+}$  is the other one near  $e$ , so  $\lambda_2 = \overline{\lambda_1}$ .

(2)  $k_0(u) = (e^u - 1 - u)/u^2 = \frac{1}{2} + \sum_{n \geq 1} u^n/(n+2)!$  differs from  $\frac{1}{2}$  by less than 0.1 on  $D_{1/2}$ . Division of  $f_\varepsilon(u) - u$  by  $(u - u_1)(u - u_2)$  (Cauchy's formula on  $|u| = 1$ ) gives joint holomorphy and continuity at  $\varepsilon = 0$ . Differentiating the product at  $u_j$  gives the multipliers.

(3)  $\text{Log}(e^{\mu(1+u)-1}) = \mu(1+u) - 1$  when  $|\text{Im}(\mu(1+u))| < \pi$ .  $\square$

**9.2. The logarithmic coordinate.** Put  $q = (u - u_1)/(u - u_2)$  and  $\chi = \log q$ , so that

$$u = u(\chi) = u_2 + \frac{u_1 - u_2}{1 - e^\chi}.$$

The points  $\chi \in 2\pi i\mathbb{Z}$  (the *holes*) correspond to  $u = \infty$ . Let  $\mathcal{G}^c = \{\chi : |\chi - 2\pi ik| \geq \frac{1}{2} \text{ for all } k \in \mathbb{Z}\}$ , and put  $A_1 = 1/\text{Log } \lambda_1 = -\varepsilon^{-1}(1 + O(\varepsilon))$  and  $Z = A_1\chi$ .

**Lemma 9.2** (The step in the logarithmic coordinate). *For  $0 < |\varepsilon| < r_0$  the following hold (the proofs of (1)–(5) do not use the sector).*

- (1) On  $\mathcal{G}^c$ ,  $|1 - e^\chi| \geq \frac{1}{6} \max(1, |e^\chi|)$ ; hence  $|u(\chi)| \leq C|\varepsilon|$  and  $|u - u_1||u - u_2| \leq C|\varepsilon|^2 \min(|e^\chi|, |e^{-\chi}|)$ . On  $\{|\chi| \leq 1\}$ ,  $|1 - e^\chi| \geq 0.63|\chi|$ .
- (2) For  $u \in D_{\rho_0} \setminus \{u_1, u_2\}$ ,

$$\frac{q(f(u))}{q(u)} = 1 + \beta(u), \quad \beta = \frac{(u_1 - u_2)k(u)}{1 + (u - u_1)k(u)},$$

and  $\Delta(u) := \text{Log}(1 + \beta(u)) = -\varepsilon(1 + \vartheta(u))$  with  $|\vartheta| \leq C(|u| + |\varepsilon|)$  and  $|\Delta'(u)| \leq C|\varepsilon|$  on  $D_{\rho_0/2}$ . Thus  $\chi(f(u)) = \chi(u) + \Delta(u)$  on the branch continued along the short segment from  $\chi(u)$ , and  $A_1\Delta = 1 + O(|u| + |\varepsilon|)$ .

- (3) On  $\mathcal{G}^c$ , regarded as functions of  $\chi$ ,  $|\vartheta| \leq C|\varepsilon|$  and  $|\partial_\chi \Delta| \leq C|\varepsilon|^2$ . On  $\{|\chi| \leq 1\} \cap \{|u(\chi)| \leq \rho_0/2\}$ ,  $|\partial_\chi \Delta| \leq C|\varepsilon|^2/|\chi|^2$ .
- (4)  $\Delta(u_1) = \text{Log } \lambda_1$  and  $\Delta(u_2) = -\text{Log } \lambda_2$ . For  $\chi \in \mathcal{G}^c$ ,  $|\Delta - \text{Log } \lambda_1| + |\partial_\chi \Delta| \leq C|\varepsilon|^2|e^\chi|$  if  $\text{Re } \chi \leq 0$ , and  $|\Delta + \text{Log } \lambda_2| + |\partial_\chi \Delta| \leq C|\varepsilon|^2|e^{-\chi}|$  if  $\text{Re } \chi \geq 0$ .
- (5)  $|\partial_\varepsilon \Delta| \leq 1 + C|\varepsilon|$  at fixed  $\chi \in \mathcal{G}^c$ .

All functions here are jointly holomorphic in  $\varepsilon$  and  $u$  (or  $\chi$ ).

*Proof.* (1) On the circle  $|\chi| = \frac{1}{2}$  the minimum of  $|e^\chi - 1|$  is  $1 - e^{-1/2} > 0.39$ , attained at  $\chi = -\frac{1}{2}$ . The function  $1/(e^\chi - 1)$  is holomorphic on the part of the strip  $|\text{Im } \chi| \leq \pi$  outside the disc, is bounded by 1 on its horizontal edges (where  $|e^\chi - 1| = 1 + e^{\text{Re } \chi}$ ), and tends to  $-1$  and 0 at the two ends; the maximum principle and periodicity give  $|e^\chi - 1| \geq 0.39$  on  $\mathcal{G}^c$ . For

$\operatorname{Re} \chi \geq \log 2$ ,  $|e^\chi - 1| \geq \frac{1}{2}|e^\chi|$ , and for  $0 \leq \operatorname{Re} \chi \leq \log 2$ ,  $0.39 \geq \frac{1}{6}|e^\chi|$ . The bounds on  $u$  follow from  $u - u_2 = (u_1 - u_2)/(1 - e^\chi)$  and  $u - u_1 = e^\chi(u - u_2)$ . The last bound is the minimum of  $|(e^\chi - 1)/\chi|$  on the closed unit disc,  $1 - e^{-1}$ .

(2) The identity follows from  $(f(u) - u_j)/(u - u_j) = 1 + (u - u_{3-j})k(u)$ . By Lemma 9.1(2) and  $u_1 - u_2 = -2\varepsilon(1 + O(\varepsilon))$ ,  $\beta = -\varepsilon(1 + O(|u| + |\varepsilon|))$ , and  $\operatorname{Log}(1 + \beta) = \beta(1 + O(|\beta|))$ . Cauchy's estimate on  $D_{\rho_0}$ , where  $\Delta = O(|\varepsilon|)$ , bounds  $\Delta'$ .

(3) The first bound is (2) with (1). For the derivatives,  $\partial_\chi \Delta = \Delta'(u) du/d\chi$  with  $du/d\chi = (u_1 - u_2)e^\chi/(1 - e^\chi)^2$ , which is  $O(|\varepsilon|)$  on  $\mathcal{G}^c$  by (1) and  $O(|\varepsilon|/|\chi|^2)$  on  $|\chi| \leq 1$ .

(4)  $1 + \beta(u_1) = 1 + (u_1 - u_2)k(u_1) = f'(u_1) = \lambda_1$ , and  $1 + \beta(u_2) = 1/(1 + (u_2 - u_1)k(u_2)) = 1/\lambda_2$ ; all logarithms are principal and close to 0. Since  $\Delta'(u) = O(|\varepsilon|)$ ,  $|\Delta(u) - \Delta(u_1)| \leq C|\varepsilon||u - u_1| \leq C|\varepsilon|^2|e^\chi|$  by (1) when  $\operatorname{Re} \chi \leq 0$ , and  $|\partial_\chi \Delta| = |\Delta'(u)| |u_1 - u_2| |e^\chi|/|1 - e^\chi|^2 \leq C|\varepsilon|^2|e^\chi|$ . The case of  $u_2$  is symmetric, with  $e^{-\chi}$  in place of  $e^\chi$ .

(5) At fixed  $\chi$ ,  $\Delta = -\varepsilon(1 + \vartheta)$  with  $\vartheta$  holomorphic in  $\varepsilon$  and  $O(|\varepsilon|)$ ; Cauchy's estimate on a disc of radius  $|\varepsilon|/2$  gives  $\varepsilon \partial_\varepsilon \vartheta = O(|\varepsilon|)$ .  $\square$

**9.3. Tilted initial regions.** Write  $\omega = e^{i\alpha}$ . For  $x_0 \in [-\frac{1}{2}, \frac{1}{2}]$  let

$$\ell = \ell_{\alpha, x_0} = \{i\pi + x_0 + t\omega : t \in \mathbb{R}\},$$

and use the coordinates  $\chi = i\pi + x_0 + (t + i\varsigma)\omega$ ,  $t, \varsigma \in \mathbb{R}$ . We call  $\ell$  *admissible* for  $\varepsilon$  if

$$(16) \quad \cos \alpha \geq \frac{3}{5}, \quad -\operatorname{Im}(\varepsilon \bar{\omega}) \geq \frac{1}{2}|\varepsilon|.$$

For  $\varepsilon \in \mathcal{S}_r^c$  put  $\alpha(\varepsilon) = \frac{1}{2}(\arg \varepsilon + \frac{\pi}{2}) \in (-\frac{\pi}{24}, \frac{7\pi}{24})$ . Then  $\cos \alpha(\varepsilon) > \cos \frac{7\pi}{24} > 0.6$  and  $-\operatorname{Im}(\varepsilon \bar{\omega})/|\varepsilon| = \sin(\frac{\pi}{4} - \frac{1}{2} \arg \varepsilon) > \sin \frac{5\pi}{24} > 0.6$ . Hence, for  $\varepsilon_0 \in \mathcal{S}_r^c$ , the line  $\ell_{\alpha, x_0}$  with  $|\alpha - \alpha(\varepsilon_0)| \leq \frac{1}{20}$  is admissible for every  $\varepsilon \in \mathcal{S}_r^c$  with  $|\arg \varepsilon - \arg \varepsilon_0| \leq \frac{1}{20}$ : then  $\alpha \in (-\frac{\pi}{24}, \frac{7\pi}{24})$ , so  $\cos \alpha > \cos \frac{7\pi}{24} > \frac{3}{5}$ , and  $-\operatorname{Im}(\varepsilon \bar{\omega})/|\varepsilon| \geq \sin(\frac{5\pi}{24} - \frac{1}{10}) > \frac{1}{2}$ . At  $\varepsilon(\theta)$  (real  $b > \eta$ ),  $\alpha = 0$  is admissible; at real  $\varepsilon > 0$ ,  $\alpha(\varepsilon) = \frac{\pi}{4}$ . For fixed  $\varepsilon$  the admissible pairs  $(\alpha, x_0)$  form a rectangle. The distance from  $2\pi i k$  to  $\ell$  is  $|(2k - 1)\pi \cos \alpha + x_0 \sin \alpha| \geq \frac{3}{5}\pi - \frac{2}{5} > 1.48$ , so the strip  $\{|\varsigma| \leq \frac{1}{4}\}$  lies in  $\mathcal{G}^c$ .

**Lemma 9.3** (Tilted initial region). *Let  $\varepsilon \in \mathcal{S}_{r_0}^c$ , let  $\ell$  be admissible for  $\varepsilon$ , and let  $f_\chi(\chi) = \chi + \Delta(\chi)$  be  $f_\varepsilon$  in the coordinate  $\chi$ .*

- (1) *On  $\mathcal{G}^c$  the normal component  $\operatorname{Im}(\Delta \bar{\omega})$  lies in  $[\frac{2}{5}|\varepsilon|, \frac{6}{5}|\varepsilon|]$ , and  $f_\chi$  is injective on every convex subset of  $\mathcal{G}^c$ .*
- (2)  *$f_\chi(\ell)$  is the graph  $\varsigma = g(t)$  of a real-analytic function with  $\frac{2}{5}|\varepsilon| \leq g \leq \frac{6}{5}|\varepsilon|$  and  $|g'| \leq C|\varepsilon|^2$ .*
- (3)  *$M(t, s) = i\pi + x_0 + t\omega + s\Delta(i\pi + x_0 + t\omega)$  is an orientation-preserving real-analytic diffeomorphism of  $\mathbb{R} \times [0, 1]$  onto the closed strip  $\hat{H} = \{0 \leq \varsigma \leq g(t)\}$  between  $\ell$  and  $f_\chi(\ell)$ .*
- (4)  *$\chi \mapsto u(\chi)$  is injective on  $\hat{H}$ . Its image  $H = H_\varepsilon^\ell$  together with  $u_1, u_2$  is a closed Jordan region in  $D_{C|\varepsilon|}$ , bounded by the arcs  $\gamma = u(\ell)$  and  $f(\gamma) = u(f_\chi(\ell))$ , which join  $u_2$  ( $t \rightarrow +\infty$ ) to  $u_1$  ( $t \rightarrow -\infty$ ) and meet only there.*
- (5)  *$f_\varepsilon$  maps a neighbourhood of  $\gamma$  biholomorphically onto a neighbourhood of  $f(\gamma)$ ; in the coordinate  $\chi$  it is  $f_\chi$ , which is injective on the convex strip  $\{|\varsigma| \leq \frac{1}{4}\}$ .*

*Proof.* (1) By Lemma 9.2(3),  $\operatorname{Im}(\Delta \bar{\omega}) = -\operatorname{Im}(\varepsilon \bar{\omega}) - \operatorname{Im}(\varepsilon \vartheta \bar{\omega}) \in [\frac{1}{2}|\varepsilon| - C|\varepsilon|^2, |\varepsilon| + C|\varepsilon|^2]$ . If  $\chi, \chi'$  lie in a convex subset of  $\mathcal{G}^c$ , then  $|f_\chi(\chi) - f_\chi(\chi') - (\chi - \chi')| \leq C|\varepsilon|^2|\chi - \chi'| < |\chi - \chi'|$ .

(2) The  $t$ -coordinate of  $f_\chi(i\pi + x_0 + t\omega)$  is  $t + \operatorname{Re}(\Delta \bar{\omega})$ , whose derivative is  $1 + O(|\varepsilon|^2) > 0$ ; it is therefore a real-analytic bijection of  $\mathbb{R}$ , and (1) and  $|\partial_\chi \Delta| \leq C|\varepsilon|^2$  bound  $g$  and  $g'$ .

(3)  $\partial_t M = \omega(1 + O(|\varepsilon|^2))$  and  $\partial_s M = \Delta$ , so the Jacobian is  $\operatorname{Im}(\Delta \bar{\omega})(1 + O(|\varepsilon|^2)) \geq \frac{1}{3}|\varepsilon| > 0$ .  $M$  maps  $s = 0$  onto  $\ell$  and  $s = 1$  onto  $f_\chi(\ell)$  bijectively, each segment  $t = \text{const}$  to a segment of

length  $\leq 2|\varepsilon|$  joining them, and is proper. A proper local diffeomorphism that is a bijection on the boundary lines is a diffeomorphism onto the region between them.

(4) Points of the strip  $\{|\varsigma| \leq \frac{1}{4}\}$  that differ by  $2\pi ik$ ,  $k \neq 0$ , have  $\varsigma$ -coordinates differing by  $2\pi|k|\cos\alpha > 1$ ; so  $\exp$  is injective on  $\hat{H}$ , and the Möbius map  $q \mapsto u$  is injective. Along  $\ell$ ,  $\operatorname{Re}\chi = x_0 + t\cos\alpha - \varsigma\sin\alpha$  tends to  $\mp\infty$  as  $t \rightarrow \mp\infty$ , so  $q \rightarrow 0$  ( $u \rightarrow u_1$ ) and  $q \rightarrow \infty$  ( $u \rightarrow u_2$ ). The two boundary curves of  $\hat{H}$  are disjoint by (2), and their images are disjoint arcs with common endpoints  $u_1, u_2$ ; the image of  $\hat{H}$  is the closure of the complementary component they bound that contains  $u(\operatorname{int}\hat{H})$ . The inclusion in  $D_{C|\varepsilon|}$  is Lemma 9.2(1).

(5) By Lemma 9.2(2),  $f_\chi$  represents  $f_\varepsilon$  in  $\chi$ , and (1) applies to the convex strip.  $\square$

Let  $X = X_\varepsilon^\ell$  be the Riemann surface obtained from  $H$  by identifying  $u \in \gamma$  with  $f_\varepsilon(u) \in f(\gamma)$ , with charts near the seam given by Lemma 9.3(5). Since  $\chi$  is injective on  $\hat{H}$  we use it (or  $Z = A_1\chi$ ) as the coordinate on  $H$ . We call the branch of  $\chi$  obtained by applying  $f_\chi^{\pm 1}$  to points of  $\hat{H}$  the *continued branch*.

**Lemma 9.4** (Flat model). *Let  $\ell$  be admissible for  $\varepsilon \in \mathcal{S}_{r_0}^c$ . In the coordinate  $Z$  let  $\hat{H}_0$  be the straight strip between the line  $A_1\ell$  and  $A_1\ell + 1$ , and let  $C_0 = \hat{H}_0/(\zeta \sim \zeta + 1) = \mathbb{C}/\mathbb{Z}$ . The map*

$$\mathcal{H}(A_1(i\pi + x_0 + t\omega) + s) = A_1M(t, s), \quad t \in \mathbb{R}, 0 \leq s \leq 1,$$

*induces a quasiconformal homeomorphism  $\mathcal{H} : C_0 \rightarrow X$  with  $|\mathcal{H}(\zeta) - \zeta| \leq C|\varepsilon|$  on  $\hat{H}_0$  and Beltrami coefficient  $|\nu_{\mathcal{H}}| \leq C|\varepsilon|$ . Consequently:*

- (1)  *$X$  is biholomorphic to  $\mathbb{C}^*$ , the ends at  $u_1$  and  $u_2$  corresponding to the punctures. Let  $Z_X : X \rightarrow \mathbb{C}^*$  be a uniformisation with the  $u_1$ -end at 0, and  $z = (2\pi i)^{-1} \log Z_X$ . Crossing the seam from  $\gamma$  to  $f(\gamma)$  increases  $z$  by 1. The inverse  $z \mapsto u$ , extended by  $\tilde{F}(z + 1) = f_\varepsilon(\tilde{F}(z))$  and by one application of  $f_\chi^{-1}$  on the strip  $\{|\varsigma| \leq \frac{1}{4}\}$ , is a holomorphic solution of this equation on the region to the right of the lift of  $f_\chi^{-1}(\ell)$ .*
- (2) *If  $\mathbf{p} \in X$  and the lift  $z$  is normalised at  $\mathbf{p}$ , then for  $|\operatorname{Im}(z - z(\mathbf{p}))| \leq 1$  and  $|\operatorname{Re}(z - z(\mathbf{p}))| \leq 2$  the  $Z$ -coordinate  $Z(z)$  of  $\tilde{F}(z)$  on the continued branch satisfies  $|Z(z) - Z(\mathbf{p}) - (z - z(\mathbf{p}))| \leq C|\varepsilon|$ .*

*Proof.*  $A_1\Delta = 1 + \vartheta_1$  with  $|\vartheta_1| \leq C|\varepsilon|$  on  $\mathcal{G}^c$  (Lemma 9.2(2),(3)). In the source coordinate  $\zeta = A_1(i\pi + x_0 + t\omega) + s$  one has  $\mathcal{H}(\zeta) = \zeta + s\vartheta_1$ , so  $|\mathcal{H}(\zeta) - \zeta| \leq C|\varepsilon|$ . The partial derivatives are  $\partial_t \mathcal{H} = A_1\omega(1 + O(|\varepsilon|^2))$  and  $\partial_s \mathcal{H} = 1 + \vartheta_1$ , against  $A_1\omega$  and 1 for the identity. The unit vectors  $A_1\omega/|A_1\omega|$  and 1 make an angle whose sine is  $|\operatorname{Im}(\omega \overline{\operatorname{Log} \lambda_1})|/|\operatorname{Log} \lambda_1| \geq \frac{1}{2} - C|\varepsilon|$ , by  $\operatorname{Log} \lambda_1 = -\varepsilon(1 + O(\varepsilon))$  and (16). So, in this basis,  $D\mathcal{H} = I + O(|\varepsilon|)$  and  $|\nu_{\mathcal{H}}| \leq C|\varepsilon|$ .  $\mathcal{H}$  is a homeomorphism of the strips by Lemma 9.3(3); it maps the edge  $s = 0$  to  $A_1\ell$  and  $s = 1$  to  $A_1f_\chi(\ell)$  with the same  $t$ , and hence respects the identifications  $\zeta \sim \zeta + 1$  and  $u \sim f_\varepsilon(u)$ . Across  $k$  seams,  $\mathcal{H}(\zeta + k)$  is represented on the continued branch by  $A_1f_\chi^k(\mathcal{H}(\zeta)/A_1)$ , and  $A_1f_\chi^k(\chi) - A_1\chi - k = O(|k||\varepsilon|)$  for  $|k| \leq 3$  by Lemma 9.2(2).

(1)  $C_0 \cong \mathbb{C}^*$ . Each end of  $X$  is the image of an end of  $C_0$ , a punctured disc; quasiconformal maps preserve the infinite modulus of an annular end [14], so each end of  $X$  is a punctured disc and  $X \cong \mathbb{C}^*$ . In  $C_0$  the passage from the edge  $s = 0$  to  $s = 1$  adds 1. The  $u_1$ -end is  $t \rightarrow -\infty$ , where  $\operatorname{Im} \zeta = t \operatorname{Im}(A_1\omega) + O(1) \rightarrow +\infty$  because  $\operatorname{Im}(A_1\omega) = -\operatorname{Im}(\operatorname{Log} \lambda_1 \bar{\omega})/|\operatorname{Log} \lambda_1|^2 < 0$ . Since  $\mathcal{H}$  preserves orientation and ends, the uniformisation with the  $u_1$ -end at 0 has the same properties. The last assertion is the usual lift of the uniformisation; the extra step  $f_\chi^{-1}$  is allowed by Lemma 9.3(5).

(2) Put  $\xi = \exp(2\pi i(\zeta - \zeta_{\mathbf{p}}))$  on  $C_0$ ,  $\zeta_{\mathbf{p}} = \mathcal{H}^{-1}(\mathbf{p})$ , and let  $v$  be the Beltrami coefficient of  $\mathcal{H}$  transported to  $\hat{\mathbb{C}}$ . For complex  $\mathbf{t}$  with  $|\mathbf{t}| < R := 1/(2\|v\|_\infty)$  let  $\mathcal{W}_{\mathbf{t}}$  be the solution with coefficient  $\mathbf{t}v$  fixing  $0, 1, \infty$ . By [2, Thm. 11],  $\mathbf{t} \mapsto \mathcal{W}_{\mathbf{t}}(\xi)$  is holomorphic;  $\mathcal{W}_0 = \operatorname{id}$ ; and the  $\mathcal{W}_{\mathbf{t}}$

are 3-quasiconformal and normalised, hence uniformly bounded, by some  $M_0$ , on the annulus  $A = \{e^{-3\pi} \leq |\xi| \leq e^{3\pi}\}$  [14, Ch. II]. Schwarz's lemma applied to  $t \mapsto \mathcal{W}_t(\xi) - \xi$  on  $|t| < R$  gives  $|\mathcal{W}_1(\xi) - \xi| \leq 2(M_0 + e^{3\pi})\|v\|_\infty \leq C|\varepsilon|$  on  $A$ , since  $R \geq 1$  for small  $|\varepsilon|$ . By a degree argument on the boundary circles of  $A$ ,  $\mathcal{W}_1^{-1}$  also moves points of  $\{e^{-2\pi} \leq |\xi| \leq e^{2\pi}\}$  by  $O(|\varepsilon|)$ .  $\mathcal{W}_1 \circ \exp(2\pi i(\mathcal{H}^{-1} - \zeta_p))$  is a uniformisation of  $X$  normalised at  $p$ , so it equals  $\exp(2\pi i(z - z(p)))$ . Taking logarithms on the annulus, and combining with the bounds on  $\mathcal{H}$  across at most three seams, gives (2).  $\square$

**Lemma 9.5** (The base orbit and the marked point). *There are  $r_2 \leq r_0$ , a small  $\rho_1 > 0$  and  $N_1$  such that the following holds for  $\varepsilon \in \mathcal{S}_{r_2}^c$  and every line  $\ell$  admissible for  $\varepsilon$ . Let  $\chi_n^* = \chi(u_n^*)$ ,  $n \geq N_1$ , be the branch continued from the principal value at  $n = N_1$  (so  $\chi_{n+1}^* = f_\chi(\chi_n^*)$ ), and  $Z_n^* = A_1 \chi_n^*$ .*

- (1) *For  $N_1 \leq n \leq N_1 + 10/|\varepsilon|$ ,  $Z_n^* = Z_{N_1}^* + (n - N_1) + \mathcal{E}_n$  with  $|\mathcal{E}_n| \leq C \log n + C|\varepsilon|n$ , and  $Z_{N_1}^*$  is within relative distance  $\frac{1}{5}$  of a real number  $\geq 4/\rho_1$ .*
- (2) *On that range the normal coordinate  $\varsigma(\chi_n^*)$  is strictly increasing. Let  $n_0$  be the first index with  $\varsigma(\chi_{n_0}^*) \geq 0$ . Then  $n_0 \leq N_1 + 8/|\varepsilon|$  and  $\chi_{n_0}^* \in \hat{H} \setminus f_\chi(\ell)$ .*
- (3) *For  $N_1 \leq n \leq n_0 + 1$ , the  $3|\varepsilon|$ -neighbourhood of  $\chi_n^*$  (equivalently, approximately, the 3-neighbourhood of  $Z_n^*$ ) lies in  $\mathcal{G}^c \cup \{|\chi| < 1\}$  and in  $\{|u(\chi)| \leq 2\rho_1\}$ .*

The marked point of  $X_\varepsilon^\ell$  is the point represented by  $\chi_{n_0}^* \in \hat{H}$ .

*Proof.* At  $\varepsilon = 0$  the orbit  $f_0^n(u^*)$  increases to 0 along the negative axis, with  $f_0^n(u^*) = -2/n + O(\log n/n^2)$ . Choose  $N_1$  with  $|f_0^{N_1}(u^*)| \leq \rho_1/2$ , and then  $r_2$  so small that  $|u_{N_1}^* - f_0^{N_1}(u^*)| \leq \frac{1}{10}|f_0^{N_1}(u^*)|$  for  $|\varepsilon| < r_2$ . Since  $q(u) = 1 + (u_2 - u_1)/(u - u_2)$ ,  $Z_{N_1}^* = -2/u_{N_1}^* (1 + O(|\varepsilon|/\rho_1) + O(\varepsilon))$ , which gives the statement about  $Z_{N_1}^*$ . By Lemma 9.2(2),  $Z_{n+1}^* - Z_n^* = A_1 \Delta(u_n^*) = 1 + e_n$  with  $|e_n| \leq C(|u_n^*| + |\varepsilon|)$  while  $u_n^* \in D_{\rho_0}$ . While  $|\chi| \leq 1$ , Lemma 9.2(1) gives  $|u| \leq |u_2| + |u_1 - u_2|/(0.63|\chi|) \leq 3.2/|Z| + C|\varepsilon|$ ; on  $\mathcal{G}^c$ ,  $|u| \leq C|\varepsilon|$ . By induction  $\text{Re } Z_n^*$  increases by at least  $\frac{1}{2}$  per step, and  $|Z| \geq |Z_{N_1}^*| - 3 \geq 3/\rho_1$  on the 3-neighbourhoods, so  $|u| \leq 1.1\rho_1 + C|\varepsilon|$  there; this gives (1) and the last part of (3). Multiplying by  $1/A_1 = -\varepsilon(1 + O(\varepsilon))$ ,  $\chi_n^*$  lies within  $C|\varepsilon| \log(1/|\varepsilon|) + C|\varepsilon|/\rho_1$  of the ray  $\{-\varepsilon\tau : \tau \geq 0\}$  in the range of (1).

The origin has  $\varsigma(0) = -\pi \cos \alpha + x_0 \sin \alpha \leq -1.48$  and  $|\varsigma(0)| \leq \sqrt{\pi^2 + 1/4}$ , and along the ray  $\varsigma$  increases at rate  $-\text{Im}(\varepsilon\bar{\omega}) \geq |\varepsilon|/2$ ; so the ray meets  $\ell$  at  $\tau \leq 6.4/|\varepsilon|$ . Before that crossing the ray stays at distance  $\geq 2\pi \sin \frac{5\pi}{12} > 6$  from  $2\pi i k$ ,  $k \leq -1$ , and in  $\{\varsigma < 0\}$ , at distance  $\geq 1.48$  from  $2\pi i k$ ,  $k \geq 1$ . With the smallness of the deviation this gives (3). The normal component of each step is at least  $|\varepsilon|(\frac{1}{2} - C\rho_1 - C|\varepsilon|) > 0$ , so  $\varsigma(\chi_n^*)$  is strictly increasing, and  $n_0$  exists with the stated bound. Then  $-\frac{6}{5}|\varepsilon| \leq \varsigma(\chi_{n_0-1}^*) < 0$ , so  $\chi_{n_0-1}^*$  lies in the convex strip  $\{|\varsigma| \leq \frac{1}{4}\}$ , on which each  $\text{id} + s\Delta$ ,  $0 \leq s \leq 1$ , is injective. This isotopy carries  $\ell$  to  $f_\chi(\ell)$  and the side  $\{\varsigma < 0\}$  of  $\ell$  into the corresponding side of  $f_\chi(\ell)$ ; hence  $\chi_{n_0}^* \in \hat{H} \setminus f_\chi(\ell)$ .  $\square$

The physical point  $u_n^*$  may lie in  $H$  for several  $n$ , because  $H$  spirals around  $u_1$  and  $u_2$  when  $\alpha \neq 0$ . These other visits are points of  $X$  different from the marked point, and correspond to the translated roots  $c_j$  discussed after Definition 8.1. Only the first crossing on the continued branch is used.

**Lemma 9.6** (Shadowing the base orbit). *Let  $W_0 = \{z : \text{dist}(z, [0, 1]) < \frac{1}{4}\}$ . Under the hypotheses of Lemma 9.5, let  $z$  be the lift of Lemma 9.4 normalised by  $z = n_0$  at the marked point. Then for  $0 \leq k \leq n_0 - N_1$  the branch  $g_\varepsilon^k$  is holomorphic on  $\tilde{F}(n_0 + W_0)$ , and the  $Z$ -coordinates of  $g_\varepsilon^k(\tilde{F}(n_0 + w))$  lie within  $\frac{1}{10}$  of  $Z_{n_0-k}^* + w$  ( $w \in W_0$ ). For  $|\varepsilon|$  small the  $N_1$  further applications of the same branch  $g_\varepsilon$  (principal logarithm) are defined and jointly holomorphic on  $g_\varepsilon^{n_0-N_1}(\tilde{F}(n_0 + W_0))$  and its images, which stay in  $\{\text{Re}(1+v) > 0\}$  near the real segment  $[u^*, 0]$ .*

*Proof.* By Lemma 9.4(2) with  $\mathbf{p}$  the marked point, the  $Z$ -coordinates of  $\tilde{F}(n_0 + w)$  lie within  $C|\varepsilon|$  of  $Z_{n_0}^* + w$ . In the coordinate  $Z$ ,  $f_\varepsilon$  is  $Z \mapsto Z + A_1\Delta$ , whose derivative is  $1 + \partial_\chi\Delta$ , and  $g_\varepsilon$  is its inverse  $g_Z$ , with  $|g'_Z - 1| \leq 2|\partial_\chi\Delta|$  evaluated at  $g_Z(Z)$ . By Lemma 9.2(3) and  $|\chi| = |Z|/|A_1|$ ,  $|g'_Z - 1| \leq \delta(Z) := C|\varepsilon|^2 + C/|Z|^2$  on the 3-neighbourhood of the orbit segment, where Lemma 9.5(3) applies and  $|Z| \geq |Z_{N_1}^*| - 3$ . Since  $g_Z^k(Z_{n_0}^*) = Z_{n_0-k}^*$ , the mean-value inequality gives, as long as the images stay in that neighbourhood,

$$|g_Z^k(Z) - Z_{n_0-k}^* - (Z - Z_{n_0}^*)| \leq \left( \prod_{j < k} (1 + 2\delta(Z_{n_0-j}^*)) - 1 \right) |Z - Z_{n_0}^*| \leq (e^{C|\varepsilon| + C\rho_1} - 1) \cdot 2,$$

because on the 3-neighbourhood of  $Z_{n_0-j}^*$  one has  $\delta \leq 2\delta(Z_{n_0-j}^*)$ , using  $\sum_j |\varepsilon|^2 \leq C|\varepsilon|$  over  $n_0 \leq N_1 + 8/|\varepsilon|$  steps and  $\sum_j |Z_{n_0-j}^*|^{-2} \leq C/|Z_{N_1}^*| \leq C\rho_1$  (the real parts grow by at least  $\frac{1}{2}$  per step). For small  $\rho_1$  and  $|\varepsilon|$  the right side is below  $\frac{1}{20}$ , so by induction the images stay in the neighbourhood and the claim holds. These sets lie in  $\{|u| \leq 2\rho_1\} \subset D_{\rho_0}$ , where  $g_\varepsilon \circ f_\varepsilon = \text{id}$  (Lemma 9.1(3)).

At  $k = n_0 - N_1$  the set lies within  $\frac{1}{10}$ , in  $Z$ , of  $Z_{N_1}^* + W_0$ . As  $\varepsilon \rightarrow 0$ ,  $Z = A_1\chi$  converges on this region to  $-2/u$  locally uniformly, so the set lies, for small  $|\varepsilon|$ , in a neighbourhood of the compact set  $\overline{K_0}$ ,  $K_0 = \{u : -2/u \in Z_0 + W_0^{1/5}\}$ , where  $Z_0 = -2/f_0^{N_1}(u^*)$  and  $W_0^{1/5}$  is the  $\frac{1}{5}$ -neighbourhood of  $W_0$ . At  $\varepsilon = 0$  the attracting Fatou coordinate is  $\Phi_{\text{att}}(u) = -2/u + \frac{1}{3}\log(-u) + O(u)$ ; on  $\overline{K_0}$  the term  $\frac{1}{3}\log(-u)$  varies by  $O(\rho_1)$ , so  $\Phi_{\text{att}}(\overline{K_0}) \subset c_0 + N_1 + W_0^{1/5 + C\rho_1}$  for a real constant  $c_0$ . The Fatou parametrisation  $P_0 = \Phi_{\text{att}}^{-1}$  extends to an entire function with  $P_0(z+1) = f_0(P_0(z))$ , and  $g_0(P_0(z)) = P_0(z-1)$  near the real axis, where  $1 + P_0(z-1)$  is close to  $(0, 1]$ . Hence  $g_0^{N_1}$  is holomorphic on a neighbourhood of  $\overline{K_0}$ , and all intermediate images lie in a compact subset of  $\{\text{Re}(1+v) > 0\}$  near  $[u^*, 0]$ . This is an open condition, and  $g_\varepsilon \rightarrow g_0$  uniformly on compact subsets of  $\{\text{Re}(1+v) > 0\}$ ; by compactness it persists for small  $|\varepsilon|$ , jointly holomorphically in  $\varepsilon$ .  $\square$

**Definition 9.7.** For  $\varepsilon \in \mathcal{S}_{r_2}^c$  and an admissible line  $\ell$ , the *normalised tilted solution* is

$$F_\varepsilon^\ell(z) = g_\varepsilon^{n_0}(\tilde{F}(z + n_0)), \quad z \in W_0,$$

with  $\tilde{F}$  lifted so that the marked point has  $z = n_0$  and  $g_\varepsilon$  the principal branch of Lemma 9.1(3), which is holomorphic along the way by Lemma 9.6. It is holomorphic on  $W_0$ ,  $F_\varepsilon^\ell(0) = u^*$ , and  $F_\varepsilon^\ell(z+1) = f_\varepsilon(F_\varepsilon^\ell(z))$  where both sides are defined; in  $w = e(1+u)$  it satisfies  $F(z+1) = b^{F(z)}$  and  $F(0) = 1$ .

**Lemma 9.8** (Change of line). *Let  $\ell, \ell'$  be admissible for the same  $\varepsilon \in \mathcal{S}_{r_2}^c$ . Then  $F_\varepsilon^\ell = F_\varepsilon^{\ell'}$ .*

*Proof.* Since the admissible pairs form a rectangle, it suffices to treat  $|\alpha - \alpha'| \leq \frac{1}{100}$  and  $|x_0 - x'_0| \leq \frac{1}{100}$ . Let  $\chi = i\pi + x_0 + (t + i\varsigma)\omega \in \hat{H}$  and let  $\varsigma'$  be its normal coordinate for  $\ell'$ . Then  $|\varsigma'| \leq \frac{1}{100}(1 + |t|) + O(|\varepsilon|)$ . Each application of  $f_\chi^{\pm 1}$  changes  $\varsigma'$  by at least  $\frac{2}{5}|\varepsilon|$  in the same direction (Lemma 9.3(1)) and moves the point by at most  $2|\varepsilon|$ . So after at most  $(1 + |t|)/(40|\varepsilon|)$  steps, forward or backward, the orbit of  $\chi$  on the continued branch reaches the strip  $\hat{H}'$  of  $\ell'$ , within distance  $(1 + |t|)/20$  of  $\chi$ . The distance from  $\chi$  to the holes is at least  $\max(1.48, 0.6|t| - 0.5)$ , which exceeds  $(1 + |t|)/20 + \frac{1}{2}$ ; so this part of the orbit stays in a disc about  $\chi$  contained in  $\mathcal{G}^c$ , on which  $f_\chi$  is injective. As in the proof of Lemma 9.5(2), exactly one iterate  $f_\chi^{n(\chi)}(\chi)$  lies in  $\hat{H}' \setminus f_\chi(\ell')$ . The map  $\chi \mapsto f_\chi^{n(\chi)}(\chi)$  is locally a fixed iterate of  $f_\chi$ , it jumps by one iterate exactly across  $\ell'$ , where the two values are identified in  $X^{\ell'}$ , and it satisfies  $n(f_\chi(\chi)) = n(\chi) - 1$  on  $\ell$ . It therefore induces a holomorphic map  $X^\ell \rightarrow X^{\ell'}$ , and the same construction from  $\ell'$  gives its inverse. It preserves the two ends. The continued base orbit  $\chi_n^*$  is common to both

lines, and both marked points are its first crossings; the orbit segment between them lies in the region just described, so the map sends the marked point  $\chi_{n_0}^*$  of  $\ell$  to  $f_\chi^{n'_0 - n_0}(\chi_{n_0}^*) = \chi_{n'_0}^*$ , the marked point of  $\ell'$ . In the lifts normalised at the marked points ( $z = n_0$  and  $z = n'_0$ ), and since  $f_\chi$  corresponds to  $z \mapsto z + 1$ , one gets  $\tilde{F}'(z + n'_0) = f_\varepsilon^{n'_0 - n_0}(\tilde{F}(z + n_0))$  near  $z = 0$ . Applying the inverse branches of Definition 9.7 gives  $F_\varepsilon^{\ell'} = F_\varepsilon^\ell$  on  $W_0$ .  $\square$

We write  $F_\varepsilon$  for the common normalised tilted solution.

#### 9.4. Holomorphic dependence.

**Proposition 9.9** (The tilted family is holomorphic). *There is  $r_3 \leq \min(r_2, 1/240)$  such that  $(\varepsilon, z) \mapsto F_\varepsilon(z)$  is jointly holomorphic on  $\mathcal{S}_{r_3}^c \times W_0$ .*

*Proof.* Fix  $\varepsilon_0 \in \mathcal{S}_{r_3}^c$  and the disc  $U_0 = \{|\varepsilon - \varepsilon_0| < |\varepsilon_0|/40\}$ ; it lies in  $\mathcal{S}_{r_2}^c$  and  $|\arg \varepsilon - \arg \varepsilon_0| < \frac{1}{20}$  on it. Choose  $\alpha \neq 0$  with  $|\alpha - \alpha(\varepsilon_0)| \leq \frac{1}{20}$ , and then  $x_0 \in [-\frac{1}{2}, \frac{1}{2}]$ , such that  $\chi_{n_0}^*(\varepsilon_0)$  is not on  $\ell$  or  $f_\chi(\ell)$ ; for  $\alpha \neq 0$  a change of  $x_0$  moves the line in its normal direction by  $x_0 \sin \alpha$ , so all but finitely many small values of  $x_0$  will do. By the admissibility remark above  $\ell = \ell_{\alpha, x_0}$  is admissible for all  $\varepsilon \in U_0$ . Shrink  $U_0$  so that  $n_0$  is constant and  $\chi_{n_0}^*(\varepsilon)$  stays in the interior of  $\hat{H}_\varepsilon$ . By Lemma 9.8 this choice does not affect  $F_\varepsilon$ .

The local-triviality argument of Proposition 3.5 is not available here, because at neutral parameters the ends of  $X_\varepsilon$  have no a priori holomorphic puncture coordinate; we use the Beltrami equation instead. Let  $M_\varepsilon$  be the diffeomorphism of Lemma 9.3(3) for  $\ell$ . The map  $\mathcal{J}_\varepsilon = M_\varepsilon \circ M_{\varepsilon_0}^{-1} : \hat{H}_{\varepsilon_0} \rightarrow \hat{H}_\varepsilon$  respects the seam identifications and induces a homeomorphism  $X_{\varepsilon_0} \rightarrow X_\varepsilon$ . At each point its derivative sends  $v_1 = \partial_t M_{\varepsilon_0} = \omega(1 + O(|\varepsilon_0|^2))$  to  $w_1 = \partial_t M_\varepsilon = \omega(1 + O(|\varepsilon_0|^2))$  and  $v_2 = \Delta_{\varepsilon_0}$  to  $w_2 = \Delta_\varepsilon$ , with  $|w_2 - v_2| \leq 1.1|\varepsilon - \varepsilon_0| \leq 0.03|\varepsilon_0|$  by Lemma 9.2(5). Writing it as  $\zeta \mapsto a\zeta + b\bar{\zeta}$ ,

$$b = \frac{(w_1 - v_1)v_2 - (w_2 - v_2)v_1}{2i \operatorname{Im}(v_2 \bar{v}_1)}, \quad |\operatorname{Im}(v_2 \bar{v}_1)| \geq \frac{2}{5}|\varepsilon_0|(1 - C|\varepsilon_0|),$$

so  $|b| \leq 0.05$ ,  $|a - 1| \leq 0.05$ , and the Beltrami coefficient  $v_\varepsilon = b/a$  of  $\mathcal{J}_\varepsilon$  satisfies  $|v_\varepsilon| \leq \frac{1}{10}$ . It is holomorphic in  $\varepsilon$  at each point, since  $a, b$  are affine in  $w_1, w_2$ ; being uniformly bounded, it is holomorphic as an  $L^\infty$ -valued function. Transport  $v_\varepsilon$  to  $\hat{\mathbb{C}}$  by a fixed uniformisation  $Z_{\varepsilon_0}$  of  $X_{\varepsilon_0}$ , and let  $\mathcal{W}_\varepsilon$  be the normalised solution fixing  $0, \infty$  (the ends) and 1. By [2, Thm. 11],  $\mathcal{W}_\varepsilon(\xi)$  is holomorphic in  $\varepsilon$  for each  $\xi$ , and  $Z_\varepsilon = \mathcal{W}_\varepsilon \circ Z_{\varepsilon_0} \circ \mathcal{J}_\varepsilon^{-1}$  is a uniformisation of  $X_\varepsilon$  with the ends at  $0, \infty$ .

In the interior of  $\hat{H}_{\varepsilon_0}$ ,  $\mathcal{J}_\varepsilon$  is a real-analytic diffeomorphism, holomorphic in  $\varepsilon$  at fixed source points, and  $v_\varepsilon$  is real-analytic in the source point and holomorphic in  $\varepsilon$ . By the interior Schauder estimates for the Beltrami equation [1, Ch. V], applied to difference quotients in  $\varepsilon$ ,  $(\varepsilon, \xi) \mapsto \mathcal{W}_\varepsilon(\xi)$  is  $C^1$  there. The maps  $\mathcal{W}_\varepsilon$  and  $\mathcal{J}_\varepsilon \circ Z_{\varepsilon_0}^{-1}$  have the same Beltrami coefficient  $v$  (transported), i.e.  $Z_\varepsilon$  is conformal. Put  $\xi_\varepsilon(\chi) = Z_{\varepsilon_0}(\mathcal{J}_\varepsilon^{-1}(\chi))$ . Differentiating  $\mathcal{J}_\varepsilon(Z_{\varepsilon_0}^{-1}(\xi_\varepsilon(\chi))) = \chi$  in  $\bar{\varepsilon}$  at fixed  $\chi$ , and using that  $\mathcal{J}_\varepsilon$  is holomorphic in  $\varepsilon$  at fixed source points, gives  $\partial_{\bar{\varepsilon}} \xi_\varepsilon + v \partial_{\bar{\varepsilon}} \bar{\xi}_\varepsilon = 0$ . With  $\partial_{\bar{\varepsilon}} \mathcal{W}_\varepsilon = v \partial_{\bar{\varepsilon}} \mathcal{W}_\varepsilon$  and  $\partial_{\bar{\varepsilon}} \mathcal{W}_\varepsilon = 0$  this gives  $\partial_{\bar{\varepsilon}}[\mathcal{W}_\varepsilon(\xi_\varepsilon(\chi))] = \partial_{\bar{\varepsilon}} \mathcal{W}_\varepsilon(\partial_{\bar{\varepsilon}} \xi_\varepsilon + v \partial_{\bar{\varepsilon}} \bar{\xi}_\varepsilon) = 0$ . So  $(\varepsilon, \chi) \mapsto Z_\varepsilon(\chi)$  is jointly holomorphic in the interior of the strips.

With  $\chi_{n_0}^*(\varepsilon)$  in the interior,  $\zeta_0(\varepsilon) = (2\pi i)^{-1} \log Z_\varepsilon(\chi_{n_0}^*(\varepsilon))$  is holomorphic, and  $\tilde{F}_\varepsilon(z + n_0) = u_\varepsilon(Z_\varepsilon^{-1}(e^{2\pi i(z + \zeta_0(\varepsilon))}))$  is jointly holomorphic wherever the point lies in the interior of  $\hat{H}_\varepsilon$ , by the holomorphic inverse function theorem. Points of  $n_0 + W_0$  whose images lie on the seam, or in the extra step of Lemma 9.4(1), are handled in the same way with finitely many further admissible angles  $\alpha$  (by compactness of  $\bar{W}_0$ ), for which they are interior; Lemma 9.8 shows that the two descriptions define the same function. Composing with the inverse branches of Lemma 9.6,

which are jointly holomorphic, gives joint holomorphy on  $U_0 \times W_0$ . On overlapping discs the local families coincide by Lemma 9.8.  $\square$

### 9.5. Abel coordinates on the tilted region.

**Lemma 9.10** (Koenigs times along orbits). *Let  $b < \eta$  be real, so  $\varepsilon > 0$ , and put  $\Psi = A_1 \log(u - u_1) + A_2 \log(u - u_2)$ ,  $A_2 = 1/\log \lambda_2$ , and  $F_b = \Psi \circ f_b - \Psi - 1$  as in Lemma 4.2. Let  $P$  be a path in the  $\chi$ -plane such that for each  $\chi \in P$  the forward orbit  $f_\chi^k(\chi)$  stays in  $\mathcal{G}^c$  and  $\operatorname{Re} f_\chi^k(\chi)$  decreases by at least  $\varepsilon/2$  per step. Then, along  $P$ , the branch of  $R^{-1}$  continued along  $P$  satisfies*

$$R^{-1} = \Psi + \sum_{k \geq 0} F_b \circ f^k + c_+, \quad \sum_{k \geq 0} |F_b(f^k(u))| \leq C\varepsilon,$$

with a constant  $c_+$ . The same holds for  $S^{-1}$  with backward orbits ( $f_\chi^{-k}$ ,  $\operatorname{Re}$  increasing by at least  $\varepsilon/2$  per step) and  $\Psi - \sum_{k \geq 1} F_b \circ f^{-k} + c_-$ .

*Proof.* By Lemma 4.2 and Lemma 9.2(1),  $|F_b(f^k(u))| \leq MC\varepsilon^2 \min(|e^{\chi_k}|, |e^{-\chi_k}|) \leq C\varepsilon^2 e^{-|\operatorname{Re} \chi_k|}$ ; since  $\operatorname{Re} \chi_k$  changes by at least  $\varepsilon/2$  per step, the sum is at most  $C\varepsilon^2 \cdot \sum_k e^{-|\operatorname{Re} \chi_0 - k\varepsilon/2|} \leq C\varepsilon$ . The orbit converges to  $u_1$  because  $\operatorname{Re} \chi_k \rightarrow -\infty$ . As in the proof of Theorem 4.4(1),  $\lim_n [\Psi(f^n u) - n] = \Psi(u) + \sum_{k \geq 0} F_b(f^k u)$  exists and equals  $A_1 \log \sigma_1(u)$  plus a constant, because  $\sigma_1(f^n u) = \lambda_1^n \sigma_1(u)$  and  $A_1 \log \lambda_1 = 1$ ; the constant depends only on the branches, which are continued along  $P$ . The backward case is identical with the reflected inverse family of Theorem 4.4(2).  $\square$

**9.6. Kneser's initial region.** The exterior end of the corridor uses the following parameter-dependent form of Kneser's construction.

**Proposition 9.11** (Local base analyticity on the real exterior ray). *For every real  $b_0 > \eta$ , Kneser's real-normalised solution admits a jointly holomorphic extension in  $(b, z)$  to  $\Delta \times W$ , where  $\Delta$  is a complex disc about  $b_0$  and  $W$  is a complex neighbourhood of  $[0, 1]$ . These local extensions agree on overlap, so the classical exterior branch has a jointly holomorphic germ along the entire real ray  $(\eta, \infty)$ . In particular, every compact subinterval of that ray has a common complex base neighbourhood and a common complex height neighbourhood of  $[0, 1]$  on which this family is jointly holomorphic.*

*Proof.* Write  $a = \log b$  and choose the principal fixed-point pair  $L_+(b)$  and  $L_-(b)$ , conjugate when  $b > \eta$  and characterised there by  $0 < a \operatorname{Im} L_+ < \pi$ . They are simple fixed points and hence holomorphic in  $b$  near  $b_0$ . Set  $u_b = (L_+ + L_-)/2$ ,  $v_b = (L_+ - L_-)/(2i)$ , and  $\gamma_b(t) = u_b + iv_b t$  for  $-1 \leq t \leq 1$ . At a real base,  $u_b, v_b \in \mathbb{R}$ ,  $v_b > 0$ , and

$$E_b(\gamma_b(t)) = r_b e^{i\theta_b t}, \quad r_b = e^{au_b} = |L_+|, \quad \theta_b = av_b \in (0, \pi).$$

Since  $v_b = r_b \sin \theta_b$ , the multipliers satisfy  $|E'_b(L_\pm)| = ar_b = \theta_b / \sin \theta_b > 1$ . The straight chord  $\gamma_b$  and this circular arc meet only at  $L_\pm$  and bound the initial region  $H_b = \{\operatorname{Re} w \geq u_b, |w| \leq r_b\} \setminus \{L_+, L_-\}$ . This is Kneser's initial region as described in [23, §4, Lem. 4]. The arcs depend holomorphically on  $b$  through their real-analytic parameter  $t$ . They remain disjoint away from their endpoints after shrinking a disc  $\Delta$  about  $b_0$ ; at the endpoints their tangents remain distinct because the fixed-point multipliers are nonreal. The bounded Jordan interiors vary continuously, so their union in  $\Delta \times \mathbb{C}$  is open; the ambient coordinates  $(b, w)$  make its projection a holomorphic family. Identify thin analytic collars of the two arcs by  $(b, w) \mapsto (b, E_b(w))$ . The map is injective on the collar after shrinking  $\Delta$ , since it is injective on the compact chord at  $b_0$  and its derivative never vanishes. This gives a Hausdorff holomorphic family of cylinders with one end at each  $L_\pm(b)$ .

The ends can be filled holomorphically. Indeed, parameter-dependent Koenigs coordinates  $\sigma_{\pm,b}$ , normalised by  $\sigma_{\pm,b}(L_{\pm}) = 0$  and  $\sigma'_{\pm,b}(L_{\pm}) = 1$ , linearise  $E_b$  near the two repelling fixed points. If  $\ell_{\pm,b} = \log E'_b(L_{\pm}(b))$  is chosen continuously, then  $\exp(2\pi i \log \sigma_{\pm,b}/\ell_{\pm,b})$  (or its reciprocal) is a puncture coordinate on the corresponding glued end. Filling both punctures yields a proper holomorphic submersion of compact genus-zero surfaces over  $\Delta$ , just as in the collar construction of Proposition 3.5.

We also have a holomorphic marked section represented by  $w = 1$ . For real  $b > \eta$ ,  $E_b(x) > x$  for every real  $x$ : the difference is positive at  $x \rightarrow \pm\infty$  and has no real zero. The adjacent real intervals between the chord and its image are  $(u_b, r_b)$ , with  $r_b = E_b(u_b) > u_b$ . The forward endpoints  $E_b^n(u_b)$  increase to  $+\infty$ , since a finite limit would be a real fixed point. For any  $x > 0$  below  $u_b$ , forward iteration eventually enters  $(u_b, r_b)$  or one of its boundary seams; for  $x > r_b$ , repeated real logarithms eventually do the same. Indeed, a monotone orbit staying on either side would have a finite real fixed-point limit. Thus these dynamical translates and their seam collars cover the positive real axis. In particular, 1 is in the interior of one such interval or on a seam. A finite chain of holomorphic dynamical charts represents it near  $b_0$ ; at a seam the analytic collar gives the same section. The entire compact real segment from 1 to  $b = E_b(1)$  is covered by finitely many of these charts.

The two filled ends and this marked point are three disjoint holomorphic sections. Fischer–Grauert local triviality [8] and the fibrewise three-point Möbius normalisation give a jointly holomorphic coordinate  $q_b$  on the compactified cylinder, taking the upper end to 0, the lower end to  $\infty$ , and  $w = 1$  to 1. The upper puncture coordinate  $\exp(2\pi i \log \sigma_{+,b}/\ell_{+,b})$  shows that the logarithmic lift  $z = (2\pi i)^{-1} \log q_b$ , with  $z(1) = 0$ , gains +1 under the edge identification by  $E_b$ . On the real slice, reflection exchanges the two ends and fixes the marked point, so  $q_b(\bar{w}) = 1/\overline{q_b(w)}$  in the real dynamical charts. Thus  $z$  is real there; its local injectivity and  $z(E_b(w)) = z(w) + 1$  make it strictly increasing along the positive axis. Inverting on the finite chart chain covering the real segment from 1 to  $b$  gives a jointly holomorphic superfunction on  $\Delta \times W$ , after shrinking both.

For real  $b$ , the lift restricted to  $H_b$  is injective and its integer translates cover the universal-cover plane. Choosing any  $d \in H_b$  and subtracting its value gives the initial-region criterion of [23, Thm. 1]; the injectivity and tiling assertions for the Kneser generalisation are proved in [23, Thm. 5]. We normalise at  $d$  here, since 1 need not itself belong to  $H_b$ . Their Abel coordinates differ only by a constant on  $H_b$ , and continuation along the real dynamical charts to  $w = 1$  makes that constant zero. Thus the inverse just constructed is the classical real-normalised Kneser branch. The marking fixes the local family uniquely, and the identity theorem patches the parameter discs on overlaps. A finite subcover of any compact real base interval, followed by intersection of its finitely many height neighbourhoods, gives the final uniformity assertion.  $\square$

### 9.7. The two ends of the corridor.

**Proposition 9.12** (Exterior end). *For  $0 < \theta < \theta_1$  (real  $b \in (\eta, \eta + r_1)$ ),  $F_{\varepsilon(\theta)}$  is Kneser’s real-normalised solution on  $W_0$ .*

*Proof.* The line  $\ell = i\pi + \mathbb{R}$  ( $\alpha = 0$ ,  $x_0 = 0$ ) is admissible for  $\varepsilon(\theta)$ , because  $-\operatorname{Im} \varepsilon(\theta) = \theta$ . Since  $\chi \in i\pi + \mathbb{R}$  means  $q < 0$  and  $u_2 = \overline{u_1}$ ,  $\gamma = u(\ell)$  is the open segment between the conjugate fixed points, i.e. in  $w = e(1+u)$  the chord  $\gamma_b$  of Proposition 9.11, and  $f(\gamma)$  is its image  $E_b(\gamma_b)$ . Hence  $H$  is the initial region  $H_b$  and  $X$  is the cylinder of that proposition, with the upper end at  $L_1 = L_+$ . There the uniformisation takes the upper end to 0 and is normalised so that  $z(1) = 0$ , with  $z$  real and increasing along the positive axis. For conjugate  $u_1, u_2$  the real axis is  $\{|q| = 1\}$ , i.e.  $\operatorname{Re} \chi = 0$ , and the continued branch of the real base orbit moves up the imaginary  $\chi$ -axis from 0; it crosses the chord between  $u_{n_0-1}^*$  and  $u_{n_0}^*$ , and  $u_{n_0}^* = E_b^{n_0}(1)$  has Kneser coordinate  $n_0$ .

So our lift is Kneser's, and our inverse branches are the real logarithms along the orbit, as in Kneser's normalisation. By Lemma 9.8 the admissible line used in the family may be replaced by this one.  $\square$

**Proposition 9.13** (Interior end). *There is  $r_4 > 0$  such that  $F_\varepsilon = K_{b(\varepsilon)}^W$  on  $W_0$  for real  $\varepsilon \in (0, r_4)$ . Here  $r_4$  is chosen so that  $b(\varepsilon) \in (b_1, \eta)$  with the closeness conditions of Theorem 6.3, and so that  $0.39h + C_{\text{im}} < d - D$ , where  $C_{\text{im}}$  is the constant in (17) below and  $d, D$  are as in Lemma 6.2 and Theorem 6.3.*

*Proof.* Let  $\varepsilon \in (0, r_4)$  be real, so  $b < \eta$  is real,  $\lambda_1 \in (0, 1)$  and  $u_1 < 0 < u_2$ . Take  $\alpha = \frac{\pi}{4}$ ,  $x_0 = 0$ , so that  $\chi = i\pi + (t+i\varsigma)e^{i\pi/4}$  has  $\text{Re } \chi = (t-\varsigma)/\sqrt{2}$  and  $\text{Im } \chi = \pi + (t+\varsigma)/\sqrt{2}$ . Put  $h = 2\pi/|\log \lambda_1|$ . The real line of the  $u$ -plane appears as follows:  $u < u_1$  is  $\{\text{Im } \chi \in 2\pi\mathbb{Z}, \text{Re } \chi < 0\}$ , the segment  $(u_1, u_2)$  is  $\{\text{Im } \chi \in \pi + 2\pi\mathbb{Z}\}$ , and  $u > u_2$  is  $\{\text{Im } \chi \in 2\pi\mathbb{Z}, \text{Re } \chi > 0\}$ ; the lower half-plane is  $\{0 < \text{Im } \chi < \pi\}$  modulo  $2\pi$ , because  $q \mapsto u$  has negative determinant  $u_1 - u_2$ . Since  $f_\varepsilon$  is real and  $\Delta$  is real at  $u_1, u_2$ , Lemma 9.2 gives  $\text{Im } \Delta = O(\varepsilon^2)$  on  $\mathcal{G}^c$  and  $\text{Im } \Delta = O(\varepsilon^2|e^{\mp\chi}|)$  near the two ends.

*Marked point.* The base orbit is real,  $u_n^* < u_1$ , so  $\chi_n^*$  is real and moves to the left. On the real axis  $\varsigma = -(\text{Re } \chi + \pi)/\sqrt{2}$ , so the marked point  $p_c := \chi_{n_0}^*$  lies in  $[-\pi - 2\varepsilon, -\pi]$ , with  $t \approx -\pi\sqrt{2}$ .

*The two charts.* Let  $U \subset X$  and  $V \subset X$  be the images of  $M(\{t < \pi\sqrt{2} - 1\})$  and  $M(\{t > -\pi\sqrt{2} + 1\})$ . They are open, and saturated for the seam identification, since  $M(t, 0) \sim M(t, 1)$  preserves  $t$ . For  $\chi \in M(\{t < \pi\sqrt{2} - 1\})$  the forward orbit  $f_\chi^k(\chi)$  moves left by at least  $\varepsilon/2$  per step, and its imaginary part drifts by  $O(\varepsilon)$  in total (the sum of  $O(\varepsilon^2)$  over  $O(1/\varepsilon)$  steps in  $|\text{Re } \chi| \leq 4$  plus the geometric tail at the end). Such points with  $\text{Re } \chi \geq -\frac{1}{2}$  have  $\text{Im } \chi \in (\pi - 1, 2\pi - 0.7)$ , so these orbits stay in  $\mathcal{G}^c$ . Symmetrically, backward orbits of points of  $M(\{t > -\pi\sqrt{2} + 1\})$  move right, stay in  $\mathcal{G}^c$  and converge to  $u_2$ . So Lemma 9.10 applies on both sets, and the attracting and repelling Koenigs coordinates  $\sigma_1$  and  $\tau$  of Section 3 are defined and nonzero along these orbits. Let  $r_+$  be the branch of  $R^{-1} = \log(\sigma_1/\sigma_1(u^*))/\log \lambda_1$  on  $M(\{t < \pi\sqrt{2} - 1\})$  continued from  $p_c$ , where  $r_+(p_c) = n_0$  is real ( $\sigma_1(u_{n_0}^*)/\sigma_1(u^*) = \lambda_1^{n_0} > 0$ ). By Lemma 9.10,  $\partial_\chi r_+ = A_1 + O(1)$  on convex subsets at distance  $\geq \frac{1}{2} + \frac{1}{10}$  from the holes. For  $\chi \in \ell$ ,  $\sigma_1 \circ f = \lambda_1 \sigma_1$  gives  $r_+(f_\chi(\chi)) - r_+(\chi) \in 1 + i\mathbb{Z}$  for the continuation along  $s \mapsto M(t, s)$ , a segment of length  $O(\varepsilon)$  on which  $r_+$  changes by  $O(1)$ , far less than  $h$ . Hence  $r_+ \circ f_\chi = r_+ + 1$  on  $\ell$ , and  $r_+$  descends to a holomorphic map  $U \rightarrow \mathbb{C}/\mathbb{Z}$ .

Let  $(t_c, \varsigma_c)$  be the coordinates of  $p_c$ , so  $0 \leq \varsigma_c < g(t_c)$ , and follow the curve  $\varsigma = \varsigma_c g(t)/g(t_c)$ , which lies in  $\hat{H}$ , in the direction of increasing  $t$ . Since  $|g'| \leq C\varepsilon^2$  and  $g \geq \frac{2}{5}\varepsilon$ ,  $\text{Im } \chi = \pi + (t+\varsigma)/\sqrt{2}$  increases strictly along it, from 0 at  $p_c$ . Let  $u_m \in (u_1, u_2)$  be the point where it equals  $\pi$ . Let  $r_-$  be the branch of  $S^{-1}$  on  $M(\{t > -\pi\sqrt{2} + 1\})$  that is real at  $u_m$ ; it descends to  $V$  in the same way.

*Imaginary parts.* In Lemma 9.10,  $\Psi = A_1\chi + (A_1 + A_2)\log(u - u_2)$ , with  $A_1 = -h/2\pi$ ,  $A_2 = \frac{h}{2\pi}(1 + O(\varepsilon))$  and  $A_1 + A_2 \rightarrow \frac{1}{3}$  (Lemma 4.1), and  $\arg(u - u_2) = \arg(u_1 - u_2) - \arg(1 - e^\chi)$ . On  $U$  the continued  $\arg(1 - e^\chi)$  is bounded, because  $\text{Im } \chi < 2\pi - 0.7$  wherever  $\text{Re } \chi \geq -\frac{1}{2}$  and  $|e^\chi| \leq e^{-1/2}$  elsewhere; on  $U \cap V$  the range of  $\text{Im } \chi$  is bounded. Hence, with the branches fixed above,

$$(17) \quad |\text{Im } r_+ + \frac{h}{2\pi} \text{Im } \chi| \leq C_{\text{im}} \text{ on } U, \quad |\text{Im } r_- + \frac{h}{2\pi} (\text{Im } \chi - \pi)| \leq C_{\text{im}} \text{ on } U \cap V.$$

On  $V \setminus U$ ,  $\text{Im } \chi > 2\pi - 1/\sqrt{2}$ . There  $\arg(1 - e^\chi) = \text{Im } \chi - \pi + O(1)$  where  $\text{Re } \chi \geq 0$ , and  $A_1 - (A_1 + A_2) = -A_2$ , so  $\text{Im } r_- \leq -\min(|A_1|, A_2)(\text{Im } \chi - \pi) + C$ . Along the curve from  $p_c$  to  $u_m$ ,  $\text{Im } \chi$  increases from 0 to  $\pi$ , so the path runs through the lower half-plane:  $\arg(u - u_1)$  increases by  $\pi$  and  $\arg(u - u_2)$  returns to its initial value. Hence  $\text{Im } r_+(u_m) = A_1\pi + O(\varepsilon) = -h/2 + O(\varepsilon)$ . By Lemma 3.1,  $\sigma_1(u_m)/\sigma_1(u^*) < 0$ , so  $\text{Im } r_+(u_m) \in -h/2 + h\mathbb{Z}$ ; therefore  $\text{Im } r_+(u_m) = -h/2$

exactly. This is the seam branch of Proposition 3.4 and Definition 8.1: by item (c) in Section 4, the path from the base point to the segment passing above  $u_1$  gives  $+h/2$ , and the one passing below gives  $-h/2$ . (The reflected branch  $\overline{K_b^W(\bar{z})}$ , denoted  $K_b^{W,\#}$  in Corollary 9.51 below, would require the path above  $u_1$ , i.e. a line of negative slope at real  $\varepsilon$ , which is not admissible.)

*Transition.* On  $U \cap V$ ,  $R(r_+) = u = S(r_-)$ , so  $r_+ - T(r_-) \in ih\mathbb{Z}$  is constant, with  $T$  the map of Section 3 (branch  $\text{Im } t_0 = h/2$ ). At  $u_m$ ,  $r_-$  is real and  $\text{Im } T(r_-) = h/2$  (Lemma 3.1), while  $\text{Im } r_+ = -h/2$ ; hence  $r_+ = T \circ r_- - ih$  on  $U \cap V$ . By (17) and  $|\text{Im } \chi - \pi| \leq \pi - 1/\sqrt{2}$  on  $U \cap V$ ,  $r_-(U \cap V)$  lies in  $\{|\text{Im } w'| < 0.39h + C_{\text{im}}\}$ , which is inside the strip  $\Sigma_q = \{|\text{Im } w'| < d - D\}$  of the proof of Theorem 8.2 by the choice of  $r_4$ . There  $T$  is univalent,  $T(\mathbb{R}) = \{\text{Im } w = h/2\}$ , and  $T$  maps  $\{\text{Im } w' \leq 0\} \cap \Sigma_q$  below and above that line.

*The map to the sewn surface.* Let  $X^W$  be the surface of Proposition 3.4:  $C_+ = \{\text{Im } w \geq -h/2\}/\mathbb{Z}$  and  $C_- = \{\text{Im } w' \leq 0\}/\mathbb{Z}$  glued by  $w = T(w') - ih$ . Define  $\Theta : X \rightarrow X^W$  by  $\Theta = r_+$  where  $\text{Im } r_+ \geq -h/2$  and  $\Theta = r_-$  where  $\text{Im } r_- \leq 0$ . By (17) and the estimate on  $V \setminus U$ , points of  $U \setminus V$  have  $\text{Im } \chi < 1/\sqrt{2}$ , hence  $\text{Im } r_+ > -0.12h - C > -h/2$ , and points of  $V \setminus U$  have  $\text{Im } r_- < -0.38h + C < 0$ . On  $U \cap V$ ,  $\text{Im } r_+ \geq -h/2$  if and only if  $\text{Im } r_- \geq 0$ , by the previous paragraph. So every point of  $X$  is covered, and on the curve  $\text{Im } r_- = 0$  both definitions give the same point of  $X^W$ . Thus  $\Theta$  is continuous, holomorphic off an analytic curve, hence holomorphic. As a point of  $X$  tends to the  $u_1$ -end,  $t \rightarrow -\infty$  and  $\text{Im } r_+ \rightarrow +\infty$ ; as it tends to the  $u_2$ -end,  $\text{Im } r_- \rightarrow -\infty$ . So  $\Theta$  extends to a holomorphic map of the compactified spheres with  $\Theta^{-1}(\text{end}) = \text{end}$ .

It has local degree one at the  $u_1$ -end; in fact  $r_+$  is injective on  $U$ . Let  $x, x' \in U$  be represented by  $\chi, \chi' \in \hat{H}$  with  $r_+(\chi') - r_+(\chi) = m \in \mathbb{Z}$ . By Lemma 9.10,  $r_+ = A_1\chi + \phi_+(\chi)$  where  $\phi_+$  is bounded with bounded derivative on the  $\frac{1}{10}$ -neighbourhood of  $\hat{H} \cap U$ , which lies in  $\mathcal{G}^c$ . Equal imaginary parts give  $|\text{Im } \chi - \text{Im } \chi'| \leq C\varepsilon$ ; since  $\hat{H}$  is a strip of normal width  $\leq \frac{6}{5}\varepsilon$  along a line of slope 1, this forces  $|\chi - \chi'| \leq C_1\varepsilon$ , and then  $|m| \leq |A_1||\chi - \chi'|(1 + C\varepsilon) + C \leq C_2$ . On the convex disc  $N$  of radius  $(C_1 + 3C_2)\varepsilon$  about  $\chi$ ,  $\partial_\chi r_+ = A_1(1 + O(\varepsilon))$ , so  $r_+$  is injective on  $N$ ; the points  $\chi'$  and  $f_\chi^m(\chi)$  lie in  $N$ , and  $r_+ \circ f_\chi^m = r_+ + m$  gives  $r_+(f_\chi^m(\chi)) = r_+(\chi')$ , hence  $\chi' = f_\chi^m(\chi)$ . Since  $\hat{H}$  is a fundamental strip for  $f_\chi$ ,  $x = x'$  in  $X$ . Hence  $\Theta$  has degree one and is biholomorphic.

*Normalisation.*  $\Theta$  maps the marked point  $p_c$  to the point of  $C_+$  with attracting time  $r_+(p_c) = n_0$ . In Proposition 3.4 the lift is fixed by  $K^W(0) = R(0) = 1$  on the branch  $w(0) = 0$ , i.e. attracting time 0 at  $z = 0$ , hence time  $n_0$  at  $z = n_0$ . So the lifts agree, and  $\tilde{F}(z + n_0) = R(r_+) = K_b^W(z + n_0)$  near  $z = 0$ . Near  $z = 0$  the points  $K_b^W(z + k)$ ,  $0 \leq k \leq n_0$ , stay near  $u_k^*$ , where the inverse branches of Definition 9.7 invert  $f_\varepsilon$  (Lemma 9.1(3)); hence  $F_\varepsilon = K_b^W$  near  $z = 0$ , and by the identity theorem on  $W_0$ .  $\square$

## 9.8. The continuation theorem.

**Theorem 9.14** (Continuation around the cusp). *There are  $r_5 > 0$  and an open set  $\Omega \supset D^+ \cup (\eta - r_5, \eta) \cup (\eta, \eta + r_5)$ , where  $D^+ = \{b : |b - \eta| < r_5, \text{Im } b > 0\}$ , with the following properties.*

- (1) *Kneser's real-normalised solution  $\kappa_b$ ,  $b \in (\eta, \eta + r_5)$ , continues analytically in the base to a family on  $\Omega$  that is jointly holomorphic on  $\Omega \times W_0$ ,  $W_0 = \{z : \text{dist}(z, [0, 1]) < \frac{1}{4}\}$ . In particular it continues across the arc of the Shell–Thron boundary contained in  $D^+$ , whatever the arithmetic of the neutral multiplier there.*
- (2) *On  $(\eta - r_5, \eta)$  the continued family equals  $K_b^W$  on  $W_0$ .*
- (3) *Let  $N_W$  be the neighbourhood of  $(1, \eta)$  of Proposition 3.5. Continuing the family from a neighbourhood of  $(\eta - r_5, \eta)$  inside  $N_W$ , on a  $b$ -dependent neighbourhood of  $z = 0$  contained in the regular domain of  $K_b^W$ , one obtains  $K^W$ . Hence for every  $b \in (1, \eta)$ , if*

$\kappa_{b+iy}$  is continued first through  $D^+$  and then inside  $N_W \cap \{\operatorname{Im} b > 0\}$ , the limit  $\lim_{y \downarrow 0} \kappa_{b+iy}$  exists near  $z = 0$  and equals  $K_b^W$ ; by the functional equation and continuation in  $z$  it equals  $K_b^W$  on its whole regular domain.

*Proof.* (1) By Lemma 9.1,  $\varepsilon \mapsto b(\varepsilon)$  maps  $\mathcal{S}_{r_3}^{c_f}$  biholomorphically onto an open set  $\Omega$  that contains  $D^+$  and the two real segments, for  $r_5$  small. By Proposition 9.9,  $(b, z) \mapsto F_{\varepsilon(b)}(z)$  is jointly holomorphic on  $\Omega \times W_0$ . By Proposition 9.12 it equals  $\kappa_b$  on  $(\eta, \eta + r_5)$ . Since  $\Omega$  is connected, it is the analytic continuation of  $\kappa$  along every path in  $\Omega$ . The Shell–Thron boundary near  $\eta$  consists of  $b = \exp(\lambda e^{-\lambda})$  with  $|\lambda| = 1$ . Its arc in the upper half-plane is  $\varepsilon = 1 - e^{i\phi} = -i\phi + O(\phi^2)$  with small  $\phi > 0$ , since then  $b - \eta$  is a positive multiple of  $-\varepsilon^2 = \phi^2 + i\phi^3 + O(\phi^4)$ ; it lies in  $\mathcal{S}_{r_3}^{c_f}$ .

(2) is Proposition 9.13.

(3) By Proposition 3.5,  $K^W$  is jointly holomorphic on the set of  $(b, z)$  with  $b \in N_W$  and  $z$  in a neighbourhood of 0 that depends continuously on  $b$ ; the upper chart contains  $w = 0$ , where  $R$  is regular. Shrinking the discs  $\Delta_{b_0}$  of that proposition, we may assume  $\Delta_{b_0} \subset \{|b - b_0| < (\eta - b_0)/2\}$ ; then  $N_W \cap \{\operatorname{Im} b > 0\}$  is a union of upper half-discs centred on  $(1, \eta)$ , which is connected since consecutive ones overlap. Each  $\Delta_{b_0} \cap D^+$  is convex, and if it is nonempty its closure meets  $(\eta - r_5, \eta)$ . On a neighbourhood of  $(\eta - r_5, \eta)$  times a small disc about  $z = 0$ , the two families agree by (2) and the identity theorem; hence they agree on every  $\Delta_{b_0} \cap D^+$ , and the continuation from  $D^+$  into  $N_W \cap \{\operatorname{Im} b > 0\}$  is unambiguous and equals  $K^W$ . At every  $b \in (1, \eta)$  the boundary value is therefore  $K_b^W$  near  $z = 0$ . Both sides satisfy  $F(z + 1) = b^{F(z)}$ , so they agree on the whole regular domain.  $\square$

**Corollary 9.15** (Separation limits for the continued Kneser family). *For the continuation of Kneser's solution in Theorem 9.14,  $K_b = \lim_{y \downarrow 0} \kappa_{b+iy}$  exists for  $b \in (1, \eta)$  and equals  $K_b^W$ . Its separation coefficients satisfy  $c_n/\Lambda^n \rightarrow B_n e^{2\pi i n a}$  for every  $n \geq 1$  as  $b \rightarrow \eta^-$ . In particular  $|c_1|/\Lambda \rightarrow |B_1| = 0.0890584\dots$ , with  $B_1 \neq 0$ . This confirms the limit suggested by the numerics of Section 10.1: the first separation mode of the continued Kneser family, scaled by  $\Lambda$ , tends to the first coefficient of the inverse horn map of  $e^u - 1$ .*

*Proof.* Theorem 9.14 gives  $K_b = K_b^W$ ; apply Corollary 1.1 and Proposition 2.4.  $\square$

**Remark 9.16** (Lower half-plane, and what is not claimed). (1) Complex conjugation of the whole construction uses the sector  $\{\bar{\varepsilon} : \varepsilon \in \mathcal{S}_r^c\}$ , the lines  $\ell_{\alpha, x_0}$  through  $-i\pi \equiv i\pi \pmod{2\pi i}$  with slope  $-\alpha$ , and the conjugate admissibility condition  $+\operatorname{Im}(\varepsilon\bar{\omega}) \geq \frac{1}{2}|\varepsilon|$ . It shows that continuation of Kneser's solution through the lower half-disc yields the reflected branch  $K_b^{W, \#}$  of Corollary 9.51 on  $(\eta - r_5, \eta)$ . By Corollary 9.51 below, the two continuations differ by at least  $2c\Lambda$  at the height  $x_*$  of Corollary 9.49; this is the monodromy of Kneser's family around  $\eta$ .

(2) The proof does not use any arithmetic property of the multiplier on the Shell–Thron boundary. The ends of the tilted region are punctures for Siegel, Cremer and parabolic multipliers alike, because the only inputs are the step estimate  $A_1\Delta = 1 + O(\varepsilon)$  and the transversality  $\operatorname{Im}(\operatorname{Log} \lambda_1 \bar{\omega}) > 0$ . In particular no Brjuno condition, Siegel-disc estimate or boundary-value (Painlevé-type) argument is needed near  $\eta$ .

(3) The theorem asserts holomorphy in  $(b, z)$  for heights near  $[0, 1]$ , and by the functional equation on the regular domain reached from there. No claim is made about the behaviour of the continued solution as  $\operatorname{Im} z \rightarrow \pm\infty$  at parameters with a neutral multiplier; the construction only uses that each end of the tilted region is a puncture of the quotient.

(4) The theorem concerns continuation along the stated paths. The width of  $N_W$  is not quantified; the restriction to paths near the real axis is removed in Theorem 9.22, and the relation to Paulsen's family is discussed in Remark 9.25.

**9.9. The whole upper Shell–Thron region.** Theorem 9.14 continues Kneser's solution into a corridor about the cusp. Here we show that the continued family extends, single-valued, to

the whole upper half of the Shell–Thron region. The extension is a quasiconformal transport of one tilted initial region by a holomorphic motion that conjugates the dynamics; the new point is a comparison lemma identifying the transported quotient with the tilted one near the base parameter. In this subsection  $\lambda$  denotes the multiplier of the attracting fixed point, so  $\lambda = \lambda_1 = 1 - \varepsilon$  in the notation above, and

$$\mathbb{H}_\lambda = \{\lambda : |\lambda| < 1, \operatorname{Im} \lambda > 0\}, \quad b(\lambda) = \exp(\lambda e^{-\lambda}), \quad E_\lambda = E_{b(\lambda)}.$$

**Lemma 9.17** (Global parameter).  *$b$  is injective on the unit disc,  $E_\lambda$  has the fixed point  $L_1(\lambda) = e^\lambda$  with multiplier  $\lambda$ , and  $b(\{|\lambda| < 1\})$  is the set of bases for which  $E_b$  has an attracting fixed point. The set  $\mathcal{U}^+ = b(\mathbb{H}_\lambda)$  lies in  $\{\operatorname{Im} b > 0\}$  and is simply connected.*

*Proof.* For  $g(\lambda) = \lambda e^{-\lambda}$  one has  $\operatorname{Re}(\lambda g'/g) = \operatorname{Re}(1 - \lambda) > 0$  on the unit disc, so  $g$  is starlike and injective. Since  $|g| < e < \pi$ , equality  $e^{g(\lambda_1)} = e^{g(\lambda_2)}$  forces  $g(\lambda_1) = g(\lambda_2)$ ; so  $b$  is injective. If  $L$  is a fixed point of  $E_b$  with multiplier  $\lambda = L \log b$  of modulus  $< 1$ , then  $L = e^\lambda$  and  $\log b = g(\lambda)$  (Lemma 2.1). By symmetry and injectivity  $g$  maps the upper half-disc into  $\{\operatorname{Im} > 0\}$  (check  $\lambda = iy$ ,  $y > 0$  small); since  $|\operatorname{Im} g| < \pi$ ,  $\operatorname{Im} b > 0$  there. An injective image of a simply connected domain is simply connected.  $\square$

Fix a base parameter  $\lambda_0 = 1 - \varepsilon_0$  with  $\varepsilon_0 \in \mathcal{S}_{r_3}'$ ,  $\operatorname{Im} \varepsilon_0 < 0$  and  $|\lambda_0| < 1$ , such that  $E_{\lambda_0}^n(1) \neq L_1(\lambda_0)$  for all  $n$ . Each of these equations is holomorphic in  $\lambda$  on the unit disc and not identically satisfied (for real  $\lambda \in (0, 1)$  the orbit of 1 increases strictly to  $L_1$ ), so this excludes a countable set. Only  $\sigma_0(p_0) \neq 0$  for the marked point  $p_0$  below is used. We take  $r_3 \leq 1/240$  in Proposition 9.9, which only shrinks the corridor. Write  $E_0 = E_{\lambda_0}$ , let  $B_0$  be its attracting basin and  $\sigma_0$  its Koenigs coordinate at  $L_1(\lambda_0)$ , normalised by  $\sigma_0'(L_1) = 1$  and extended to  $B_0$  by  $\sigma_0 = \lambda_0^{-n} \sigma_0 \circ E_0^n$ . Since  $E_0$  has no critical points,  $\sigma_0' \neq 0$  on  $B_0$ .

**Proposition 9.18** (Transport of the dynamics). *For  $\lambda \in \mathbb{H}_\lambda$  put  $s = \operatorname{Log} \lambda$ ,  $s_0 = \operatorname{Log} \lambda_0$  (so  $\operatorname{Re} s, \operatorname{Re} s_0 < 0$ ),*

$$b(\lambda) = \frac{s - s_0}{2 \operatorname{Re} s_0}, \quad c(\lambda) = \frac{b}{1 + b} = \frac{s - s_0}{s + \overline{s_0}}, \quad \phi_\lambda(\zeta) = \zeta |\zeta|^{2b(\lambda)}.$$

*Then  $|c(\lambda)| < 1$  on  $\mathbb{H}_\lambda$ . Let  $\nu_\lambda = c(\lambda) \frac{\sigma_0}{\sigma_0'} \frac{\overline{\sigma_0'}}{\sigma_0}$  on  $B_0 \setminus \sigma_0^{-1}(0)$  and  $\nu_\lambda = 0$  elsewhere, and let  $h_\lambda$  be the quasiconformal homeomorphism of  $\hat{\mathbb{C}}$  with Beltrami coefficient  $\nu_\lambda$  fixing  $0, 1, \infty$ . Then:*

- (1)  $\lambda \mapsto h_\lambda(w)$  is holomorphic for each  $w$ ,  $h_{\lambda_0} = \operatorname{id}$ , and  $h_\lambda \rightarrow \operatorname{id}$  locally uniformly on  $\mathbb{C}$  as  $\lambda \rightarrow \lambda_0$ ;
- (2)  $h_\lambda \circ E_0 = E_\lambda \circ h_\lambda$ , hence  $h_\lambda(E_0^n(1)) = E_\lambda^n(1)$  for all  $n \geq 0$ ;
- (3)  $h_\lambda(B_0)$  is the basin of  $L_1(\lambda) = h_\lambda(L_1(\lambda_0))$ , and  $\sigma_\lambda \circ h_\lambda = k_\lambda \phi_\lambda \circ \sigma_0$  on  $B_0$ , with  $k_\lambda \neq 0$  holomorphic in  $\lambda$  and  $k_{\lambda_0} = 1$ ;
- (4) for  $\lambda$  near  $\lambda_0$ ,  $h_\lambda$  maps the repelling fixed point  $L_2(\lambda_0)$  of Section 9 to  $L_2(\lambda)$ .

*Proof.*  $|s + \overline{s_0}|^2 - |s - s_0|^2 = 4 \operatorname{Re} s \operatorname{Re} s_0 > 0$ , so  $|c| < 1$ , uniformly on compact sets. In  $L = \log \zeta$ ,  $\phi_\lambda$  is the real-linear map  $L \mapsto L + 2b \operatorname{Re} L$ , with determinant  $\operatorname{Re} s / \operatorname{Re} s_0 > 0$ ; it fixes  $2\pi i$  and sends  $s_0$  to  $s$ . So  $\phi_\lambda$  is a quasiconformal homeomorphism with  $\phi_\lambda(\lambda_0 \zeta) = \lambda \phi_\lambda(\zeta)$  and Beltrami coefficient  $c(\lambda) \zeta / \overline{\zeta}$ ;  $\nu_\lambda$  is its pull-back by  $\sigma_0$ , hence  $E_0$ -invariant (the basin is completely invariant and  $\sigma_0 \circ E_0 = \lambda_0 \sigma_0$ ). Item (1) is the Ahlfors–Bers theorem [2, Thm. 11], since  $\nu_\lambda$  depends holomorphically on  $\lambda$  through the scalar  $c(\lambda)$ , and continuity of normalised solutions in  $\nu$ . The map  $h_\lambda \circ E_0$  has Beltrami coefficient  $E_0^* \nu_\lambda = \nu_\lambda$ , so  $G_\lambda = h_\lambda \circ E_0 \circ h_\lambda^{-1}$  has vanishing dilatation and is holomorphic on  $\mathbb{C}$  by Weyl’s lemma; it is a covering of  $\mathbb{C}^*$  (it omits  $h_\lambda(0) = 0$ ), so  $G_\lambda(w) = e^{Aw+D}$ , and  $G_\lambda(0) = h_\lambda(1) = 1$  gives  $D \in 2\pi i \mathbb{Z}$ , which we drop. Near  $L_1(\lambda_0)$ ,  $h_\lambda \circ \sigma_0^{-1} \circ \phi_\lambda^{-1}$  is conformal, so  $G_\lambda$  is conformally conjugate to  $\zeta \mapsto \lambda \zeta$  at

$h_\lambda(L_1(\lambda_0))$ : that point is a fixed point of  $e^{Aw}$  with multiplier  $\lambda$ , which forces it to be  $e^\lambda$  and  $A = \lambda e^{-\lambda}$ . So  $G_\lambda = E_\lambda$ , proving (2). For (3), let  $V = \sigma_0^{-1}(D_\varrho)$  be a small forward-invariant disc about  $L_1(\lambda_0)$ . The map  $\psi = \sigma_\lambda \circ h_\lambda \circ \sigma_0^{-1} \circ \phi_\lambda^{-1}$  is defined on  $\phi_\lambda(D_\varrho) = D_{\varrho^{\operatorname{Re}s/\operatorname{Re}s_0}}$ , conformal off 0 (equal Beltrami coefficients) and continuous at 0, hence conformal; it fixes 0 and satisfies  $\psi(\lambda\zeta) = \lambda\psi(\zeta)$ . Comparing Taylor coefficients ( $\lambda^{n-1} \neq 1$  for  $n \geq 2$ ) gives  $\psi(\zeta) = k_\lambda\zeta$ . The identity extends to  $B_0$  by  $\sigma_\lambda = \lambda^{-n}\sigma_\lambda \circ E_\lambda^n$ , and  $k_\lambda = \sigma_\lambda(h_\lambda(w))/\phi_\lambda(\sigma_0(w))$  is holomorphic in  $\lambda$ , because  $\phi_\lambda(\zeta) = \zeta \exp(b(\lambda) \log|\zeta|^2)$  is. In particular, on  $B_0$  near any point where  $\sigma_\lambda$  is locally injective,  $h_\lambda = \sigma_\lambda^{-1} \circ (k_\lambda \phi_\lambda \circ \sigma_0)$  is jointly real-analytic in the space variable, holomorphic in  $\lambda$ , with nonvanishing Jacobian. For (4),  $h_\lambda(L_2(\lambda_0))$  is a fixed point of  $E_\lambda$  depending continuously on  $\lambda$  and equal to  $L_2(\lambda_0)$  at  $\lambda_0$ ; fixed points are isolated.  $\square$

Let  $H_0 \subset \mathbb{C}$  be the tilted region of Lemma 9.3 for  $E_0$  and an admissible line  $\ell$ , written in  $w = e(1+u)$ , and  $p_0 = E_0^{n_0}(1)$  its marked point; choose  $\ell$  with  $p_0$  in the interior. For  $\lambda \in \mathbb{H}_\lambda$  let  $X_\lambda^{\operatorname{tr}}$  be  $h_\lambda(H_0)$  with its edges  $h_\lambda(\gamma)$  and  $h_\lambda(E_0(\gamma)) = E_\lambda(h_\lambda(\gamma))$  identified by  $E_\lambda$ , and  $p_\lambda = h_\lambda(p_0) = E_\lambda^{n_0}(1)$ .

**Proposition 9.19** (The transported family).  *$X_\lambda^{\operatorname{tr}}$  is biholomorphic to  $\mathbb{C}^*$ , its ends at  $L_{1,2}$ . Let  $z$  be the lift of its uniformisation with the  $L_1$ -end at  $+i\infty$  and  $z = n_0$  at  $p_\lambda$ , and let  $F_\lambda^{\operatorname{tr}}$  be the germ at  $z = 0$  given by the local inverse of  $E_\lambda^{n_0}$  at  $p_\lambda$  with value 1, applied to the inverse of  $z$  at  $z + n_0$ . Then  $F_\lambda^{\operatorname{tr}}(0) = 1$ , the lifted inverse uniformisation  $\tilde{F}$  satisfies  $\tilde{F}(z+1) = E_\lambda(\tilde{F}(z))$ , and  $(\lambda, z) \mapsto F_\lambda^{\operatorname{tr}}(z)$  is jointly holomorphic near  $\mathbb{H}_\lambda \times \{0\}$  and single-valued there.*

*Proof.*  $h_\lambda$  induces a quasiconformal homeomorphism  $J_\lambda : X_0 \rightarrow X_\lambda^{\operatorname{tr}}$ , because it conjugates the edge identifications; each end of  $X_\lambda^{\operatorname{tr}}$  is the image of a punctured-disc end of  $X_0 \cong \mathbb{C}^*$  (Lemma 9.4), hence a punctured disc. Near the seam the complex structure is given by local inverses of  $E_\lambda$ , which is injective there because  $E_0$  is. The Beltrami coefficient of  $J_\lambda$  is the descent of  $\nu_\lambda$ ; transported to  $\hat{\mathbb{C}}$  by a uniformisation  $Z_0$  of  $X_0$  normalised at the ends and  $p_0$ , it equals  $c(\lambda)\nu$  for a fixed measurable  $\nu$  with  $|\nu| \leq 1$ . Let  $\mathcal{W}_\lambda$  be the normalised solution; by Ahlfors–Bers it is holomorphic in  $\lambda$  at each point, and  $Z_\lambda = \mathcal{W}_\lambda \circ Z_0 \circ J_\lambda^{-1}$  is the uniformisation of  $X_\lambda^{\operatorname{tr}}$  normalised at the ends and  $p_\lambda$ . Near  $p_0$  the coefficient  $\nu_\lambda$  is real-analytic in  $w$ , because  $\sigma_0(p_0) \neq 0$ . The  $\partial_{\bar{\lambda}}$ -cancellation of Proposition 9.9, with  $h_\lambda$  in place of  $\mathcal{J}_\varepsilon$ , shows that  $(\lambda, w) \mapsto Z_\lambda(w)$  is jointly holomorphic near  $(\lambda, p_\lambda)$ . The inverse function theorem and the local inverse of  $E_\lambda^{n_0}$ , which is determined by the value 1 at  $p_\lambda$  and depends holomorphically on  $\lambda$ , give the germ. The  $C^1$ -dependence needed in the cancellation is explicit here: near  $p_0$ ,  $h_\lambda$  is given by the formula at the end of the proof of Proposition 9.18, and  $\mathcal{W}_\lambda$  is  $C^1$  in  $(\lambda, \xi)$  by interior Schauder estimates [1, Ch. V]. All objects are defined by a single formula for every  $\lambda \in \mathbb{H}_\lambda$ , so the family is single-valued.  $\square$

**Lemma 9.20** (The lifted motion near the ends). *For  $\lambda$  near  $\lambda_0$  let  $\chi_\lambda = \log \frac{u-u_1(\lambda)}{u-u_2(\lambda)}$  be the coordinate of Section 9 for  $E_\lambda$ , and let  $\tilde{h}_\lambda$  be the lift of  $h_\lambda$  to these coordinates (defined off the holes, commuting with  $\chi \mapsto \chi + 2\pi i$ , continuous in  $\lambda$  and equal to the identity at  $\lambda_0$ ). The lift satisfies  $\tilde{h}_\lambda \circ f_{\chi, \lambda_0} = f_{\chi, \lambda} \circ \tilde{h}_\lambda$  exactly (no deck shift), and  $\tilde{h}_\lambda \rightarrow \operatorname{id}$  uniformly on compact subsets of  $\mathcal{G}^c$ . Moreover, for every  $\delta > 0$  there are  $T$  and a neighbourhood  $U_\delta$  of  $\lambda_0$  such that for  $\lambda \in U_\delta$  there are real-linear maps  $A_\lambda^\pm$  and constants  $a_\lambda^\pm$  with  $\|A_\lambda^\pm - I\| + |a_\lambda^\pm| \leq \delta$  and*

$$|\tilde{h}_\lambda(\chi) - A_\lambda^+ \chi - a_\lambda^+| \leq \delta \quad (\operatorname{Re} \chi \leq -T), \quad |\tilde{h}_\lambda(\chi) - A_\lambda^- \chi - a_\lambda^-| \leq \delta \quad (\operatorname{Re} \chi \geq T).$$

*The same holds for  $\tilde{h}_\lambda^{-1}$ .*

*Proof.* This is a statement for  $\lambda$  near  $\lambda_0$ , where  $u_{1,2}(\lambda)$  and  $\chi_\lambda$  are defined.  $h_\lambda$  maps  $u_1(\lambda_0), u_2(\lambda_0), \infty$  to  $u_1(\lambda), u_2(\lambda), \infty$  (Proposition 9.18(3),(4)), so in the cross-ratio  $q = (u - u_1)/(u - u_2)$  it is an orientation-preserving homeomorphism of  $\mathbb{C}^* \setminus \{1\}$  fixing  $0, \infty$ . Such a map acts trivially on

the winding number about 0, so it lifts to  $\mathbb{C} \setminus 2\pi i\mathbb{Z}$  commuting with  $2\pi i$ ; joint continuity of  $(\lambda, w) \mapsto h_\lambda(w)$  (the  $\lambda$ -lemma) lets us choose the lift continuously with  $\tilde{h}_{\lambda_0} = \text{id}$ , and gives local uniform convergence. The lifted conjugacy  $\tilde{h}_\lambda \circ f_{\chi, \lambda_0} = f_{\chi, \lambda} \circ \tilde{h}_\lambda + 2\pi i k$  holds with an integer  $k$  that is continuous in  $(\lambda, \chi)$  and 0 at  $\lambda_0$ , hence  $k = 0$ .

Attracting end. In  $L = \log \sigma_0$  and  $L' = \log \sigma_\lambda$ , Proposition 9.18(3) says that  $h_\lambda$  is the affine map  $L' = L + 2b \operatorname{Re} L + \log k_\lambda$ . Near  $u_1$ ,  $\sigma_\lambda$  and  $q$  are both local coordinates vanishing at  $u_1$ , so  $\chi_\lambda = L' + \psi_\lambda(e^{L'})$  with  $\psi_\lambda$  holomorphic near 0 and continuous in  $\lambda$ ; likewise for  $\lambda_0$ . Composing, with  $A_\lambda^+ = I + 2b \operatorname{Re}$  and  $a_\lambda^+ = \log k_\lambda + \psi_\lambda(0) - A_\lambda^+ \psi_{\lambda_0}(0)$ , the remainder  $\tilde{h}_\lambda(\chi) - A_\lambda^+ \chi - a_\lambda^+$  equals  $[\psi_\lambda(e^{L'}) - \psi_\lambda(0)] - A_\lambda^+[\psi_{\lambda_0}(e^L) - \psi_{\lambda_0}(0)]$ . Since  $|e^{L'}| \leq C|e^L|^{\operatorname{Re} s / \operatorname{Re} s_0}$  and  $\operatorname{Re} s / \operatorname{Re} s_0 \geq \frac{1}{2}$  near  $\lambda_0$ , it is at most  $Ce^{-T/2}$  on  $\operatorname{Re} \chi \leq -T$ , uniformly for  $\lambda$  near  $\lambda_0$ . Choose  $T$  with  $Ce^{-T/2} \leq \delta$ , and then  $U_\delta$  so that  $\|A_\lambda^+ - I\| + |a_\lambda^+| \leq \delta$ .

Repelling end. Let  $M = \log \tau_0$  and  $M' = \log \tau_\lambda$  be logarithms of the repelling Koenigs coordinates at  $L_2$ , in which  $E_0, E_\lambda$  are the translations by  $m_0 = \log \mu_0$ ,  $m_\lambda = \log \mu_\lambda$ ,  $\operatorname{Re} m > 0$ . For  $\lambda$  near  $\lambda_0$ ,  $h_\lambda$  maps a linearising disc of  $E_0$  into one of  $E_\lambda$ , and its lift  $G_\lambda$  to  $\{\operatorname{Re} M < -R\}$  satisfies  $G_\lambda(M - m_0) = G_\lambda(M) - m_\lambda$  and  $G_\lambda(M + 2\pi i) = G_\lambda(M) + 2\pi i$  (the first because  $h_\lambda$  conjugates the local inverse branches at  $L_2$ , with the integer fixed by continuity from  $\lambda_0$ ; the second by degree). Let  $A'_\lambda$  be the real-linear map with  $A'_\lambda m_0 = m_\lambda$ ,  $A'_\lambda(2\pi i) = 2\pi i$ . Then  $D_\lambda = G_\lambda - A'_\lambda$  is invariant under  $M \mapsto M - m_0$  and  $M \mapsto M \pm 2\pi i$ , which preserve  $\{\operatorname{Re} M < -R\}$ . Every point of that half-plane is carried by these maps into the compact set  $\{-R - \operatorname{Re} m_0 \leq \operatorname{Re} M \leq -R, 0 \leq \operatorname{Im} M \leq 2\pi\}$ , so  $\sup |D_\lambda|$  is its supremum there, which tends to 0 as  $\lambda \rightarrow \lambda_0$ . As at the other end,  $\chi = -M + \text{const} + \psi^-(e^M)$  with  $\psi^-$  holomorphic and vanishing at 0, for both parameters, and the same two-step choice of  $T$  and  $U_\delta$  gives the second estimate. The inverse is treated in the same way.  $\square$

**Lemma 9.21** (Comparison). *There is a neighbourhood  $U$  of  $\lambda_0$  such that for  $\lambda \in U$  the transported quotient  $X_\lambda^{\text{tr}}$  and the tilted quotient  $X_{1-\lambda}^\ell$  of Section 9 (same admissible line  $\ell$ ) are biholomorphic by a map that preserves the two ends and sends  $p_\lambda$  to the marked point. Consequently  $F_\lambda^{\text{tr}} = F_{1-\lambda}$  near  $z = 0$  for  $\lambda \in U$ .*

*Proof.* Let  $\ell$  be the admissible line fixed in the proof of Proposition 9.9 for the disc about  $\varepsilon_0$ ; it is the line used for  $H_0$ . Take  $U$  inside that disc, so that  $\ell$  is admissible,  $n_0$  is constant, and the marked point of  $X_{1-\lambda}^\ell$  is interior. Take also  $U \subset U_\delta$  of Lemma 9.20 with  $\delta = 1/2000$ , and  $T' \geq (T+1)/0.6$ , so that  $|t| \geq T'$  implies  $|\operatorname{Re} \chi| \geq T$  on the strips; for fixed  $\lambda_0$  this is a genuine neighbourhood, although its size may be small compared with  $\operatorname{Re} \varepsilon_0$ . Work in the  $\chi$ -plane of  $E_\lambda$ , where  $E_\lambda$  is  $f_\chi = \text{id} + \Delta$  and the tilted strip is  $\hat{H}_\lambda$  between  $\ell$  and  $f_\chi(\ell)$ , and put  $\hat{T}_\lambda = \tilde{h}_\lambda(\hat{H}_0)$ , the transported strip, bounded by  $\Gamma = \tilde{h}_\lambda(\ell)$  and  $f_\chi(\Gamma)$ . Let  $\varsigma, t$  be the coordinates of Section 9 relative to  $\ell$ , and  $W_\rho = \{|\varsigma| \leq \rho(1+|t|)\}$ .

*Step 1:*  $\hat{T}_\lambda \subset W_{1/100} \cap \mathcal{G}^c$  for  $\lambda \in U$ , after shrinking  $U$ .  $\hat{H}_0$  lies in  $\{|\varsigma| \leq \frac{6}{5}|\varepsilon_0|\}$ . On  $\{|t| \leq T'\}$  this is compact and  $\tilde{h}_\lambda \rightarrow \text{id}$  uniformly. For  $|t| \geq T'$  (so  $|\operatorname{Re} \chi| \geq T$ ), Lemma 9.20 gives  $|\tilde{h}_\lambda(\chi) - \chi| \leq \|A_\lambda^\pm - I\| |\chi| + |a_\lambda^\pm| + \delta$ , and  $|\chi| < |t| + 4 \leq 5(1+|t|)$  on  $\hat{H}_0$ ; with  $\delta = 1/2000$  the displacement is at most  $6\delta(1+|t|) = 0.003(1+|t|)$  (also on  $|t| \leq T'$ , after shrinking  $U$ ). The parameter  $t$  of the image point differs from that of the source by at most this amount. Since  $|\varepsilon| \leq 1.025|\varepsilon_0| \leq 0.005$  on  $U$  (using  $|\varepsilon_0| \leq 1/240$ ), every point of  $\hat{T}_\lambda$  has  $|\varsigma| \leq 0.005 + \frac{0.003}{0.997}(1+|t|) < 0.0082(1+|t|) \leq \frac{1}{100}(1+|t|)$ , and its distance to the holes is at least  $\max(1.48, 0.6|t| - 0.5) - 0.003(1+|t|) \geq \frac{1}{20}(1+|t|) + \frac{1}{2} + 0.2$ , so that the orbit discs of Step 2, of radius  $(1+|t|)/20$ , stay in  $\mathcal{G}^c$ . Symmetrically,  $\tilde{h}_\lambda^{-1}(\hat{H}_\lambda) \subset W_{1/100}$  in the  $\chi$ -plane of  $E_0$ .

*Step 2: orbit correspondence.* Let  $P \in \hat{T}_\lambda$  with parameter  $t$ . As in the proof of Lemma 9.8, each application of  $f_\chi^{\pm 1}$  changes  $\varsigma$  by at least  $\frac{2}{5}|\varepsilon|$  in the same direction and moves the point by at

most  $2|\varepsilon|$ , and  $|\varsigma(P)| \leq \frac{1}{100}(1+|t|)$ ; so the orbit of  $P$  reaches  $\hat{H}_\lambda$  after at most  $(1+|t|)/(40|\varepsilon|)$  steps, forward or backward, inside a disc about  $P$  contained in  $\mathcal{G}^c$ , on which  $f_\chi$  is injective. Exactly one iterate  $f_\chi^{n(P)}(P)$  lies in  $\hat{H}_\lambda \setminus f_\chi(\ell)$ , by the monotonicity of  $\varsigma$  and the isotopy argument of Lemma 9.5(2). (The edges of  $\hat{T}_\lambda$  are  $\Gamma$  and  $f_\chi(\Gamma)$  by the exact lifted conjugacy of Lemma 9.20.) The map  $P \mapsto f_\chi^{n(P)}(P)$  is locally a fixed iterate, jumps by one iterate exactly across  $\ell$ , where the two values are identified, and satisfies  $n(f_\chi(P)) = n(P) - 1$  for  $P \in \Gamma$ , matching the identification of the edges of  $\hat{T}_\lambda$ . It therefore induces a holomorphic map  $\Theta : X_\lambda^{\text{tr}} \rightarrow X_{1-\lambda}^\ell$ . For the inverse, apply the same construction in the  $\chi$ -plane of  $E_0$  to the strips  $\hat{H}_0$  and  $\tilde{h}_\lambda^{-1}(\hat{H}_\lambda) \subset W_{1/100} \cap \mathcal{G}^c$  (Step 1), and conjugate back by  $\tilde{h}_\lambda$ : this gives a map  $\Theta'$  from  $X_{1-\lambda}^\ell$  to  $X_\lambda^{\text{tr}}$ , locally a fixed iterate of  $E_\lambda$ . Both  $\Theta' \circ \Theta$  and  $\Theta \circ \Theta'$  send each point to a point of its own orbit in the same half-open strip, so they are the identity. Since orbit segments have length at most  $(1+|t|)/20$  in  $\chi$ ,  $\Theta$  preserves the ends ( $\text{Re } \chi \rightarrow \mp\infty$ ).

*Step 3: marked points.* By Proposition 9.18(2) and continuity of the lift,  $\tilde{h}_\lambda$  maps the continued base orbit  $\chi_n^*(\lambda_0)$  of Lemma 9.5 to  $\chi_n^*(\lambda)$ . So  $p_\lambda$  is represented by  $\chi_{n_0}^*(\lambda)$ , which lies in  $\hat{H}_\lambda \setminus f_\chi(\ell)$  because  $n_0$  is the first crossing for all  $\lambda \in U$ . Hence  $n(p_\lambda) = 0$  and  $\Theta(p_\lambda)$  is the marked point.

A biholomorphism of  $\mathbb{C}^*$  preserving both ends and a marked point is the identity in the normalised coordinates, so the two lifts  $z$  agree; both germs at  $z = 0$  are obtained by the local inverse of  $E_\lambda^{n_0}$  with value 1 at the marked point, so they agree.  $\square$

**Theorem 9.22** (Continuation to the upper Shell–Thron region). *Let  $\Omega = b(\mathcal{S}_{r_3}^c)$  be the corridor of Theorem 9.14. The continuation of Kneser’s solution from  $(\eta, \eta + r_5)$  into  $\Omega \cap \mathcal{U}^+$  extends analytically along every path in  $\mathcal{U}^+ = b(\mathbb{H}_\lambda)$ , the upper half of the Shell–Thron region, to a single-valued family of germs at  $z = 0$ , jointly holomorphic in  $(b, z)$ . For every  $b_0 \in (1, \eta)$  there is a disc  $D$  about  $z = 0$  such that along every path in  $\mathcal{U}^+$  ending at  $b_0$  these germs converge uniformly on  $D$  to  $K_{b_0}^W$ ; hence the boundary value is  $K_{b_0}^W$  on its regular domain, by the functional equation.*

*Proof.* By Proposition 9.19,  $\lambda \mapsto F_\lambda^{\text{tr}}$  is a single-valued holomorphic family of germs on  $\mathbb{H}_\lambda$ . By Lemma 9.21 it coincides with the cusp family  $F_{1-\lambda}$  near  $\lambda_0$ . The part of the parameter domain of Theorem 9.14 that lies in  $\mathbb{H}_\lambda$ ,  $\{1 - \varepsilon : \varepsilon \in \mathcal{S}_{r_3}^c, \text{Im } \varepsilon < 0, |1 - \varepsilon| < 1\}$ , is connected (it is the part of a small sector lying inside the disc  $|\varepsilon - 1| < 1$  and below the real axis) and contains  $\lambda_0$ , so the two families agree on it by the identity theorem. Hence  $F^{\text{tr}}$ , transferred to  $b$  by Lemma 9.17, is the analytic continuation of Kneser’s solution along every path in  $\mathcal{U}^+$  starting in  $\Omega \cap \mathcal{U}^+$ , and it is single-valued there.

For the boundary values, shrink the discs  $\Delta_{b_0}$  of Proposition 3.5 to radius at most  $(\eta - b_0)/2$ , as in the proof of Theorem 9.14(3). For  $b$  in such a disc,  $|\log b| \leq \log b_0 - \log(1 - x/2) < \log b_0 + \log(1 + x) = e^{-1}$  with  $x = (\eta - b_0)/b_0$ , and  $g(\{|\lambda| < 1\}) \supset \{|y| < e^{-1}\}$  because  $\min_{|\lambda|=1} |\lambda e^{-\lambda}| = e^{-1}$ ; so  $E_b$  has an attracting fixed point and  $N_W \cap \{\text{Im } b > 0\} \subset \mathcal{U}^+$  (Lemma 9.17). This set is connected and meets  $\Omega \cap \mathcal{U}^+$  near  $\eta$ , where  $K^W$  agrees with the cusp family (Theorem 9.14(3)); by the identity theorem  $K^W = F^{\text{tr}}$  on all of it. The  $z$ -domain of  $K^W$  in Proposition 3.5 contains a disc  $D$  about 0 for all  $b$  in a neighbourhood of  $b_0$ . Every path in  $\mathcal{U}^+$  ending at  $b_0$  eventually lies in  $\Delta_{b_0} \cap \{\text{Im } b > 0\}$ , where  $F^{\text{tr}} = K^W$  is jointly holomorphic on a neighbourhood of the closure of that half-disc times  $D$ ; so the germs converge uniformly on  $D$  to  $K_{b_0}^W$ .  $\square$

**Proposition 9.23** (Crossing the attracting real slice). *The family of Theorem 9.22 extends jointly holomorphically in  $(b, z)$  across every  $b \in (e^{-e}, 1)$ . The local extensions agree on overlaps and hence give a single-valued family of germs on a neighbourhood of this interval.*

*Proof.* Fix  $x \in (-1, 0)$  and choose a simply connected neighbourhood  $V$  of  $x$  in  $\{0 < |\lambda| < 1\}$  which meets  $\mathbb{H}_\lambda$ . Continue the branch  $s = \text{Log } \lambda$  from  $V \cap \mathbb{H}_\lambda$  to  $V$ . The proof of Proposition 9.18 uses only  $\text{Re } s = \log |\lambda| < 0$  to obtain  $|\mathbf{c}(\lambda)| < 1$ . Thus its Beltrami coefficient, normalised solution  $h_\lambda$ , and all four conclusions hold on  $V$  by exactly the same formulas. The marked quotient and its uniformisation in Proposition 9.19 consequently give a jointly holomorphic family of germs on  $V \times \{z : |z| < \delta\}$  for some  $\delta > 0$ . On  $V \cap \mathbb{H}_\lambda$  this is the original transported family, since both use the same normalised Beltrami solution and marked quotient. Any two such local extensions agree on an upper-half-plane overlap, hence agree wherever both are defined by the identity theorem. Finally  $b'(x) = b(x)(1-x)e^{-x} \neq 0$ , and  $b$  maps  $(-1, 0)$  increasingly onto  $(e^{-e}, 1)$ ; local inversion transfers these germs to the base coordinate.  $\square$

**Remark 9.24** (The punctured interior). The same argument works globally in  $s$ . Every object in Propositions 9.18 and 9.19 is given by one formula in  $s = \log \lambda$ , and only  $\text{Re } s < 0$  and  $|e^s| < 1$  are used. So  $s \mapsto F_s^{\text{tr}}$  is holomorphic and single-valued on the whole half-plane  $\{\text{Re } s < 0\}$ . This half-plane is the universal cover of the punctured interior  $\mathcal{ST}^\circ \setminus \{1\}$  via  $s \mapsto \exp g(e^s)$ . Hence the continuation of Kneser's family continues along every path in  $\mathcal{ST}^\circ \setminus \{1\}$ , and  $b = 1$  is its only possible singularity in the interior. A loop about  $b = 1$  acts by  $s \mapsto s + 2\pi i$ . Whether this monodromy is trivial is open. Numerically, the normalised modes  $|c_1|/\Lambda$  of the boundary values  $K_b^W$  stay near 0.08 as  $b \downarrow 1$  (0.0816 at  $b = 1.01$ ), while  $\Lambda \rightarrow 1$ . So near  $b = 1$  the separation from regular iteration is of order one.

**Remark 9.25** (Relation to Paulsen's family). Theorem 9.22 removes the restriction to paths near the real axis in Theorem 9.14: starting from  $\kappa_b$  on  $(\eta, \eta + r_5)$  and entering  $\mathcal{U}^+$  inside  $\Omega$ , every further path in  $\mathcal{U}^+$  gives  $K^W$  on all of  $(1, \eta)$ . The family of [18] may enter  $\mathcal{U}^+$  through a boundary arc far from  $\eta$ . It coincides with ours if its germ at one base in  $\mathcal{U}^+$  equals  $F^{\text{tr}}$  there. Theorems 9.33 and 9.34 show that Kneser's family crosses the entire open upper boundary arc holomorphically and enters with  $F^{\text{tr}}$ , and Theorem 9.40 shows that the whole continuation over the upper half-plane is single-valued, so this is Paulsen's family (Corollary 9.41). The nine-digit agreement at  $b = \sqrt{2}$  (Section 10.1) is consistent with this.

**9.10. Crossing the boundary away from the cusp.** Theorem 9.14 crosses the Shell–Thron boundary only near  $\eta$ . Here we cross it along arcs far from the cusp, up to the multiplier angle  $\alpha_1 = 0.9353 \dots \pi$ . The tool is Kneser's own initial region: the straight chord between the two fixed points and its image. We show that it stays a crescent, with the same marked point, on a collar that crosses the boundary arc. This needs no linearisation of the neutral fixed point, as in Section 9.

*Notation.* For complex  $\theta$  with  $0 < \text{Re } \theta < \pi$  put

$$\mu(\theta) = \theta \cot \theta, \quad \lambda_\pm = \mu \pm i\theta, \quad r = e^\mu, \quad a(\theta) = \frac{\theta e^{-\mu}}{\sin \theta}, \quad b(\theta) = e^{a(\theta)}.$$

We write  $E_\theta(w) = e^{a(\theta)w}$ ; this is  $E_{b(\theta)}$  with the logarithm of the base continued along paths in  $\theta$ , and it is the map that enters the analytic continuation of  $F(z+1) = b^{F(z)}$  in the base.  $L_\pm = re^{\pm i\theta}$  are fixed points of  $E_\theta$  with multipliers  $aL_\pm = \lambda_\pm$ , and  $a = \lambda_+ e^{-\lambda_+}$ . So if  $\lambda_+ \in \mathbb{H}_\lambda$ , then  $a = \text{Log } b$  is principal ( $|\text{Im } a| < \pi$ ) and  $b(\theta) = b(\lambda_+(\theta)) \in \mathcal{U}^+$  in the notation of Lemma 9.17. For real  $\theta \in (0, \pi)$ ,  $b(\theta) > \eta$  is real,  $L_\pm$  is the conjugate pair of Proposition 9.11 with  $\theta_b = \theta$ , and  $|\lambda_+| = \theta / \sin \theta > 1$ . The affine coordinate

$$\zeta(w) = \frac{w/r - \cos \theta}{i \sin \theta}$$

sends  $L_\pm$  to  $\pm 1$  and the chord  $\gamma_\theta = [L_-, L_+]$  to  $[-1, 1]$ , and a direct computation ( $aw = \mu + i\theta\zeta$ ) gives

$$\zeta(E_\theta(w)) = s_\theta(\zeta(w)), \quad s_\theta(\zeta) = \frac{e^{i\theta\zeta} - \cos \theta}{i \sin \theta}.$$

$s_\theta$  is entire,  $s_\theta(\pm 1) = \pm 1$  and  $s'_\theta(\pm 1) = \lambda_\pm$ , so  $g_\theta(t) = (s_\theta(t) - t)/(1 - t^2)$  is entire in  $t$ . Since  $s_\theta(\zeta_1) = s_\theta(\zeta_2)$  only if  $\theta(\zeta_1 - \zeta_2) \in 2\pi\mathbb{Z}$ ,  $s_\theta$  is injective on every set of diameter  $< 2\pi/|\theta|$ .

*The parameter region.* Let  $P_0 = \frac{1}{10}$ ,  $P_1 = \frac{11}{5}$ , let  $V(p) = \frac{1}{4}$  for  $p \leq \frac{8}{5}$  and  $V(p) = \frac{1}{4} - \frac{7}{60}(p - \frac{8}{5})$  for  $p \geq \frac{8}{5}$ , and

$$\Theta_C = \{p(1 + ic) : 0 < p \leq P_0, 0 \leq c \leq \frac{1}{10}\}, \quad \Theta_K = \{p + ivp^2 : P_0 \leq p \leq P_1, 0 \leq v \leq V(p)\},$$

$$\Theta = \Theta_C \cup \Theta_K. \text{ Its vertical fibre over } p \text{ is a segment } \{p + iq : 0 \leq q \leq Q(p)\}.$$

**Lemma 9.26** (Certified inequalities). *For  $\theta = p + iq \in \Theta$  and  $-1 \leq t \leq 1$ :*

- (1)  $\text{Im } g_\theta(t) < 0$  and  $\text{Im}(\overline{s'_\theta(t)} g_\theta(t)) < 0$ ;
- (2)  $\partial_q |\lambda_+(p + iq)|^2 < 0$ ,  $|\lambda_+| < 1$  on the top edge  $q = Q(p)$ , and  $a'(\theta) \neq 0$ ;
- (3) on  $\Theta_C$ ,  $\text{Re } \cot \theta > 0$ ;
- (4) for  $\text{Re } \theta \leq P_2 = \frac{19}{10}$ ,  $|\text{Im } a(\theta)| < \pi$ .

*Proof.* See Appendix A.1. □

By (1) at  $t = 1$ ,  $\text{Im } \lambda_+ = -2 \text{Im } g_\theta(1) > 0$ . By (2), on each fibre of  $\Theta$  the multiplier  $\lambda_+$  of the continued fixed point  $L_+$  crosses the unit circle exactly once, at  $q = q_c(p)$ , and

$$\Theta^+ = \{\theta \in \Theta : |\lambda_+(\theta)| < 1\} = \{p + iq \in \Theta : q > q_c(p)\}$$

is connected, with  $a$  principal and  $b(\Theta^+) \subset \mathcal{U}^+$ . (Below  $q_c$  the logarithm  $a$  need not be principal: for  $p$  near  $P_1$ ,  $\text{Im } a$  exceeds  $\pi$  on the fibre. By (4) this does not happen for  $p \leq P_2$ .) The boundary point  $\beta(p) = b(p + iq_c(p))$  has  $\lambda_+ = e^{i\alpha(p)}$  with  $\alpha$  continuous,  $\alpha(p) \rightarrow 0$  as  $p \rightarrow 0$  and  $\alpha(P_1) = \alpha_1 = 0.9353 \dots \pi$ ; so  $\beta$  runs through the whole upper boundary arc  $\{\exp(e^{i\alpha} e^{-e^{i\alpha}}) : 0 < \alpha \leq \alpha_1\}$ .

**Lemma 9.27** (The crescent). *Let  $\theta$  lie in a sufficiently small open neighbourhood  $\Theta'$  of  $\Theta \setminus \{0\}$ .*

- (1)  $A_\theta = s_\theta([-1, 1])$  is a simple arc from  $-1$  to  $1$ , in the open lower half-plane except for its ends. Let  $H_\theta$  be the closed Jordan region bounded by  $[-1, 1] \cup A_\theta$ .
- (2)  $N_\theta(t, \sigma) = t + \sigma(s_\theta(t) - t)$  is a real-analytic diffeomorphism of  $(-1, 1) \times [0, 1]$  onto  $H_\theta \setminus \{\pm 1\}$ , with  $N(t, 0) = t$  and  $N(t, 1) = s_\theta(t)$ .
- (3)  $s_\theta$  maps a neighbourhood of  $[-1, 1]$  biholomorphically onto a neighbourhood of  $A_\theta$ , and the upper side of  $[-1, 1]$  to the side of  $A_\theta$  facing  $H_\theta$ .
- (4) Let  $X_\theta$  be  $H_\theta \setminus \{\pm 1\}$  with  $t \in (-1, 1)$  glued to  $s_\theta(t)$ . In the coordinates  $x = \log \frac{1-t}{1+t}$  and  $\chi = \log \frac{\zeta-1}{\zeta+1}$  the map  $\Phi_\theta(x + i\sigma) = \chi(N_\theta(t, \sigma))$  induces a quasiconformal homeomorphism from  $C_0 = \mathbb{C}/i\mathbb{Z}$  onto  $X_\theta$ . Its Beltrami coefficient is holomorphic in  $\theta$  at each fixed  $x + i\sigma$  and bounded by some  $k < 1$  locally uniformly in  $\theta$ . Hence  $X_\theta \cong \mathbb{C}^*$ , with the ends at  $\pm 1$  as punctures.

*Proof.* All conditions below are strict inequalities that hold on  $\Theta$  by Lemma 9.26, uniformly on compact parts, so they persist on a neighbourhood.  $|\theta| < 2.45$  there (also on the rectangle used in Theorem 9.34), so  $s_\theta$  is injective on every set of diameter  $< 2\pi/2.45$ , in particular on the disc of radius 1.28.

- (1)  $\text{Im } s_\theta(t) = (1 - t^2) \text{Im } g_\theta(t) < 0$  for  $|t| < 1$ , and  $s_\theta$  is injective on  $[-1, 1]$ .
- (2)  $\partial_t N = (1 - \sigma) + \sigma s'$  and  $\partial_\sigma N = (1 - t^2)g$ , so the Jacobian is

$$\text{Im}(\overline{\partial_t N} \partial_\sigma N) = (1 - t^2)[(1 - \sigma) \text{Im } g + \sigma \text{Im}(\overline{s'g})] < 0$$

for  $|t| < 1$ .  $N$  is continuous on the square  $[-1, 1] \times [0, 1]$  and maps its boundary once around the Jordan curve  $\partial H_\theta$ , the vertical sides to the points  $\pm 1$ . For  $\zeta_0 \notin \partial H_\theta$  all preimages are interior and count with the same sign, so their number is the winding number of  $\partial H_\theta$  about  $\zeta_0$ , namely 1 in  $H_\theta$  and 0 outside. An interior point cannot map to  $\partial H_\theta$ , since its image would have nearby points outside  $H_\theta$  with preimages.

(3)  $s'_\theta \neq 0$ , and  $s_\theta$  is injective near  $[-1, 1]$ . At  $t \in (-1, 1)$  the image of the upward direction is  $is'(t)$ , and the direction from  $A_\theta$  into  $H_\theta$  is  $-\partial_\sigma N(t, 1) = -(1 - t^2)g(t)$ . Both lie on the same side of the tangent  $s'(t)$ , because  $\text{Im}(\overline{s'}is') = |s'|^2 > 0$  and  $\text{Im}(\overline{s'}(-g)) > 0$ .

(4) Near  $t = 1$  write  $N - 1 = (t - 1)m$  with  $m = 1 - \sigma(1 + t)g(t)$ . For  $t < 1$ ,  $m \neq 0$  by (2); at  $t = 1$ ,  $m = 1 + \sigma(\lambda_+ - 1)$ , whose imaginary part is  $\sigma \text{Im } \lambda_+ > 0$  for  $\sigma > 0$ . Hence

$$\Phi_\theta = x + i\pi + R_\theta(t, \sigma), \quad R_\theta = \log m + \log(1 + t) - \log(N + 1),$$

with  $R$  real-analytic in  $(t, \sigma, \theta)$  up to  $t = 1$ ; similarly at  $t = -1$ . Since  $dt/dx = -(1 - t^2)/2$ , one has  $\partial_x \Phi = 1 - \frac{1}{2}(1 - t^2)\partial_t R \rightarrow 1$  and  $\partial_\sigma \Phi \rightarrow (\lambda_+ - 1)/(1 + \sigma(\lambda_+ - 1))$  as  $t \rightarrow 1$ , whose imaginary part is positive. So  $D\Phi_\theta$  is continuous and nonsingular on  $[-\infty, \infty] \times [0, 1]$ , with constant orientation. Its Beltrami coefficient  $(\partial_x \Phi + i\partial_\sigma \Phi)/(\partial_x \Phi - i\partial_\sigma \Phi)$  is continuous there and in  $\theta$ , of modulus  $< 1$ , and holomorphic in  $\theta$  because  $\Phi_\theta(x + i\sigma)$  is.  $\Phi(x) = \chi(t)$  and  $\Phi(x + i) = \chi(s_\theta(t))$ , so  $\Phi$  respects the two identifications. The rest is as in Lemma 9.4(1).  $\square$

*The marked point.* Let  $w_{-1} = 0$  and  $w_n = E_\theta^n(1)$ ,  $n \geq 0$ , and write  $\zeta_n = \zeta(w_n)$ . We say  $w$  lies *above the chord* if  $\text{Im } \zeta(w) > 0$ ; such points are not in  $H_\theta$ . If  $w_{-1}, \dots, w_{n-1}$  lie above the chord and  $\zeta_n \in H_\theta \setminus A_\theta$ , we call  $n = n_\theta$  the *entry index* and the class  $\mathfrak{m}_\theta \in X_\theta$  of  $\zeta_n$  the *marked point*. It is defined intrinsically.

**Lemma 9.28** (First entry on the cone). *For  $\theta \in \Theta_C$  the entry index exists.*

*Proof.* An analytic estimate of the orbit in the coordinate  $\log \frac{\zeta - 1}{\zeta + 1}$ ; see Appendix A.1.  $\square$

**Lemma 9.29** (The marked point on  $\Theta$ ). *The entry index exists for every  $\theta \in \Theta$ , and  $\theta \mapsto \mathfrak{m}_\theta$  is a holomorphic section of the family  $X_\theta$  near  $\Theta$ : near each  $\theta_0$  it is represented by a fixed iterate  $\zeta_n(\theta)$ , either in the interior of  $H_\theta$  or in the seam chart given by  $s_\theta$ .*

*Proof.* On  $\Theta_C$  this is Lemma 9.28. On  $\Theta_K$  the orbit is enclosed on parameter boxes and the entry is certified box by box (Appendix A.1).

Near  $\theta_0$ : if  $\zeta_n(\theta_0)$  is in the interior of  $H$ ,  $n_\theta = n$  nearby. If  $\zeta_n(\theta_0) \in (-1, 1)$ , then  $n_\theta \in \{n, n+1\}$  nearby, according to the side of the chord, and  $\zeta_n$  and  $\zeta_{n+1} = s_\theta(\zeta_n)$  are the same point of  $X_\theta$  in the seam chart. The conditions  $\text{Im } \zeta_m > 0$ ,  $m < n$ , are open. So  $\mathfrak{m}_\theta$  is locally the class of the single holomorphic function  $\zeta_n(\theta)$ .  $\square$

**Definition 9.30.** For  $\theta$  near  $\Theta$  let  $Z_\theta : X_\theta \rightarrow \mathbb{C}^*$  be the uniformisation taking the  $L_+$ -end to 0, the  $L_-$ -end to  $\infty$  and  $\mathfrak{m}_\theta$  to 1, and let  $z = (2\pi i)^{-1} \log Z_\theta$  be lifted so that  $\mathfrak{m}_\theta$  has  $z = n_\theta$ . With  $\tilde{F}_\theta$  the inverse, extended by  $\tilde{F}(z + 1) = E_\theta(\tilde{F}(z))$  as in Lemma 9.4(1), put

$$F_\theta^{\text{far}}(z) = G_\theta(\tilde{F}_\theta(z + n_\theta)),$$

where  $G_\theta$  is the branch of  $E_\theta^{-n_\theta}$  along the orbit  $w_{n_\theta} \mapsto \dots \mapsto w_0 = 1$  (for  $n_\theta = -1$ ,  $G_\theta = E_\theta$ ). Then  $F_\theta^{\text{far}}(0) = 1$ , and  $F(z + 1) = e^{a(\theta)F(z)}$  holds as an identity of germs, with the corresponding branches along  $w_{n_\theta+1} \mapsto \dots \mapsto w_1$ .

The sign of the lift, i.e. that crossing the seam from the chord to  $A_\theta$  adds  $+1$ , is locally constant in  $\theta$ . At real  $\theta$  it is Kneser's, by Proposition 9.32 below.

**Proposition 9.31** (Holomorphic dependence).  *$(\theta, z) \mapsto F_\theta^{\text{far}}(z)$  is jointly holomorphic on a neighbourhood of  $(\Theta \setminus \{0\}) \times \{0\}$ .*

*Proof.* This is the proof of Proposition 9.9, with  $M_\varepsilon$  replaced by  $N_\theta$  in the coordinates of Lemma 9.27(4). Fix  $\theta_0$ . Then  $\mathcal{J}_\theta = \Phi_\theta \circ \Phi_{\theta_0}^{-1}$  is quasiconformal, respects the seams, and is holomorphic in  $\theta$  at fixed source points. Its Beltrami coefficient  $v_\theta$  is holomorphic in  $\theta$  and, by the uniform continuity in Lemma 9.27(4), is bounded by  $\frac{1}{10}$  for  $\theta$  near  $\theta_0$ . Ahlfors–Bers and the  $\partial_{\bar{\theta}}$ -cancellation give a uniformisation  $Z_\theta$  that is jointly holomorphic in  $(\theta, \zeta)$  in the interior of  $H_\theta$ . Near a point  $\zeta_0$  of the chord, which does not move with  $\theta$  in these coordinates,  $Z_\theta$  is given by  $Z_\theta$  below the chord and by  $Z_\theta \circ s_\theta$  above it. Both are continuous up to the chord, and they agree there. So  $\theta \mapsto Z_\theta(\zeta)$  is a locally uniform limit of holomorphic functions for each  $\zeta$ , and  $Z_\theta(\zeta)$  is holomorphic in  $\zeta$  by Morera's theorem. Hartogs' theorem then gives joint holomorphy near the chord. The marked point is holomorphic by Lemma 9.29. The inverse branches  $G_\theta$  are finitely many logarithms along an orbit that avoids 0, holomorphic in  $\theta$ .  $\square$

**Proposition 9.32** (The real slice). *For real  $\theta \in (0, P_1]$ ,  $F_\theta^{\text{far}}$  is Kneser's real-normalised solution  $\kappa_{b(\theta)}$  near  $z = 0$ .*

*Proof.* For real  $\theta$ ,  $\zeta^{-1}([-1, 1])$  is the chord  $\gamma_b$  and  $\zeta^{-1}(A_\theta) = E_b(\gamma_b)$ , so  $H_\theta$  is Kneser's initial region  $H_b$  and  $X_\theta$  is the cylinder of Proposition 9.11, with the upper end at  $L_+$  sent to 0 in both. For real  $w$ ,  $\text{Im } \zeta(w) = (\cos \theta - w/r)/\sin \theta$ , which is positive exactly when  $w < u_b = r \cos \theta$ , and  $H_b \cap \mathbb{R} = [u_b, r_b)$  with  $r_b = E_b(u_b)$ . The real orbit  $0 = w_{-1} < 1 = w_0 < w_1 < \dots$  increases to  $+\infty$ . Let  $n$  be the first index with  $w_n \geq u_b$ . If  $n \geq 0$  then  $w_{n-1} < u_b$ , so  $w_n < E_b(u_b) = r_b$ . If  $n = -1$  then  $u_b \leq 0 < r_b$ . Either way  $w_n \in [u_b, r_b)$ , so  $n = n_\theta$ , and  $\mathfrak{m}_\theta$  is the point that Proposition 9.11 represents by  $w = 1$  through the real dynamical charts. Its Kneser coordinate is  $n_\theta$ . The normalisations of the lifts agree, and the branches  $G_\theta$  are the real logarithms.  $\square$

**Theorem 9.33** (Crossing the upper boundary away from the cusp). *Let  $p \in (0, P_1]$ , and continue Kneser's real-normalised solution  $\kappa_b$  at  $b = b(p) > \eta$  analytically in the base along  $q \mapsto b(p + iq)$ ,  $0 \leq q \leq Q(p)$ , with  $\log b = a(p + iq)$  continued along the path.*

- (1) *The continuation exists along the whole path and is holomorphic in a neighbourhood of it. The multiplier  $\lambda_+$  of the continued fixed point crosses the unit circle once, at  $q = q_c(p)$ , where  $b = \beta(p)$  lies on the upper Shell–Thron boundary arc; near  $\beta(p)$  the continuation is a holomorphic family on a full neighbourhood of  $\beta(p)$ .*
- (2) *For  $q > q_c(p)$  the base lies in  $\mathcal{U}^+$ ,  $a = \text{Log } b$ , and the continuation equals  $F_{b(p+iq)}^{\text{tr}}$ , the family of Theorem 9.22.*

*In particular, along these paths Kneser's family crosses every point of the upper boundary arc  $\{\exp(e^{i\alpha}e^{-e^{i\alpha}}) : 0 < \alpha \leq \alpha_1\}$ ,  $\alpha_1 = 0.9353\dots\pi$ , holomorphically, and it enters the upper Shell–Thron region with the germ  $F^{\text{tr}}$ . Its boundary values on  $(1, \eta)$  are therefore  $K_b^W$ , by Theorem 9.22. For  $p \leq P_2$  ( $\alpha \leq 0.7531\dots\pi$ ) the whole path lies on the principal sheet,  $|\text{Im } \log b| < \pi$ .*

*Proof.* By Proposition 9.31 and Lemma 9.26(2),  $b$  is locally biholomorphic on a neighbourhood  $\Theta'$  of  $\Theta \setminus \{0\}$  (as is  $a$ ), and  $F^{\text{far}}$  is a holomorphic family of germs there. By Proposition 9.32 it equals  $\kappa$  on the real segment. So along every path in  $\Theta'$  starting on that segment it is the analytic continuation of  $\kappa$  in the base, with the logarithm  $a$  continued along the path.

Next,  $a(\theta) = e^{-1}(1 + \theta^2/2 + O(\theta^4))$ , so  $b'(\theta) = b a'(\theta)$  has positive real part for small  $p$  on  $\Theta_C$ , and there  $a$  is principal. Hence  $\text{Im } b(p + iq) = \int_0^q \text{Re } b'(p + iq') dq' > 0$  for  $q > 0$ , and for  $p$  small the vertical fibre is carried into  $D^+ \cup (\eta, \eta + r_5) \subset \Omega$ . By Theorem 9.14(1) the continuation of  $\kappa$  along it agrees with the family on  $\Omega$ . By Theorem 9.22 that family equals  $F^{\text{tr}}$  on  $\Omega \cap \mathcal{U}^+$ . So  $F^{\text{far}} = F^{\text{tr}} \circ b$  on a nonempty open subset of the connected set  $\Theta^+$  (interior), and hence on all of it, by the identity theorem. The remaining assertions are the description of  $\Theta^+$  and  $\beta$  after Lemma 9.26.  $\square$

**Theorem 9.34** (Completion of the open upper boundary arc). *For every  $0.935\pi \leq \alpha < \pi$ , put*

$$\beta(\alpha) = \exp(e^{i\alpha}e^{-e^{i\alpha}}).$$

*Kneser's real-normalised family admits analytic continuation, on the principal logarithm sheet, from a real base  $b > \eta$  to a full neighbourhood of  $\beta(\alpha)$ . This continuation crosses the upper Shell–Thron boundary holomorphically and agrees with  $F^{\text{tr}}$  on its inner side. Consequently the same assertion holds at every point of the open upper boundary arc  $0 < \alpha < \pi$ . The endpoint  $\beta(\pi) = e^{-e}$  is excluded.*

**Corollary 9.35** (Entering from outside). *There is an open neighbourhood  $N$  of the open upper boundary arc  $\{\beta(\alpha) : 0 < \alpha < \pi\}$  and a single-valued holomorphic family of germs on  $N \cup \mathcal{U}^+$  that equals  $F^{\text{tr}}$  on  $\mathcal{U}^+$ . On  $N \setminus \overline{\mathcal{U}^+}$  it is the continuation of Kneser's solution along paths that stay outside the closed Shell–Thron region on the principal sheet. Thus for every  $0 < \alpha < \pi$  Kneser's family, continued from  $(\eta, \infty)$  through the exterior, reaches  $\beta(\alpha)$  and enters  $\mathcal{U}^+$  with the germ  $F^{\text{tr}}$ .*

*Proof.* Theorems 9.33 and 9.34 give, about each point of the open arc, a disc on which the continuation is holomorphic and equals  $F^{\text{tr}}$  on the inner side. Two such discs meet in a connected set containing inner points, so the local families agree on overlaps, and  $F^{\text{tr}}$  is single-valued on  $\mathcal{U}^+$ . Take  $N$  to be the union of the discs, shrunk so that  $N \setminus \overline{\mathcal{U}^+}$  is a connected band along the arc. For  $p$  small the vertical path of Theorem 9.33 runs from the real exterior ray to this band outside  $\mathcal{U}^+$ , on the principal sheet. Moving along the band then reaches the outer side of every  $\beta(\alpha)$ , and  $|\text{Im } a| < \pi$  near the arc because  $|a| \leq e$  there.  $\square$

*Proof.* By Appendix A.2,  $\lambda_+$  is injective on  $R = [2.19, 2.31] + i[0.71, 0.78]$ , the branch  $\theta(\alpha) \in R$  of  $\lambda_+(\theta) = e^{i\alpha}$ ,  $0.935\pi \leq \alpha < \pi$ , satisfies the inequalities of Lemma 9.26(1) on  $(-1, 1)$  on a complex neighbourhood, and  $w_{-1} = 0$  lies in the crescent, so the entry index is  $-1$ . Lemma 9.27 and Proposition 9.31 apply on a complex parameter neighbourhood of every  $\alpha < \pi$ . Also  $\lambda'_+ \neq 0$  and  $a' = e^{-\lambda_+}(1 - \lambda_+)\lambda'_+ \neq 0$ , so  $b = e^a$  is a valid local parameter. These neighbourhoods carry one holomorphic family of marked crescent quotients: the inverse branch is unique on  $R$ , and their constructions agree on overlaps.

At  $\alpha = 0.935\pi < \alpha_1$  this is exactly the crescent and the first mark of Theorem 9.33. The families therefore agree in an open parameter overlap. Analytic continuation along the connected arc, followed by the identity theorem on the inner overlaps, identifies the new family with  $F^{\text{tr}}$  wherever it meets  $\mathcal{U}^+$ .

Finally choose  $p \leq P_2$  in Theorem 9.33, follow its vertical path from the real exterior ray into  $\mathcal{U}^+$ , then follow a path in  $\mathcal{U}^+$  to the inner side of  $\beta(\alpha)$  and use the local crescent family to cross the boundary. The first segment is on the principal sheet by Lemma 9.26(4). On  $\mathcal{U}^+$  the logarithm is  $a = \lambda e^{-\lambda} = \text{Log } b$ ; at the target boundary  $|a| \leq e < \pi$ , so the last short segment is principal as well. This gives the asserted principal-sheet path and completes the proof.  $\square$

**Remark 9.36.** (1) The boundary point  $b_c \approx 1.5217 + 0.0148i$  ( $\alpha = 2\pi(\sqrt{2} - 1)/5$ ) tested numerically in Section 10.5 lies on the arc covered here. The smooth join observed there agrees with the holomorphic crossing proved here. (2) The endpoint  $b = e^{-e}$  itself is treated in Section 9.12. There the crescent keeps its angle  $\arg \lambda_+ \rightarrow \pi$  at  $L_+$ ; only the interpolation used here degenerates. (3) The identification with the family of [18] is completed in Section 9.11 (Corollary 9.41).

**9.11. The whole upper half-plane.** Theorems 9.22, 9.33 and 9.34 describe the continuation inside the upper Shell–Thron region and across its boundary arc. Here we show that Kneser's straight chord also carries the family over the whole exterior part of the upper half-plane. So the continuation is a single holomorphic family on  $\mathbb{H} = \{\text{Im } b > 0\}$ , and there is no path dependence.

Write  $\overline{\mathcal{ST}}$  for the closed Shell–Thron region,  $g(\lambda) = \lambda e^{-\lambda}$ , and

$$\mathcal{E}^+ = \mathbb{H} \setminus \overline{\mathcal{ST}}, \quad \mathbb{H} = \mathcal{E}^+ \cup \{\beta(\alpha) : 0 < \alpha < \pi\} \cup \mathcal{U}^+.$$

On  $\mathbb{H}$  we use the principal logarithm  $a = \text{Log } b$ ,  $0 < \text{Im } a < \pi$ .

**Lemma 9.37** (The exterior parameter).  *$\mathcal{E}^+$  is simply connected. The fixed point  $L_+$  of  $E_b$  continued from the real exterior ray (the one with  $\text{Im } L_+ > 0$  there) is single-valued and holomorphic on  $\mathcal{E}^+$ . Its multiplier  $\lambda(b)$  maps  $\mathcal{E}^+$  biholomorphically onto a domain  $\Lambda$ , with inverse  $\lambda \mapsto \exp g(\lambda)$ , and*

$$\Lambda \subset \{\lambda : |\lambda| > 1, 0 < \text{Im } \lambda < \pi, 0 < \text{Im } g(\lambda) < \pi, g(\lambda) \notin g(\overline{\mathbb{D}})\}.$$

*Proof.*  $K = \overline{\mathcal{ST}} \cap \overline{\mathbb{H}}$  is a closed topological disc: it is bounded by the Jordan arc  $\beta$  and the segment  $[e^{-e}, \eta]$ . The complement of  $\mathcal{E}^+$  in the Riemann sphere,  $(\hat{\mathbb{C}} \setminus \mathbb{H}) \cup K$ , is connected, so  $\mathcal{E}^+$  is simply connected. A fixed point of  $E_b$  is multiple only if its multiplier is 1, i.e.  $a = 1/e$ , which does not occur for  $0 < \text{Im } a < \pi$ . So  $L_+$  continues without monodromy.  $L_+ = e^\lambda$  and  $a = g(\lambda)$  (Lemma 2.1), so  $b = \exp g(\lambda)$  and  $\lambda$  is injective. All fixed points of  $E_b$ ,  $b \in \mathcal{E}^+$ , are repelling, so  $|\lambda| > 1$  and  $g(\lambda) = a \notin g(\overline{\mathbb{D}})$ . If  $\text{Im } \lambda = 0$  then  $a$  is real; if  $\text{Im } \lambda = \pi$  then  $\text{Im } a = -\pi e^{-\text{Re } \lambda} < 0$ . Neither happens on  $\mathcal{E}^+$ , and near the real ray  $\lambda = \theta \cot \theta + i\theta$  with small  $\theta > 0$ ; by connectedness  $0 < \text{Im } \lambda < \pi$ .  $\square$

For  $\lambda \in \Lambda$  let  $\theta = \theta(\lambda) \neq 0$  be a root of  $(\lambda - 2i\theta)e^{2i\theta} = \lambda$ . (The root  $\theta = 0$  is always present and is excluded.) For  $\theta \neq 0$  the equation is equivalent to  $\lambda_+(\theta) = \lambda$ . Then  $\lambda - 2i\theta$  is the multiplier of a second fixed point  $L_- = e^{\lambda - 2i\theta}$ , so the chord construction of Section 9.10 is defined with this  $\theta$ .  $g$  has no critical point on  $\Lambda$ , so  $\lambda$  is a local holomorphic parameter.

**Lemma 9.38** (Scope of Section 9.10). *Let  $\Theta''$  be an open set of  $\theta$  with  $0 < |\theta| < \pi$  on which the inequalities of Lemma 9.26(1) hold for  $-1 < t < 1$ ,  $\text{Im } \lambda_+ > 0 > \text{Im } \lambda_-$ , and the entry index of the marked point exists. Then Lemma 9.27, the local statement of Lemma 9.29, Definition 9.30 and Proposition 9.31 hold on  $\Theta''$ .*

*Proof.* Those proofs use only these properties and the injectivity of  $s_\theta$  on a neighbourhood of  $[-1, 1]$ . The injectivity follows from  $|\theta| < \pi$ :  $s_\theta(\zeta_1) = s_\theta(\zeta_2)$  forces  $\theta(\zeta_1 - \zeta_2) \in 2\pi\mathbb{Z}$ , and  $|2\pi/\theta| > 2$ . The bound  $|\theta| < 2.45$  in Lemma 9.27 was used only for this. At the ends,  $\text{Im } \lambda_\pm$  play the roles of  $\text{Im } g(\pm 1)$  in part (4).  $\square$

**Proposition 9.39** (Certified covering of  $\Lambda$ ). *There is a continuous choice of  $\theta(\lambda)$  on  $\Lambda$ , real on the image of the real exterior ray, such that for every  $\lambda \in \Lambda$  the inequalities of Lemma 9.26(1) hold on  $(-1, 1)$ ,  $\text{Im}(\lambda - 2i\theta) < 0$ ,  $|\theta| < \pi$ , and the entry index of the marked point exists. The certificate `certify_upper_halfplane.py` has four parts (tails, core, cusp, interfaces), described in Appendix A.3.*

*Proof.* The listed checks are outward-rounded Arb computations. Every point of  $\Lambda$  lies in a certified box, because of the inclusion in Lemma 9.37. The roots are simple, since  $\partial_\theta$  of the equation is  $2ie^{2i\theta}(\lambda - 2i\theta - 1)$  and  $\lambda - 2i\theta \neq 1$  ( $L_-$  is repelling). So the certified local roots, which agree on all overlaps, define one continuous branch.  $\square$

**Theorem 9.40** (Kneser's tetration on the upper half-plane). *Kneser's real-normalised solution  $\kappa_b$ ,  $b > \eta$ , continues analytically along every path in  $\mathbb{H} = \{\text{Im } b > 0\}$ , with the principal logarithm. The result is a single-valued family of germs at  $z = 0$ , jointly holomorphic in  $(b, z)$ . It is given on  $\mathcal{E}^+$  by the straight-chord construction with the marked point of Section 9.10, on  $\mathcal{U}^+$  by  $F^{\text{tr}}$ , and it is holomorphic across the open upper boundary arc. In particular, its boundary values on  $(1, \eta)$  are  $K_b^W$ , whatever path in  $\mathbb{H}$  is used. The continuation through the lower half-plane is the complex conjugate family.*

*Proof.* By Proposition 9.39 and Lemma 9.38 (with  $\lambda$  as parameter), the chord construction with  $\theta(\lambda)$  and the first-entry mark gives a holomorphic family on  $\Lambda$ , hence on  $\mathcal{E}^+$ . Near the image of the real ray it is Kneser's solution, by Proposition 9.32, since there  $\theta$  is real. Since  $\mathcal{E}^+$  is connected, the family is the analytic continuation of  $\kappa$  along every path in  $\mathcal{E}^+$ . Here the image  $\Gamma$  of the real ray lies in  $\partial\Lambda$ . The certified cells straddle it,  $\theta(\lambda)$  extends holomorphically across  $\Gamma$ , it is real there by the anchor in (I), and Proposition 9.32 applies. Shrink the neighbourhood  $N$  of Corollary 9.35 into  $\mathbb{H}$ , so that  $N \cap \mathcal{E}^+$  is the connected outer band. By that corollary, near the open arc this continuation extends holomorphically across the arc and equals  $F^{\text{tr}}$  on the inner side;  $F^{\text{tr}}$  is single-valued on  $\mathcal{U}^+$  by Theorem 9.22. The three pieces agree on their overlaps, so they define one holomorphic family on  $\mathbb{H}$ . Since  $\mathbb{H}$  is simply connected, analytic continuation of  $\kappa$  along any path in  $\mathbb{H}$  gives it. The boundary values on  $(1, \eta)$  are those of Theorem 9.22. For the lower half-plane use  $\overline{\kappa_b(\bar{z})} = \kappa_b(z)$ .  $\square$

**Corollary 9.41** (Paulsen's family). *Suppose the family of [18] is understood, as in its definition, as the analytic continuation in  $b$  of Kneser's solution from  $(\eta, \infty)$ , along paths in the upper half-plane and with the principal logarithm. Then it is the family of Theorem 9.40. In particular it enters the Shell-Thron region with the germ  $F^{\text{tr}}$  across every point of the open boundary arc. Its boundary values on  $(1, \eta)$  are  $K_b^W$ , and the separation limits of Corollary 9.15 hold for it.*

**9.12. The endpoint  $b = e^{-e}$  and the real segment below 1.** At  $b = e^{-e}$  the fixed point  $L_+$  has multiplier  $\lambda_+ = -1$ . In Theorem 9.34 the straight crescent seemed to degenerate there, because  $f_1(1) = -\frac{1}{2} \text{Im } \lambda_+ \rightarrow 0$ . The crescent itself does not degenerate: its angle at  $L_+$  is  $\arg \lambda_+ \rightarrow \pi$ . What degenerates is the interpolation  $t + \sigma(s_\theta(t) - t)$ , whose factor  $1 + \sigma(\lambda_+ - 1)$  vanishes at  $\sigma = \frac{1}{2}$ . In the coordinate of the pair of fixed points the end is harmless, and the crescent can be continued through  $\lambda_+ = -1$  and across the real axis.

Put  $\chi = \log \frac{\zeta-1}{\zeta+1}$ , so that the chord is  $\{\text{Im } \chi = \pi\}$ , with  $x = \log \frac{1-t}{1+t}$ . The image arc is  $\Gamma(x) = x + i\pi + \varphi(x)$ , where, with the entire function  $S(w) = \sinh w/w$ ,

$$\varphi(t) = i\theta + \log S\left(\frac{i\theta(t-1)}{2}\right) - \log S\left(\frac{i\theta(t+1)}{2}\right),$$

with principal logarithms where  $\text{Re } S > 0$ . Then  $e^{\varphi(1)} = \lambda_+$  and  $e^{-\varphi(-1)} = \lambda_-$ , and under (b) below  $\varphi(\pm 1)$  are the logarithms with imaginary part in  $(0, 2\pi)$ . This follows from  $s_\theta(t) \mp 1 = e^{\pm i\theta}(e^{i\theta(t \mp 1)} - 1)/(i \sin \theta)$ . We write  $\varphi_x = d\varphi/dx = -\frac{1}{2}(1-t^2)d\varphi/dt$ , which is regular at  $t = \pm 1$ . The mark has the closed form  $\chi(\zeta(0)) = \log \frac{i \cot \theta - 1}{i \cot \theta + 1} = 2i\theta \pmod{2\pi i}$ .

**Lemma 9.42** (The crescent in the pair coordinate). *Let  $\Theta''$  be an open set of  $\theta$  on which, for  $-1 \leq t \leq 1$ :*

- (f)  $|\theta| < \pi$ , and  $\text{Re } S > 0$  at both arguments above;
- (a)  $\text{Re}(1 + \varphi_x) > 0$ ;
- (b)  $0 < \text{Im } \varphi < 2\pi$ ;
- (c)  $\text{Im}(\overline{(1 + \varphi_x)} \varphi) > 0$ ;
- (d)  $\text{Im } \varphi < \pi$  wherever  $\text{Re } \Gamma = 0$ ;
- (e) the point  $2i\theta$  ( $= \chi(\zeta(0))$ ,  $\zeta(0) = i \cot \theta$ ), which must have imaginary part in  $(\pi, 3\pi)$ , lies in the strip between  $\text{Im } \chi = \pi$  and  $\Gamma$ .

Then the region  $H_\theta$  between the chord and  $s_\theta([-1, 1])$  is a Jordan region, and  $s_\theta$  maps the outer side of the chord to the inner side of the arc.  $M(x, \sigma) = x + i\pi + \sigma\varphi(x)$  is a flat model of the quotient with the properties of Lemma 9.27(4) (with  $M$  in place of  $N_\theta$ ),  $w_{-1} = 0$  lies in  $H_\theta$  (entry index  $-1$ ), and Definition 9.30 and Proposition 9.31 apply on  $\Theta''$ . Parts (1) and (2) of Lemma 9.27 are not asserted. The arc need not lie in the lower half-plane, and it may leave  $\zeta = 1$  on the side  $\zeta > 1$ .

*Proof.* By (a)  $\Gamma$  is a graph over  $\operatorname{Re} \chi$ . By (b) and (c) the map  $M$  is a local diffeomorphism. Its Jacobian is  $\operatorname{Im} \varphi + \sigma \operatorname{Im}(\overline{\varphi_x} \varphi)$ , linear in  $\sigma$ , and  $M$  is bijective on the two boundary lines. So, as in Lemma 9.3(3), it is a diffeomorphism onto the strip between  $\operatorname{Im} \chi = \pi$  and  $\Gamma$ . By (b) the strip embeds in  $\mathbb{C}/2\pi i\mathbb{Z}$ . By (d) it avoids  $\chi \equiv 0$ , which corresponds to  $\zeta = \infty$ . So its image under  $\zeta = (1 + e^\chi)/(1 - e^\chi)$  is a bounded Jordan region bounded by the chord and the arc.

As  $x \rightarrow \mp\infty$ ,  $\varphi \rightarrow \log \lambda_+$  and  $\varphi \rightarrow -\log \lambda_-$  exponentially fast. Both limits have imaginary part in  $(0, 2\pi)$  by (b), so  $M$  is asymptotically affine and nondegenerate at both ends. This holds also for  $\lambda_+ = -1$ , where  $\operatorname{Im} \log \lambda_+ = \pi$ . The quotient of the strip is therefore quasiconformally  $\mathbb{C}/i\mathbb{Z}$ , with punctures at both ends, and  $M$  is holomorphic in  $\theta$  at fixed  $(x, \sigma)$ .

The seam map  $s_\theta$  is injective near  $[-1, 1]$  because  $|\theta| < \pi$ . In the coordinate  $\chi$  the map  $s_\theta$  sends the outward normal  $-i$  of the chord to  $-i(1 + \varphi_x)$ . The direction from  $\Gamma$  into the strip is  $-\varphi = -\partial_\sigma M$ . By (c) both have  $\operatorname{Im}((1 + \varphi_x)\overline{v}) < 0$ , so they lie on the same side of  $\Gamma$ : the outer side of the chord goes to the inner side of the arc. By (e),  $w_{-1} = 0$  lies in the crescent. Since  $-1$  is the smallest index, the entry index is  $-1$ , and no condition on earlier points is needed. The proofs of Lemma 9.27(4), Definition 9.30 and Proposition 9.31 use only these facts. The quotient does not depend on the interpolation, so on overlaps with Section 9.10 the two constructions give the same family.  $\square$

**Theorem 9.43** (Holomorphy at  $b = e^{-e}$ ). *The family of Theorem 9.40 extends holomorphically to a neighbourhood of the real segment*

$$[\exp(-12e^{12}), \exp(-\tfrac{1}{5}e^{1/5})] \approx [e^{-1.95 \cdot 10^6}, 0.7833],$$

*which contains  $b = e^{-e}$ . At every base of this neighbourhood, including  $b = e^{-e}$ , it is given by the straight-chord construction with the mark  $w_{-1} = 0$ . In particular its boundary values on this segment are real-analytic.*

*Proof.* By Appendix A.4, (a)–(f) of Lemma 9.42 hold for all  $\lambda$  in the square  $[-1.1, -0.9] \times [-0.1, 0.1]$  and the strip  $[-12, -\tfrac{1}{5}] \times [-10^{-3}, 10^{-3}]$ , with one branch  $\theta(\lambda)$ , which lies in  $R' = [2.20, 2.45] \times [0.62, 0.95]$  on the square. All checks are strict, so they persist on an open neighbourhood of the closed square and strip. For  $\lambda = e^{i\alpha}$  near  $-1$  the root in  $R'$  is the one used in Theorem 9.34. Indeed, that theorem's contraction disc of radius 0.00019 about  $\theta_*$  lies in  $R \cap R'$ , and  $\lambda_+$  is injective on  $R'$ . So the two roots agree near  $\alpha = \pi$ , and by continuity on the whole arc inside the square. So on the upper part of the square the new family is that of Theorem 9.34. It therefore equals  $F^{\text{tr}}$  inside and the family of Theorem 9.40 outside. The whole strip cannot be compared with Theorem 9.40 in the base variable, because  $\operatorname{Im} g(x + iy) \approx y(1 - x)e^{-x}$  is large for  $x \ll 0$ . Let  $U$  be the tube of points of the strip near  $[-12, -\tfrac{1}{5}]$  with  $|\operatorname{Im} g| < \pi$ ; its width is about  $\pi e^x/(1 - x)$ . On the real segment  $g$  is real, strictly increasing and has  $g' = (1 - \lambda)e^{-\lambda} > 0$ . So  $b = \exp g$  is injective on  $U$ , after shrinking it, and maps it onto a neighbourhood of the stated  $b$ -segment, with  $b(-1) = e^{-e}$ .  $U \cap \{\operatorname{Im} \lambda > 0\}$  is connected, maps into  $\mathbb{H}$  with the principal logarithm, and contains the upper part of the square near  $\lambda = -1$ . By the identity theorem the new family agrees there with the family of Theorem 9.40. Composed with the inverse of  $\exp \circ g$  on  $U$ , it gives the holomorphic extension.  $\square$

**Remark 9.44.** (1) The extension across the segment is the continuation of the upper family. It need not agree with the continuation through the lower half-plane, which is the complex-conjugate family. At real  $b$  in the segment the partner fixed point  $L_-$  is not real, so the construction is not symmetric under conjugation. (2) For  $b \in (1, \eta)$  the partner is real:  $\theta$  is purely imaginary, the image arc falls onto the chord, and the straight crescent collapses. There the tilted regions of Section 9 are needed. Near  $b = 1$  ( $\lambda_+ \rightarrow 0^-$ ) the root  $\theta$  goes to infinity. The interval towards  $b = 0$  is covered instead by transport through the attracting two-cycle (Theorem 9.45).

**9.13. The real slice below the period-doubling point.** The interval below  $e^{-e}$  admits a different transport coordinate: the multiplier of the attracting two-cycle. This avoids a uniform straight-crescent estimate as  $b \downarrow 0$ .

**Theorem 9.45** (Continuation across the real slice below  $\eta$ ). *The family of Theorem 9.40 extends jointly holomorphically in  $(b, z)$  across every  $b \in (0, 1)$ . Together with Theorem 9.22, it extends across every point of  $(0, \eta) \setminus \{1\}$ . The extensions are locally single-valued and agree wherever their domains overlap in the upper half-plane.*

*Proof.* Proposition 9.23 covers  $(e^{-e}, 1)$ , Theorem 9.43 covers  $e^{-e}$ , and Proposition 3.5 covers  $(1, \eta)$  by the identification in the proof of Theorem 9.22. We construct the extension on  $(0, e^{-e})$ .

Write  $b = e^{-a}$ ,  $a > e$ , and let  $p < q$  be the real two-cycle of  $E_b(w) = e^{-aw}$ . Its multiplier  $m = a^2 pq$  has the parametrisation

$$ap = u(\coth u - 1), \quad aq = u(\coth u + 1), \quad a = \frac{u}{\sinh u} e^{u \coth u}, \quad m = \left( \frac{u}{\sinh u} \right)^2, \quad u > 0.$$

Indeed  $aq/ap = e^{2u}$  and  $aq - ap = 2u$ , which give  $p = e^{-aq}$  and  $q = e^{-ap}$ . Moreover

$$\frac{d \log a}{du} = \frac{1}{u} - \frac{u}{\sinh^2 u} > 0, \quad \frac{d \log m}{du} = 2 \left( \frac{1}{u} - \coth u \right) < 0.$$

Thus  $a$  increases from  $e$  to  $+\infty$ , while  $m$  decreases from 1 to 0. In particular  $m$  is a real-analytic coordinate on the whole interval  $(0, e^{-e})$ , with nonzero derivative.

Choose  $b_* = \exp(-2e^2)$ , which is in the certified segment of Theorem 9.43, and let  $m_*$  be its two-cycle multiplier. Denote by  $B_p, B_q$  the two completely invariant basins for  $E_{b_*}^2$ , distinguished by convergence to  $p, q$ , and by  $\sigma_p, \sigma_q$  their Koenigs coordinates, normalised to have derivative 1 at the cycle. They satisfy  $\sigma_j \circ E_{b_*}^2 = m_* \sigma_j$  and have nonzero derivative on their respective basins. On a simply connected domain containing  $(0, 1)$  and contained in  $\{0 < |m| < 1\}$ , take the logarithm real on  $(0, 1)$  and set

$$c(m) = \frac{\operatorname{Log} m - \operatorname{Log} m_*}{\operatorname{Log} m + \operatorname{Log} m_*}, \quad \nu_m = c(m) \frac{\sigma_j \overline{\sigma_j'}}{\overline{\sigma_j} \sigma_j'} \quad \text{on } B_j \setminus \sigma_j^{-1}(0),$$

and  $\nu_m = 0$  elsewhere. Since  $\operatorname{Re} \operatorname{Log} m < 0$ ,  $|c(m)| < 1$ . The relation  $\sigma_q \circ E_{b_*} = E_{b_*}'(p) \sigma_p$  on  $B_p$ , and its counterpart on  $B_q$ , show that  $\nu_m$  is  $E_{b_*}$ -invariant. The normalised solution  $h_m$  fixing  $0, 1, \infty$  therefore conjugates  $E_{b_*}$  to an entire universal cover of  $\mathbb{C}^*$  taking 0 to 1, hence to  $E_{b(m)}$  for a holomorphic  $b(m) = h_m(b_*)$ . The radial-stretch calculation of Proposition 9.18, applied to  $E_{b_*}^2$ , shows that the transported two-cycle has multiplier  $m$ . For real  $m \in (0, 1)$  the coefficient is conjugation-symmetric, so  $h_m$  preserves the real axis in order. In particular  $h_m(p), h_m(q) \in (0, 1)$  are a distinct real two-cycle and  $b(m) > 0$ . The parametrisation above then gives  $b(m) = \exp(-a(u(m))) \in (0, e^{-e})$  and  $b'(m) \neq 0$ .

Let  $H_*$  be the marked straight crescent at  $b_*$  from Theorem 9.43; its mark is  $w_{-1} = 0$ . The quotients of  $h_m(H_*)$ , with the edges identified by  $E_{b(m)}$ , are quasiconformally equivalent to the twice-punctured quotient of  $H_*$ . They have the transported fixed-point ends and the same mark 0. The Beltrami coefficient descends to the quotient as  $c(m)$  times a fixed coefficient. Ahlfors–Bers uniformisation and the  $\partial_{\bar{m}}$ -cancellation of Proposition 9.19 give a jointly holomorphic family of normalised germs, locally in  $(m, z)$ : give the mark height  $-1$  and compose the inverse uniformisation with  $E_{b(m)}$ , as in Definition 9.30, so that  $F_m(0) = E_{b(m)}(0) = 1$ . At the mark the coefficient is real-analytic in space because 0 belongs to a two-cycle basin and  $\sigma_j(0) \neq 0$ .

These germs agree with the straight-crescent family near  $b_*$ . Here is the local comparison. At  $m = m_*$  the two marked crescents are identical. On a compact part away from the fixed-point ends, their edges are close and the orbit-selection argument of Lemma 9.8 gives a conformal map

of quotients by locally fixed iterates of  $E_{b(m)}$ . At either end, use a repelling linearising coordinate and its logarithm. A lift of  $h_m$  commutes with the deck translation  $2\pi i$  and conjugates translation by  $\log \lambda_j(b_*)$  to translation by  $\log \lambda_j(b(m))$ . Consequently it is a real-linear map, tending to the identity as  $m \rightarrow m_*$ , plus a bounded doubly periodic remainder tending uniformly to zero on a fundamental parallelogram. In the logarithmic coordinate the transported and straight edges differ by at most  $O(|m - m_*|(1 + |\operatorname{Re} t|))$ , whereas a dynamical iterate changes their normal coordinate by a quantity bounded away from zero. At most  $O(|m - m_*|(1 + |\operatorname{Re} t|))$  iterates select the unique point in the other half-open crescent; the total tangential displacement is of the same order and remains within the linearising collar after shrinking  $|m - m_*|$ . This is the end-collar argument of Lemma 9.21, now with repelling rather than neutral ends. These orbit maps agree on overlaps, preserve the ends and the mark, and identify the normalised inverse uniformisations. Thus the two families agree on an open neighbourhood of  $b_*$ .

Finally,  $b'(m) > 0$  on the real interval. Take overlapping complex neighbourhoods of its points on which  $b(m)$  is biholomorphic and whose upper halves map into the upper base half-plane. The transported family and that of Theorem 9.40 agree on the first such neighbourhood by the preceding comparison, hence on every successive overlap by the identity theorem. This gives the asserted extension along the whole interval.  $\square$

The proof of Theorem 9.22 gives more than boundary values on  $(1, \eta)$ : on  $N_W \cap \{\operatorname{Im} b > 0\}$  it identifies the upper family with the jointly holomorphic sewn family of Proposition 3.5. Thus the upper continuation already extends holomorphically across every base in  $(1, \eta)$ .

**Proposition 9.46** (Obstructions at  $b = 1$  and  $b = 0$ ). *Let  $W \subset \mathbb{C}$  be a connected height domain with  $0 \in W$  and  $W + 1 \subset W$ . No normalised family  $F(b, z)$  satisfying  $F(b, 0) = 1$  and  $F(b, z + 1) = \exp((\operatorname{Log} b)F(b, z))$  can be jointly holomorphic on a product of  $W$  with a disc about  $b = 1$ . Moreover, for any such family on upper-half-plane bases approaching  $b = 0$ , the height value  $F(b, 2)$  has no holomorphic extension through  $b = 0$ .*

*Proof.* Suppose first that the family extended jointly holomorphically across  $b = 1$ . At  $b = 1$  the functional equation gives  $F(1, z + 1) = 1$  for  $z \in W$ . Since  $W + 1$  is a nonempty open subset of the connected domain  $W$ , the identity theorem yields  $F(1, z) \equiv 1$  on  $W$ . Put  $H(z) = \partial_b F(1, z)$ . Differentiating the functional equation at  $b = 1$  now gives  $H(z + 1) = F(1, z) = 1$ . The identity theorem again gives  $H \equiv 1$  on  $W$ , whereas  $F(b, 0) = 1$  gives  $H(0) = 0$ , a contradiction.

For the second assertion, the functional equation and normalisation give  $F(b, 1) = b$  and  $F(b, 2) = \exp(b \operatorname{Log} b)$ . Along  $b = it$ ,  $t \downarrow 0$ , this value tends to 1, but its derivative  $\exp(b \operatorname{Log} b)(1 + \operatorname{Log} b)$  is unbounded. Hence the upper-half-plane germ has no holomorphic extension through 0.  $\square$

**9.14. Consequences at the cusp.** The following statements describe the sewn family for real  $b < \eta$ , mostly close to  $\eta$ , and hence, by Theorem 9.14(3), the boundary values  $K_b$  of the continued Kneser family.

**Corollary 9.47** (Non-reality at fractional real heights). *For every real  $b < \eta$  sufficiently close to  $\eta$ , and every  $\delta > 0$  sufficiently small, some real  $x$  with  $0 < |x| < \delta$  satisfies  $\operatorname{Im} K_b^W(x) \neq 0$ . In particular the sewn solution is not real-valued on any real interval about  $z = 0$ .*

*Proof.* Suppose that  $K_b^W(x)$  were real for all real  $x$  in a neighbourhood of 0. Shrink that neighbourhood so its values lie in a real interval about 1 on which the attracting Koenigs coordinate  $\sigma_b$  is holomorphic and real, with  $\sigma_b(K_b^W(x))/\sigma_b(1) > 0$ . The upper representation of Theorem 6.3 gives

$$\frac{\sigma_b(K_b^W(x))}{\sigma_b(1)} = \lambda_b^{x+P_b(x)}, \quad P_b(0) = 0.$$

Since  $\log \lambda_b$  is real and nonzero, continuity at  $x = 0$  makes  $\operatorname{Im} P_b(x) = 0$  on this interval. The function  $P_b$  is holomorphic and periodic on a strip containing the whole real line, so reflection and the identity theorem make it real on that line. But its Fourier expansion has only non-negative modes. A real-valued periodic function has conjugate positive and negative coefficients; therefore all its nonconstant coefficients vanish. This contradicts  $c_1(b) \neq 0$  for  $b$  sufficiently close to  $\eta$ , established in Corollary 6.4.  $\square$

By Theorem 9.14, this non-reality statement applies to the continued Kneser family  $K_b = K_b^W$  near  $\eta$ . The tiny non-real part observed numerically by Paulsen at  $b = \sqrt{2}$  [18, §6] is reproduced by  $K^W$  in all nine quoted digits (Section 10.1); this is numerical, not a proof that his continuation path gives  $K^W$ .

**Corollary 9.48** (Complex cusp profile and its imaginary trace). *Let  $A = B_1 e^{2\pi i a} \neq 0$  and put  $f = F_\eta^{\text{att}}$ . On some fixed complex disk about  $z = 0$ , as  $b \rightarrow \eta^-$  through real bases,*

$$\frac{K_b^W(z) - R_b(z)}{\Lambda(b)} \longrightarrow f'(z) A (e^{2\pi i z} - 1)$$

*locally uniformly, hence with all  $z$ -derivatives. In particular, on a fixed real interval about 0,*

$$\frac{\operatorname{Im} K_b^W(x)}{\Lambda(b)} \longrightarrow f'(x) \operatorname{Im} [A (e^{2\pi i x} - 1)]$$

*uniformly on compact subintervals. The right-hand side is not identically zero on any interval about 0.*

*Proof.* By Theorem 6.3(4), with  $B = e^{2\pi(Y_0 + D + 2)}$ , there is a constant  $C$  independent of  $b$  and  $n$  such that  $|c_n(b)| \leq C(\Lambda B)^n$  for all  $n \geq 1$  and  $b$  sufficiently close to  $\eta$ . Here we used  $|\tau_n| \leq M e^{2\pi n(Y_0 + 1)}$  and absorbed the factor  $\Lambda$  in the error bound into  $C$ . Since  $P_b(0) = 0$ , its constant mode is  $-\sum_{n \geq 1} c_n$ . Choose a disk in the common upper representation supplied by Corollary 6.5. The Fourier tail is geometric on every smaller disk once  $b$  is close enough to  $\eta$ , and there

$$P_b(z) = c_1(b)(e^{2\pi i z} - 1) + O(\Lambda^2)$$

locally uniformly. Theorem 6.3(4) and Corollary 4.5 give  $c_1/\Lambda \rightarrow A$ . On a fixed disk about 0,  $R_b \rightarrow F_\eta^{\text{att}}$  with its derivatives by Corollary 6.5 and its proof. Taylor's formula gives  $K_b^W(z) = R_b(z) + R_b'(z)P_b(z) + O(\Lambda^2)$  on a smaller disk. This proves the complex limit, with derivative convergence by Cauchy's formula. For real  $x$ ,  $R_b(x)$  and  $f'(x)$  are real, giving the imaginary trace. If its last factor vanished on an interval, both the first and second derivatives at  $x = 0$  would vanish, forcing  $\operatorname{Re} A = \operatorname{Im} A = 0$ , contrary to  $A \neq 0$ .  $\square$

**Corollary 9.49** (Obstruction at the real cusp). *For every sufficiently small complex neighbourhood  $V$  of 0, one can choose a real  $x_* \in V \setminus \{0\}$  and constants  $c, \delta > 0$  such that*

$$|\operatorname{Im} K_b^W(x_*)| \geq c\Lambda(b) \quad (\eta - \delta < b < \eta, \ b \in \mathbb{R}).$$

*There is therefore no jointly holomorphic family  $F(b, z)$  on a product of a disc about  $b = \eta$  and  $V$  which is real at real heights for real  $b > \eta$  and agrees with  $K_b^W$  for all real  $b < \eta$  sufficiently close to  $\eta$ . In particular, direct holomorphic continuation of a real-normalised Kneser family through the real cusp cannot identify it with the inner sewn branch.*

*Proof.* The limiting profile in Corollary 9.48 is not identically zero on any real interval about 0. Choose  $x_*$  where it is nonzero. Since  $\Lambda(b) > 0$  for real  $b < \eta$ , the uniform limit gives the stated lower bound after reducing  $\delta$ . For fixed real  $x_*$ , Schwarz reflection in the parameter says that a holomorphic function  $b \mapsto F(b, x_*)$  which is real on a real interval above  $\eta$  is also real on the real interval below  $\eta$  in the same disc. This contradicts the lower bound if  $F = K^W$  there.  $\square$

**Proposition 9.50** (Logarithmic monodromy at the first backward singularity). *Let  $U$  be a simply connected parameter disc on which  $a(b) = \log b \neq 0$ . Suppose  $F_b(z)$  is jointly holomorphic near  $U \times \{-1\}$ ,  $F_b(-1) = 0$ , and  $F'_b(-1) \neq 0$ . Locally in  $(b, t)$  write  $F_b(-1 + t) = t u_b(t)$  with  $u_b(0) \neq 0$ . On any simply connected sector  $0 < |t| < r$ , all holomorphic backward branches satisfying  $\exp(a(b)G_b(-2 + t)) = F_b(-1 + t)$  have the form*

$$G_b(-2 + t) = \frac{\operatorname{Log} t + \log u_b(t) + 2\pi i k}{a(b)}, \quad k \in \mathbb{Z}.$$

*Continuation once around  $t = 0$  adds  $2\pi i/a(b)$ . Hence no single-valued holomorphic or meromorphic backward branch satisfying the equation exists on a full punctured disc about  $z = -2$ . For every real  $b \in (1, \eta)$ , the sewn solution  $K_b^W$  has a simple zero at  $z = -1$ , so this description applies to it and to its local holomorphic parameter family.*

*Proof.* The quotient  $u_b(t)$  extends jointly holomorphically through  $t = 0$ . After shrinking  $U$  and  $r$ , it has a joint holomorphic logarithm. Two logarithms of  $tu_b(t)$  on a connected sector differ by a constant multiple of  $2\pi i$ , giving the formula. A turn about  $t = 0$  changes  $\operatorname{Log} t$  by  $2\pi i$ . If a single-valued backward branch existed on the punctured disc, integrating the logarithmic derivative of  $\exp(aG) = tu$  around a small circle would give both 0 and  $2\pi i$ . A meromorphic branch has no pole away from the centre, since its exponential would have an essential singularity there; it is therefore subject to the same contradiction.

For the sewn family, the upper chart of Proposition 3.4 contains  $w = -1$  and its inverse uniformising coordinate  $w(z)$  is conformal, with  $w(-1) = -1$ . For real  $b$ ,  $R'_b(0) = \sigma(1) \log \lambda / \sigma'(1) \neq 0$ , since  $\sigma(1) \neq 0$  and  $\sigma' > 0$  on  $(-\infty, L)$  (forward iteration), and differentiating the functional equation at  $z = -1$  gives  $R'_b(-1) = R'_b(0)/a(b) \neq 0$ . Thus  $(K_b^W)'(-1) = R'_b(-1)w'(-1) \neq 0$ . Proposition 3.5 preserves this simple zero on a sufficiently small complex parameter disc about each real base.  $\square$

**Corollary 9.51** (Conjugate inner branches and their jump). *For real  $b \in (1, \eta)$  put  $K_b^{W,\#}(z) = \overline{K_b^W(\bar{z})}$  on the reflected regular domain. This reflected family is locally jointly holomorphic in  $(b, z)$  near every such real base. Both branches obey the tetration equation, are normalised at  $z = 0$ , agree at every nonnegative integer height where defined, and have a simple zero at  $z = -1$  with the same logarithmic monodromy  $2\pi i / \log b$  at  $z = -2$ . For  $x_*$  as in Corollary 9.49,*

$$|K_b^W(x_*) - K_b^{W,\#}(x_*)| \geq 2c\Lambda(b) \quad (\eta - \delta < b < \eta, \quad b \in \mathbb{R}).$$

*More precisely, on a fixed real interval about 0 the quotient of their difference by  $\Lambda(b)$  tends locally uniformly to  $2i(F_\eta^{\text{att}})'(x) \operatorname{Im}[A(e^{2\pi i x} - 1)]$ .*

*Proof.* For complex  $b$  near a real base, define the reflected family by  $K_b^{W,\#}(z) = \overline{K_b^W(\bar{z})}$ . Proposition 3.5 makes it jointly holomorphic. Reflection preserves the equation for real  $b$  and therefore, by the identity theorem, on its local complex parameter disc. The integer values follow from the equation and normalisation. The zero at  $-1$  and its simplicity follow from Proposition 9.50, as does the monodromy. On real heights the difference is exactly  $2i \operatorname{Im} K_b^W(x)$ , so the lower bound and the profile follow from Corollaries 9.49 and 9.48.  $\square$

**Proposition 9.52** (The cusp splitting is beyond all algebraic orders). *On a sufficiently small real interval  $J$  about 0, put*

$$M(x) = (F_\eta^{\text{att}})'(x) \operatorname{Im}[A(e^{2\pi i x} - 1)].$$

*The interval  $J$  may be chosen so that  $M(x) \neq 0$  for every  $x \in J \setminus \{0\}$ . For every such  $x$  and every integer  $q \geq 1$ , the germ along  $t > 0$  of*

$$t \mapsto K_{\eta-t^q}^W(x)$$

has no meromorphic extension through  $t = 0$ . In particular, no finite ramification of the base parameter makes this value meromorphic at the real cusp; it has no convergent Puiseux–Laurent germ there. If it admits a single-valued holomorphic continuation on a full punctured  $t$ -disc, the singularity at  $t = 0$  is essential.

*Proof.* The profile  $M$  is real analytic, vanishes at 0, and, by Corollary 9.48, is not identically zero on any interval about 0. Thus 0 is an isolated zero, and we shrink  $J$  accordingly. Suppose such a meromorphic germ  $G(t)$  existed. Its reflection  $G^\#(t) = \overline{G(\overline{t})}$  would also be meromorphic, so  $D(t) = G(t) - G^\#(t)$  would be meromorphic near 0. For positive real  $t$ , the imaginary profile and the saddle-node scale of  $\Lambda$  give

$$D(t) = 2i \operatorname{Im} K_{\eta-t^q}^W(x) = O\left(\exp(-c_0 t^{-q/2})\right) = o(t^N) \quad \text{for every } N \geq 0$$

with some  $c_0 > 0$ . A meromorphic Laurent germ with this decay on the positive axis is identically zero. This contradicts the nonzero limit  $D(t)/\Lambda(\eta - t^q) \rightarrow 2iM(x)$ .  $\square$

**Corollary 9.53** (A two-jet witness at the normalisation height). *Put  $f = F_\eta^{\text{att}}$  and  $A = B_1 e^{2\pi i a} \neq 0$ . As  $b \rightarrow \eta^-$  through real bases,*

$$\frac{1}{\Lambda(b)} \begin{pmatrix} \operatorname{Im}[(K_b^W)'(0)] \\ \operatorname{Im}[(K_b^W)''(0)] \end{pmatrix} \longrightarrow \begin{pmatrix} 2\pi f'(0) \operatorname{Re} A \\ 4\pi f''(0) \operatorname{Re} A - 4\pi^2 f'(0) \operatorname{Im} A \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

*Consequently at least one of the first two  $z$ -derivatives at 0, with the same choice of derivative for every  $q \geq 1$ , has no meromorphic base germ through  $t = 0$  after  $b = \eta - t^q$ . The corresponding derivative of the reflected inner branch differs from that of  $K_b^W$  by at least  $c\Lambda(b)$  in modulus for some  $c > 0$  and all sufficiently close real  $b < \eta$ .*

*Proof.* The complex limit of Corollary 9.48 and Cauchy’s formula give convergence of the first two derivatives. The derivatives of  $R_b$  at 0 are real for real  $b$ , giving the displayed vector. Since  $f'(0) \neq 0$  by Corollary 6.5, vanishing of its first entry forces  $\operatorname{Re} A = 0$ , and then vanishing of its second forces  $\operatorname{Im} A = 0$ , a contradiction. For a nonzero entry, the reflection argument in the proof of Proposition 9.52 applies to that derivative. Reflection doubles its imaginary part, which also gives the lower bound.  $\square$

**Remark 9.54** (Conjugate bypasses of the cusp). Thus the monodromy of Remark 9.16(1) is quantified by Corollary 9.51.

## 10. NUMERICAL RESULTS

**10.1. Ladder computations of a two-sided numerical solution.** In this numerical section,  $K_b$  denotes the extrapolated values from a pointwise two-fixed-point construction; no base continuation is assumed. The solution is computed at complex bases  $b + iy$  by a two-fixed-point  $\theta$ -mapping construction. The program represents it as  $R_{b+iy} \circ (\operatorname{id} + \theta_{\text{up}})$  in the upper half-plane and  $S_{b+iy} \circ (\operatorname{id} + \theta_{\text{dn}})$  in the lower one. The computation runs along ladders  $y \downarrow 0$  at working precisions up to 52 digits and is extrapolated to  $y = 0$ . Then  $c_1$  is read off from  $D = K - R$ . First-mode dominance is confirmed by the constancy of  $q(z) = D(z)/(R'(z)(e^{2\pi iz} - 1))$  over four points, including  $z = \frac{1}{2} - \frac{i}{2}$ , where the first mode is amplified by  $e^\pi$  (Table 2). Omitting the constant mode  $c_0 = -\sum_{n \geq 1} c_n$  produces a spread of 1.917 instead.

Table 3 lists  $|\hat{c}_1| = |c_1|/\Lambda$ . Over the range shown,  $|D|$  varies by forty-five orders of magnitude while  $|\hat{c}_1|$  varies by less than 5% and increases towards  $|B_1|$ . The row  $b = \sqrt{2}$  uses Paulsen’s value  $\operatorname{Im} K_{\sqrt{2}}(\frac{1}{2}) \approx -1.18899697 \cdot 10^{-48}$ , from a 120-digit contour computation [18]. It shares neither code nor method with ours. Solving the sewing equation directly at  $b = \sqrt{2}$  (80 digits, 32 Fourier modes, residual  $< 10^{-75}$ ; `paulsen_sqrt2_check.py`) gives  $c_1 = -2.0924123 \cdot 10^{-50} +$

TABLE 2. The ratio  $|q|$  at four points. The last column is  $\max |q|/\min |q|$ .

| base            | $z = \frac{1}{2}$         | $z = -\frac{1}{2}$        | $z = \frac{1}{4}$         | $z = \frac{1}{2} - \frac{1}{2}$ | spread |
|-----------------|---------------------------|---------------------------|---------------------------|---------------------------------|--------|
| $1.1 + 0.0005i$ | $1.987636 \times 10^{-9}$ | $1.987636 \times 10^{-9}$ | $1.987636 \times 10^{-9}$ | $1.987637 \times 10^{-9}$       | 1.0000 |
| $1.05 + 0.003i$ | $1.422053 \times 10^{-7}$ | $1.422053 \times 10^{-7}$ | $1.422053 \times 10^{-7}$ | $1.422061 \times 10^{-7}$       | 1.0000 |

$1.4780488648 \cdot 10^{-48}i$ ,  $R'_{\sqrt{2}}(\frac{1}{2}) = 0.4022184246$ , and  $\text{Im } K_{\sqrt{2}}^W(\frac{1}{2}) = \text{Im}[R'(\frac{1}{2})P(\frac{1}{2})] + O(\Lambda^2) = -1.18899697185 \cdot 10^{-48}$ . This matches Paulsen's value to relative accuracy  $1.6 \cdot 10^{-9}$ , the rounding of his quoted digits.

TABLE 3. Ladder values of  $|\hat{c}_1|$  and their ratio to  $|B_1|$ .

| $b$        | $\lambda$ | $\Lambda$              | $ \hat{c}_1 $ | $ \hat{c}_1 / B_1 $ | source            |
|------------|-----------|------------------------|---------------|---------------------|-------------------|
| 1.05       | 0.051362  | $1.68 \times 10^{-6}$  | 0.085017      | 0.95462             | ladder, 20 digits |
| 1.10       | 0.105964  | $2.30 \times 10^{-8}$  | 0.086427      | 0.97045             | ladder, 20 digits |
| 1.15       | 0.164802  | $3.10 \times 10^{-10}$ | 0.087205      | 0.97919             | ladder, 20 digits |
| 1.20       | 0.229312  | $2.28 \times 10^{-12}$ | 0.087732      | 0.98511             | ladder, 20 digits |
| 1.25       | 0.301735  | $4.91 \times 10^{-15}$ | 0.088125      | 0.98952             | ladder, 24 digits |
| 1.30       | 0.385935  | $9.82 \times 10^{-19}$ | 0.088434      | 0.99299             | ladder, 32 digits |
| $\sqrt{2}$ | 0.693147  | $1.66 \times 10^{-47}$ | 0.088941      | 0.99868             | [18]              |

The ladder values agree with the Kneser-free prediction  $\tau_1$  and  $t_2/t_1^2$ , moduli and arguments, to the precision of the extrapolation (Table 1). Solving the sewing equation of Theorem 6.3 numerically gives  $\hat{c}_1 = \tau_1(1 + \vartheta)$  with  $|\vartheta| = 0.38\Lambda$  at  $b = 1.10$  and  $0.40\Lambda$  at  $b = 1.30$ . This matches the first-order formula  $\vartheta \approx \Lambda[-4\pi^2|\tau_1|^2 - 2\pi i\tau_1 - 4\pi i\tau_2\bar{\tau}_1/\tau_1]$ .

**10.2. Approach to the parabolic limit.** The transition map needs no ladder. Its modes are measured on the line  $\text{Im } z = -h_2/2 + Y$  of repelling time, near the gate carrying the positive modes, so the working precision needed is independent of  $h$ . Results for  $Y = 1, 1.5, 2, 2.5, 3$  agree to all printed digits. Table 4 follows the invariants to  $\eta - b = 2.7 \cdot 10^{-5}$ .

TABLE 4. Kneser-free invariants near  $\eta$ . Here  $\delta_1 = 1 - |\tau_1|/|B_1|$  and  $p = |\log \lambda| \log \lambda_2$ .

| $\lambda$ | $\eta - b$            | $ \tau_1 / B_1 $ | $\delta_1/p$ | $\arg \tau_1$ | $ t_2/t_1^2 $ | $ t_3/t_1^3 - B_3/B_1^3 $ |
|-----------|-----------------------|------------------|--------------|---------------|---------------|---------------------------|
| 0.5       | $9.04 \times 10^{-2}$ | 0.99601457       | 0.0102076    | 1.581415      | 1.555683      | 0.148                     |
| 0.8       | $1.21 \times 10^{-2}$ | 0.99952725       | 0.0102002    | 1.585920      | 1.572187      | 0.0177                    |
| 0.9       | $2.84 \times 10^{-3}$ | 0.99989062       | 0.0101993    | 1.586391      | 1.573905      | 0.00410                   |
| 0.95      | $6.87 \times 10^{-4}$ | 0.99997362       | 0.0101991    | 1.586499      | 1.574298      | 0.000989                  |
| 0.98      | $1.08 \times 10^{-4}$ | 0.99999587       | 0.0101990    | 1.586528      | 1.574403      | 0.000155                  |
| 0.99      | $2.67 \times 10^{-5}$ | 0.99999897       | 0.0101990    | 1.586532      | 1.574418      | 0.0000385                 |
| limit     | 0                     | 1                | 0.0101990063 | 1.5865331     | 1.5744227     | 0                         |

**10.3. The deficit law.** The natural parameter is the product of the logarithmic multipliers,  $p = |\log \lambda| \log \lambda_2 = \varepsilon^2(1 - \varepsilon/3 + O(\varepsilon^2))$ . Up to an analytic factor, this is the canonical parameter of the Mardešić–Roussarie–Rousseau normal form [15]. Fits of degree 2, 3 and 4 in  $p$  over twelve

bases agree to eleven digits:

$$(18) \quad \log \frac{\tau_n}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^2), \quad \begin{aligned} \kappa^{(1)} &= -0.0101990063459 - 0.0132509561825 i, \\ \kappa^{(2)} &= -0.0510565562 - 0.0623157805 i, \\ \kappa^{(3)} &= -0.1400443481 - 0.1692244639 i, \end{aligned}$$

with  $p^2$ -coefficients of order  $10^{-4}$  and constant terms below  $10^{-16}$ . Hence  $\delta_1 = Cp + O(p^2)$  with  $C = 0.0101990063$ . Theorem 5.6 proves the first-order statement and computes  $\kappa^{(n)}$  independently of any fit (Section 5). By Theorem 6.3, the Kneser-side  $\hat{c}_n$  inherit (18) up to  $O(\Lambda)$ , which is beyond all orders in  $p$ . The  $\kappa^{(n)}$  are not linear in  $n$ , so the first-order effect of the unfolding is not a translation of Fatou coordinates.

**10.4. Identification with an independent Kneser-type implementation.** The sewing solution can also be compared with a completely independent two-fixed-point computation at individual bases. Sheldon Levenstein’s PARI/GP program `fatou.gp` builds the “merged” two-fixed-point solution for complex bases. At each base it constructs, independently, the solution with a  $\theta$ -mapping at each of the two fixed points: the two regular representations of [18, Lemma 1]. Its author regards this as the continuation of Kneser’s solution in the base. We use the unmodified program (sha256 70559daf...): published reference values with 100 verified digits at  $b = 0.8 + 0.4i$  and  $b = 1 + i$ , and our own runs at `\p 40` at  $b = 1.2 + 0.1i$  and  $b = 1.3 + 0.2i$ , near the interval  $(1, \eta)$ . In those runs  $\text{sexp}(0) = 1$  to  $10^{-53}$ , and the functional equation holds to  $10^{-54}$ .

We solved the sewing equation of Theorem 6.3 directly at these complex bases (script `weld_complex.py`).  $R$  is the regular solution at the attracting fixed point and  $S$  the one at the repelling fixed point; the seam lies in the band of holomorphy of  $T = R^{-1} \circ S$ , and the root is selected as in Definition 8.1. Table 5 shows the result. At the three bases where the band is not too narrow, the sewing solution agrees with the independent value *to the full working precision of 40 digits*. That is 27 to 35 orders of magnitude below the size of the separation  $|\kappa - R|$  itself. At  $1 + i$  the band is narrowest and the agreement ( $2 \cdot 10^{-25}$ ) is limited by the number of samples. The neighbouring roots  $c_{\pm 1}$  reproduce  $R$  instead, and miss by the full separation. The  $\theta$ -mapping builder used for the ladders does not converge at  $0.8 + 0.4i$ , whereas the sewing iteration converges in a few dozen steps.

TABLE 5. Sewing solution versus the independent `fatou.gp` values of  $\kappa_b(\frac{1}{2})$  (working precision 40 digits).

| $b$          | $ \Lambda $            | $ \kappa_b(\frac{1}{2}) - R_b(\frac{1}{2}) $ | $ K_b^W(\frac{1}{2}) - \kappa_b(\frac{1}{2}) $ | $ \hat{c}_1 $ |
|--------------|------------------------|----------------------------------------------|------------------------------------------------|---------------|
| $1.2 + 0.1i$ | $1.91 \times 10^{-11}$ | $6.91 \times 10^{-13}$                       | $4.6 \times 10^{-40}$                          | 0.0869956     |
| $1.3 + 0.2i$ | $2.92 \times 10^{-11}$ | $1.81 \times 10^{-12}$                       | $8.3 \times 10^{-41}$                          | 0.0876095     |
| $0.8 + 0.4i$ | $1.05 \times 10^{-3}$  | $9.86 \times 10^{-5}$                        | $5.5 \times 10^{-39}$                          | 0.0856072     |
| $1 + i$      | $2.13 \times 10^{-2}$  | $4.41 \times 10^{-3}$                        | $2.0 \times 10^{-25}$                          | 0.0873850     |

The agreement identifies the sewing solution with an independent implementation of the two-sided regular representation, at four bases, two of them near  $(1, \eta)$ . It does *not* test the analytic continuation across the Shell–Thron boundary: both programs work at one base at a time. The continuation itself is proved near the cusp in Section 9.

**10.5. A pointwise test near the Shell–Thron boundary.** Before Theorem 9.40 was available, we tested the pointwise output of the unmodified `fatou.gp` (\p 24, and 50 digits in `boundary_multiheight.py`) at six bases along the normal  $\lambda = re^{i\phi}$  through the boundary point  $b_c = \exp(\lambda_c e^{-\lambda_c}) \approx 1.5217 + 0.0148i$ ,  $\lambda_c = e^{i\phi}$ ,  $\phi = 2\pi(\sqrt{2} - 1)/5$ : three inside ( $r = 0.98, 0.99, 0.995$ ) and three outside ( $r = 1.005, 1.01, 1.02$ ). At heights  $z = \frac{k}{6}$ ,  $k = 1, \dots, 5$ , a degree-4 polynomial in  $b$  through five bases predicts the sixth to  $\leq 1.7 \cdot 10^{-12}$ , and quadratics fitted on the two sides differ at  $b_c$  by  $\leq 7.1 \cdot 10^{-9}$ , which is the size of the truncation term (raw values in `data/boundary-multiheight.json`). The two sides use structurally different representations, so this was an independent check of the smooth join proved in Theorem 9.34.

## 11. OPEN PROBLEMS

The boundary arc up to  $e^{-e}$  and the identification with Paulsen’s base-continued family, under the analytic-continuation interpretation of Corollary 9.41, are settled by Theorems 9.34 and 9.40. The following questions remain.

- (1) *Endpoint asymptotics and uniqueness.* Theorems 9.45 and 9.22 extend the upper family across every  $b \in (0, \eta) \setminus \{1\}$ . Proposition 9.46 shows that  $b = 1$  and  $b = 0$  cannot be removable singularities of a normalised jointly holomorphic family on a common forward-invariant height domain. Paulsen [19] studies selected noninteger heights as the base approaches 1. Determine uniform noninteger-height asymptotics for the present upper continuation near  $b = 1$  and  $b = 0$ . A related question is a global uniqueness theorem for Kneser-type solutions inside the Shell–Thron region. The condition by limits at  $\pm i\infty$  alone admits solutions of the form  $R \circ (z + 2\pi imz / \log \lambda + \dots)$ .
- (2) *The first-order coefficients.* Theorem 5.6 expresses  $\kappa^{(n)}$  as a convergent series over parabolic orbits. Is there a closed form, for instance in terms of the horn-map data  $B_n$ ,  $a$  and the iterative residue? Determine the radius of convergence and the optimal Gevrey order of the expansion in  $p$ . The  $\frac{1}{2}$ -summability result of [5] is in a different normalisation.
- (3) *Remaining certification.* The coordinate  $a$ , the moduli of the first three horn coefficients, and the two horn ratios used in the numerical comparisons have interval enclosures above. Give interval enclosures of the ladder and extrapolated two-fixed-point values. This would turn the remaining numerical comparisons into computer-assisted ones.

## CODE AND DATA AVAILABILITY

The exploratory computations use the Python library `kneser` (`mpmath`); the interval certificates use `python-flint/Arb`. The transition-map invariants are computed by `transition_map.py`, `eta_approach.py`, `eta_sweep.py`, `deficit_param.py` and `deficit_modes.py`. The parabolic data come from `parabolic_horn.py` and `parabolic_horn_inverse.py`. The first-order series and the second-order check are `mrr_derivative.py` and `mrr_second.py`. The boundary comparison is reproduced by `boundary_multiheight.py` from the unmodified `fatou.gp`. The confluence check is `confluence_check.py`, and the sewing solution is computed by `weld_solve.py`. The complex-base sewing solution and its comparison with `fatou.gp` are computed by `weld_complex.py`. The ladder computations of  $K_b$  use the complex-base builder of the library together with `c1_invariant.py` and `first_mode_check.py`. The signs and branches used in Section 9 (normal component of the step, the first crossing of the base orbit, and the seam height  $-h/2$  through the lower half-plane) are checked numerically by `cusp_tilted_check.py`. The interval certificates are `certify_far_collar.py` (boundary collar), `certify_last_boundary_arc.py` (last boundary arc), `certify_upper_halfplane.py` (upper half-plane), `certify_endpoint.py` (endpoint segment), `certify_parabolic_a.py` (the coordinate  $a$ ) and `certify_horn_coefficients.py` (horn coefficients).

Code and data are available from the author on request.

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## APPENDIX A. INTERVAL CERTIFICATES

All certificates are Python programs using `python-flint/Arb` ball arithmetic with outward rounding; every strict inequality is decided on enclosures, and floating point is used only to choose centres of Krawczyk boxes. Krawczyk's operator  $K = m - Yf(m) + (1 - Yf'(B))(B - m)$ ,  $K \subset \text{int } B$ , proves existence and uniqueness of a zero in  $B$  for every parameter in the given balls. Running times are on a laptop.

**A.1. The boundary collar.** *Lemma 9.26.* Interval arithmetic with outward rounding (`certify_far_collar.py` `python-flint/Arb`), on boxes in  $(p, c)$  and  $(p, v)$  and subintervals of  $t$ . On the cone the quantities in (1) are divided by  $p$ , and  $|\lambda_+|^2 - 1$  on its top edge is divided by  $p$ ; this makes them analytic at  $p = 0$ . On  $\Theta_K$  each strip of width  $(P_1 - P_0)/440$  is certified up to the larger value of  $V$  at its ends. The top-edge test is made at the smaller value; this suffices because  $|\lambda_+|$  decreases in  $q$ . They are evaluated through the entire series  $g_\theta = -i\theta(\theta/\sin\theta)h$ ,  $h = -\sum_{n \geq 2} i^n \theta^{n-2} P_n(t)/n!$  with  $P_n(t) = (t^n - t^{n \bmod 2})/(t^2 - 1)$ ,  $|P_n| \leq n/2$  on  $[-1, 1]$ , and the Bernoulli series of  $\mu$ , each with an explicit tail bound. For  $p \rightarrow 0$  the limits are  $\text{Im } g/p \rightarrow -\frac{1}{2}$  and  $\text{Im}(\bar{s}'g)/p \rightarrow -\frac{1}{2}$ .

*Proof of Lemma 9.28.*  $w_{-1} = 0$  lies above the chord:  $\zeta(0) = i \cot \theta$ , and Lemma 9.26(3) applies. Put  $q_n = (\zeta_n - 1)/(\zeta_n + 1)$ , so that  $\zeta_n = (1 + q_n)/(1 - q_n)$  and  $\text{Im } \zeta_n = 2 \text{Im } q_n / |1 - q_n|^2$ . From  $s_\theta$ ,

$$q_{n+1} = \frac{e^{i\theta\zeta_n} - e^{i\theta}}{e^{i\theta\zeta_n} - e^{-i\theta}} = e^{i\theta} \frac{\sin(\theta(\zeta_n - 1)/2)}{\sin(\theta(\zeta_n + 1)/2)} = q_n e^{i\theta} \frac{T(v_n - i\theta)}{T(v_n)}, \quad v_n = \frac{i\theta}{1 - q_n},$$

with  $T(w) = \sinh w/w$ , since  $\theta(\zeta_n + 1)/2 = -iv_n$  and  $\theta(\zeta_n - 1)/2 = -i(v_n - i\theta)$ . From the Bernoulli expansion of  $\coth w - 1/w$  and  $2\zeta(2k) \leq \pi^2/3$ : on  $|w| \leq \frac{9}{20}$ ,  $|(\log T)'(w)| \leq K|w|$  and  $|(\log T)''(w)| \leq K_2$ , with  $K = 1/(3(1-x)) < 0.341$ ,  $K_2 = (1+x)/(3(1-x)^2)$  and  $x = (9/20)^2/\pi^2$ .

A series enclosure gives  $\chi_0 = \log q_0$  with  $3p < y_0 = \text{Im } \chi_0 < 3.3p$  on the whole cone. Put  $\chi_{n+1} = \chi_n + i\theta + \log T(v_n - i\theta) - \log T(v_n)$  and  $y_n = \text{Im } \chi_n$ . The segment  $v_n - \tau i\theta = i\theta(1/(1 - q_n) - \tau)$ ,  $0 \leq \tau \leq 1$ , has modulus at most  $|\theta|(1 + 1/|1 - q_n|)$ . Now  $|1 - q| \geq \sin y$  if  $0 < y \leq \pi/2$  and  $|1 - q| \geq 1$  if  $\pi/2 \leq y \leq 3\pi/2$ , whatever  $\text{Re } \chi$  is. So while  $3p \leq y_n \leq \pi + 1.2p$  the modulus is at most  $1.005p(1 + 1/\sin 3p) < \frac{9}{20}$ , and the correction term has modulus at most  $K|\theta|^2(1 + 1/\sin 3p) < 0.16p$ . Hence  $0.84p < y_{n+1} - y_n < 1.16p$ . Let  $n_*$  be the first index with  $y_{n_*} \geq \pi$ . Then  $y_{n_*} < \pi + 1.16p$ , and  $0 < y_n < \pi$ , i.e.  $\text{Im } q_n > 0$ , for  $n < n_*$ . So  $w_0, \dots, w_{n_*-1}$  lie above the chord.

If  $y_{n_*} = \pi$ , then  $\zeta_{n_*} \in (-1, 1)$ . Otherwise let  $S = \{x_{n_*-1} + iy : y_{n_*-1} \leq y \leq \pi\}$  and  $f(\chi) = \chi + i\theta + \log T(v - i\theta) - \log T(v)$ , the map  $E_\theta$  in the coordinate  $\chi$ . On  $S$  one has  $|1 - q| \geq 1$  and  $|\zeta| \leq 1$ , and  $f' - 1 = [(\log T)'(v - i\theta) - (\log T)'(v)]vq/(1 - q)$  has modulus at most  $2K_2|\theta|^2 < \frac{1}{10}$ . So  $\text{Im } f$  increases along  $S$ , from  $y_{n_*} > \pi$  to less than  $\pi + 1.16p$ . Thus  $f(S)$  is a curve in the open lower  $\zeta$ -half-plane. It ends at the point of  $A_\theta$  that is the image of  $x_{n_*-1} + i\pi$ , and it meets  $A_\theta$  nowhere else, since  $s_\theta$  is injective on the unit disc.  $S$  approaches the chord from above, so by Lemma 9.27(3)  $f(S)$  leaves  $A_\theta$  into  $H_\theta$ . Hence  $f(S)$  minus its end lies in the interior of  $H_\theta$ , and so does  $\zeta_{n_*}$ . The constants are checked at  $p = P_0$  in `certify_far_collar.py`. The bounds  $|\theta|(1 + 1/\sin 3p)$  and  $K|\theta|^2(1 + 1/\sin 3p)/p$  increase with  $p$ , because  $p/\sin 3p$  does, so the checks at  $P_0$  cover the whole cone.

*Lemma 9.29 on  $\Theta_K$ .* On  $\Theta_C$  this is Lemma 9.28. On  $\Theta_K$  the script encloses the orbit on parameter boxes. It iterates the exact step of the previous proof in mean-value form about the midpoint parameter. For each box it certifies  $\text{Im } \zeta_m > 0$  for  $-1 \leq m < n$ , and then one of the following. Either a Krawczyk test proves  $\zeta_n = N_\theta(t, \sigma)$  with  $|t| < 1$ ,  $0 < \sigma < 1$ ; so  $\zeta_n$  lies in the interior of  $H_\theta$ , by Lemma 9.27(2). Or  $\zeta_{n'}$  ( $n' = n$  or  $n - 1$ ) lies in a rectangle  $D = [a, b] \times [-\rho, \rho]$  with  $-1 < a < b < 1$ ,  $\rho \leq \frac{1}{4}$ , for which Krawczyk tests prove the following: every point of  $D \cap \{\text{Im} \leq 0\}$  is  $N_\theta(t, \sigma)$  with  $|t| < 1$ ,  $\sigma < 1$ ; and every point of  $s_\theta(D \cap \{\text{Im} \geq 0\})$  is  $N_\theta(t, \sigma)$  with  $|t| < 1$ ,  $\sigma > 0$ , the root being unique in a Krawczyk box that contains the known roots  $(t, 1)$ ,  $t \in [a, b]$ , of the real edge. In the first case  $\sigma \geq 0$  is automatic, since  $\text{Im } N(t, \sigma) = \sigma(1 - t^2) \text{Im } g > 0$  for  $\sigma < 0$ . In the second case the root is  $(t, 1)$  on the real edge, by uniqueness; off the real axis  $\sigma \neq 1$ , since  $\sigma = 1$  would put a point of  $s_\theta(D)$  on  $A_\theta$  and contradict injectivity. Since  $D \cap \{\text{Im} > 0\}$  is connected and, by Lemma 9.27(3),  $\sigma < 1$  just above the real axis,  $\sigma < 1$  there. Hence  $\zeta_{n'} \in H_\theta$  if it is not above the chord, and  $\zeta_{n'+1} \in H_\theta \setminus A_\theta$  otherwise. All 33 027 boxes pass (about six minutes on a laptop).

**A.2. The last boundary arc.** *Theorem 9.34.* We use the chord and the coordinate  $s_\theta$  of Section 9.10. The interval certificate `certify_last_boundary_arc.py` validates a simple root

$$\theta_* = 2.29857900665128663858 \dots + 0.76604606099318995273 \dots i$$

of  $\lambda_+(\theta) = -1$ . On the rectangle  $R = [2.19, 2.31] + i[0.71, 0.78]$  it verifies  $\text{Re}(\lambda'_+(\theta)/\lambda'_+(\theta_*)) > 4/5$ . The Noshiro–Warschawski criterion makes  $\lambda_+$  injective on  $R$ . Validated contraction boxes give its unique inverse branch  $\theta(\alpha) \in R$  for  $0.935\pi \leq \alpha \leq 0.9999\pi$ ; a contraction disk of radius 0.00019 about  $\theta_*$  covers the remaining interval  $0.9999\pi \leq \alpha \leq \pi$ .

For  $-1 \leq t \leq 1$  set  $f_1 = \text{Im } g_\theta(t)$  and  $f_2 = \text{Im}(\overline{s'_\theta(t)}g_\theta(t))$ . The certificate proves  $f_1, f_2 < 0$  on  $[-1, 19/20]$  and  $\partial_t f_1, \partial_t f_2 > 0$  on  $[19/20, 1]$  along this branch. The first part uses 79 parameter boxes up to  $0.9999\pi$  and a fixed 0.0002 square about  $\theta_*$  for the rest. The evaluations use the entire series for  $g_\theta$  and its  $t$  derivative, with explicit tails and outward-rounded Arb intervals. At the endpoint of the chord there is the exact identity

$$f_1(1) = f_2(1) = -\frac{1}{2} \text{Im } \lambda_+ = -\frac{1}{2} \sin \alpha < 0 \quad (\alpha < \pi).$$

The positive derivatives therefore close the inequalities all the way to  $t = 1$ , even as their endpoint margin tends to zero when  $\alpha \rightarrow \pi$ . The same certificate gives, by Krawczyk inclusion,  $\zeta_\theta(0) = i \cot \theta \in \text{int } H_\theta$  throughout the arc. Thus the first-entry index is always  $-1$ ; no orbit tracking is needed.

**A.3. The upper half-plane.** The certificate consists of four parts (`certify_upper_halfplane.py`):

- (T) *Tails*,  $\text{Re } \lambda \leq -12$ . With  $\kappa = 1/\lambda$  and  $\theta = \pi - \delta$ ,  $D = \delta/\kappa$  is the fixed point of the contraction  $D = -(\pi - \kappa D) \ell(2i\kappa(\pi - \kappa D))$ ,  $\ell(z) = -\log(1 - z)/z$ . After multiplication by  $1/|\lambda|$  all quantities are analytic at  $\kappa = 0$ .  $F$  is a contraction on the fixed convex set  $|\text{Re } D + \pi|, |\text{Im } D| \leq \frac{3}{2}$ , so  $D$  is unique there. The quantities are checked on boxes in  $(1/|\text{Re } \lambda|, \text{Im } \lambda)$  that include the limit. The mark is  $w_{-1} = 0$ :  $\zeta(0) = i \cot \theta = N_\theta(t, \sigma)$  with  $(t, \sigma)$  near  $(0, \frac{1}{2})$ , by a scaled Krawczyk test. There are 6230 boxes.
- (K) *Core*,  $-12 \leq \text{Re } \lambda \leq 5$ ,  $|\lambda - 1| \geq 0.08$ . Boxes are discarded when they certainly violate  $|\lambda| \geq 1$  or  $0 \leq \text{Im } g \leq \pi$ , or satisfy  $g(\lambda) \in g(\mathbb{D})$ . The last test uses that  $g(\mathbb{D})$  is starlike with boundary  $\rho = e^{-\cos \phi}$  at argument  $\psi = \phi - \sin \phi$ . For  $\text{Re } \lambda \geq 5$ ,  $|g| \leq (5 + \pi)e^{-5} < e^{-1}$ , so  $g(\lambda) \in g(\mathbb{D})$ . On each remaining box a Krawczyk test gives the unique root  $\theta$  in a small box. Adjacent boxes are checked to carry the same root, by uniqueness in the hull of their  $\theta$ -boxes. The chain starts at a real  $\theta$ . Then (1) of Lemma 9.26 is checked, together with the endpoint identities  $f_1(1) = f_2(1) = -\frac{1}{2} \text{Im } \lambda$  and  $f_1(-1) = f_2(-1) = \frac{1}{2} \text{Im}(\lambda - 2i\theta)$  and a sign of  $\partial_t f$  near  $t = \pm 1$ . This handles

the degenerate edge  $\text{Im } \lambda \rightarrow 0$  ( $b \rightarrow (0, e^{-e})$ ). The mark is checked as in Lemma 9.29. A box whose  $\theta$ -box lies in  $\Theta$  or in its complex conjugate needs no new check. Indeed  $s_{\bar{\theta}}(t) = -s_{\theta}(-t)$  and  $\zeta_{\bar{\theta}}(\bar{w}) = -\zeta_{\theta}(w)$ , so the crescent, the inequalities (with  $t \mapsto -t$ ) and the entry index of  $\theta$  are those of  $\theta$  reflected in the imaginary axis.

- (P) *Cusp*,  $|\lambda - 1| < 0.08$ . Write  $\lambda_+(\theta) = 1 + i\theta + \theta^2 D(\theta^2)$  with  $|D| \leq 0.34$  for  $|\theta| \leq 0.09$  (certified). By Rouché,  $\lambda_+(\theta) = \lambda$  has exactly one root with  $|\theta| < 0.09$ . In polar form  $\theta = \rho e^{i\phi}$ ,  $\rho \leq 0.09$ , the conditions  $|\lambda| \geq 1$ ,  $\text{Im } \lambda > 0$ ,  $\text{Im } g > 0$ , scaled by  $\rho$ ,  $\rho$ ,  $\rho^2$ , force  $0 \leq \phi \leq \arctan \frac{1}{10}$ . The scaled  $\text{Im } g$  vanishes identically along the real directions  $\phi = 0$  and  $\phi = -\pi/2$ , and so does  $\text{Im } \lambda$  along  $\phi = -\pi/2$ ; there the sign of  $\partial_{\phi}$  is used. So  $\theta$  lies in the cone  $\Theta_C$ , where Lemmas 9.26 and 9.28 apply.
- (I) *Interfaces and anchor*. The core roots at  $\text{Re } \lambda = -12$  coincide with the tail roots. In the tails the root is the unique fixed point of the contraction on the fixed set  $|\text{Re } D + \pi|, |\text{Im } D| \leq \frac{3}{2}$ , so it is one continuous branch there. The core  $\theta$ -boxes on cells meeting  $|\lambda - 1| < 0.08$  lie in  $0 < |\theta| < 0.09$ , so they carry the Rouché root. At  $\lambda = \cot 1 + i$  the certified box contains the real root  $\theta = 1$  and not 0.

**A.4. The endpoint segment.** The certificate `certify_endpoint.py` (python-flint/Arb, under a minute) checks (a)–(f) of Lemma 9.42 for all  $\lambda$  in the square  $[-1.1, -0.9] \times [-0.1, 0.1]$  and in the strip  $[-12, -\frac{1}{5}] \times [-10^{-3}, 10^{-3}]$ .

- *The square*.  $\text{Re}(\lambda'_+(\theta)/\lambda'_+(\theta_*)) > 0$  on the convex rectangle  $R' = [2.20, 2.45] \times [0.62, 0.95]$  ( $\theta_*$  is the root of  $\lambda_+ = -1$ ), so  $\lambda_+$  is injective there (Noshiro–Warschawski). Krawczyk boxes inside  $R'$  give the root  $\theta(\lambda)$ . Conditions (a)–(f) are checked on  $t$ -intervals, and (e) by comparing  $\zeta(0)$  with the arc.
- *The strip*. The first strip box lies in the square, and its root enclosure lies in  $R'$ ; this anchors the strip to the square's branch. Adjacent boxes carry the same root: a Krawczyk test on the hull of their  $\theta$ -boxes, for  $\lambda$  on the common edge, proves that the root is unique there. On  $t$ -intervals touching  $\pm 1$ , where  $x = \mp\infty$ , conditions (d) and (e) use the one-sided bounds  $x \leq x_{\max}$  and  $x \geq x_{\min}$ .

**A.5. The parabolic constants.** *Proposition 2.6*. The certificate `certify_horn_coefficients.py` uses the first 12 formal Abel coefficients as exact rationals. For their truncation  $\alpha_{12}$ , the Abel defect vanishes to order 14 and has modulus  $< 4$  on  $|u| = \frac{1}{2}$ . Once the forward and backward orbits reach  $\text{Re}(-2/u) > 80$  and  $\text{Re}(2/u) > 80$ , respectively, the defect estimate and the increase of these real parts by at least  $3/4$  per step bound each Fatou-coordinate tail after 100 iterates by  $2.25 \cdot 10^{-17}$ , and after 1200 iterates by  $4.08 \cdot 10^{-31}$ .

Outward-rounded Arb arithmetic covers  $0 \leq \text{Re } z \leq 1$ ,  $1.9 \leq \text{Im } z \leq 2.1$  with  $256 \times 40$  boxes. On each box a Krawczyk test gives a unique local inverse of  $\Phi_{\text{rep}}$ ; adjacent roots agree on their common edges, and the roots at  $\text{Re } z = 0, 1$  are related by  $f(u) = e^u - 1$ . The same boxes give  $|D(z)| < 1$  for  $D = \Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1} - \text{id}$ . A chain of 400 further boxes joins this branch to  $u = 11i/20$  in the upper gate, fixing the horn-map normalisation.

At  $z_j = j/128 + 2i$  the inverse and  $D(z_j)$  are enclosed using 1200 iterates. For  $n = 1, 2, 3$ , multiply the discrete Fourier average by  $e^{4\pi n}$ . The analytic periodic trapezoid bound on the strip of half-width  $\delta = 1/10$  makes its alias error at most

$$2e^{2\pi n(2+\delta)} \frac{e^{-2\pi(128)\delta}}{1 - e^{-2\pi(128)\delta}}.$$

Adding this bound to the Arb sample enclosures gives the stated intervals.

*The coordinate a.* `certify_parabolic_a.py` computes the first ten formal Abel coefficients as exact rationals. For their truncation  $\alpha_{10}$ , the residual  $\alpha_{10}(e^u - 1) - \alpha_{10}(u) - 1$  vanishes to

order 12; Arb bounds its modulus by 0.438 on  $|u| = \frac{1}{2}$ . Put  $u_0 = 1/e - 1$ ,  $u_{k+1} = e^{u_k} - 1$ , and  $x = -u_{1000} < 0.002$ . Since  $-u_{k+1} \leq (-u_k) - (-u_k)^2/3$  for  $-u_k \leq 1$ , Cauchy's estimate bounds the tail after  $\alpha_{10}(u_{1000}) - 1000$  by

$$0.438 \, 2^{12} \left( x^{12} + \frac{3x^{11}}{11} \right) < 10^{-27}.$$

The orbit and truncated value are enclosed by Arb; together they give the stated interval.

## REFERENCES

- [1] L. V. Ahlfors, *Lectures on Quasiconformal Mappings*, 2nd ed., University Lecture Series 38, Amer. Math. Soc., Providence, RI, 2006.
- [2] L. Ahlfors and L. Bers, *Riemann's mapping theorem for variable metrics*, Ann. of Math. (2) **72** (1960), 385–404.
- [3] X. Buff and A. L. Epstein, *A parabolic Pommerenke–Levin–Yoccoz inequality*, Fund. Math. **172** (2002), 249–289. [Publisher PDF](#).
- [4] A. Chéritat, *Near parabolic renormalization for unicritical holomorphic maps*, Arnold Math. J. **8** (2022), 169–270. [doi:10.1007/s40598-020-00172-6](https://doi.org/10.1007/s40598-020-00172-6).
- [5] C. Christopher and C. Rousseau, *The moduli space of germs of generic families of analytic diffeomorphisms unfolding a parabolic fixed point*, Int. Math. Res. Not. IMRN (2014); arXiv:0809.2167.
- [6] J. Écalle, *Les fonctions résurgentes*, Publ. Math. d'Orsay, 1981.
- [7] A. L. Epstein, *Towers of finite type complex analytic maps*, Ph.D. thesis, City University of New York, 1993.
- [8] W. Fischer and H. Grauert, *Lokal-triviale Familien kompakter komplexer Mannigfaltigkeiten*, Nachr. Akad. Wiss. Göttingen Math.-Phys. Kl. II (1965), 89–94.
- [9] A. A. Glutsyuk, *Confluence of singular points and the nonlinear Stokes phenomenon*, Trans. Moscow Math. Soc. **62** (2001), 49–95.
- [10] H. Kneser, *Reelle analytische Lösungen der Gleichung  $\vartheta(\vartheta(x)) = e^x$  und verwandter Funktionalgleichungen*, J. Reine Angew. Math. **187** (1950), 56–67.
- [11] D. Kouznetsov, *Solution of  $F(z+1) = \exp(F(z))$  in complex  $z$ -plane*, Math. Comp. **78** (2009), 1647–1670.
- [12] D. Kouznetsov and H. Trappmann, *Portrait of the four regular super-exponentials to base  $\sqrt{2}$* , Math. Comp. **79** (2010), 1727–1756.
- [13] P. Lavaurs, *Systèmes dynamiques holomorphes: explosion de points périodiques paraboliques*, Thèse, Université Paris-Sud, Orsay, 1989.
- [14] O. Lehto and K. I. Virtanen, *Quasiconformal Mappings in the Plane*, 2nd ed., Grundlehren der mathematischen Wissenschaften 126, Springer, 1973.
- [15] P. Mardešić, R. Roussarie and C. Rousseau, *Modulus of analytic classification for unfoldings of generic parabolic diffeomorphisms*, Mosc. Math. J. **4** (2004), no. 2, 455–502.
- [16] R. Oudkerk, *The parabolic implosion for  $f_0(z) = z + z^{\nu+1} + O(z^{\nu+2})$* , Ph.D. thesis, University of Warwick, 1999; see also *The parabolic implosion: Lavaurs maps and strong convergence for rational maps*, in: Value Distribution Theory and Complex Dynamics, Contemp. Math. **303**, Amer. Math. Soc., 2002, 79–105.
- [17] W. Paulsen and S. Cowgill, *Solving  $F(z+1) = b^{F(z)}$  in the complex plane*, Adv. Comput. Math. **43** (2017), 1261–1282. [doi:10.1007/s10444-017-9524-1](https://doi.org/10.1007/s10444-017-9524-1).
- [18] W. Paulsen, *Tetration for complex bases*, Adv. Comput. Math. **45** (2019), 243–267. [doi:10.1007/s10444-018-9615-7](https://doi.org/10.1007/s10444-018-9615-7).
- [19] W. Paulsen, *Analyzing the function  $z \uparrow\uparrow a$  for fractional  $a$* , Adv. Comput. Math. **52** (2026), article 25. [doi:10.1007/s10444-026-10300-z](https://doi.org/10.1007/s10444-026-10300-z).
- [20] C. Rousseau and L. Teyssier, *Analytical moduli for unfoldings of saddle-node vector fields*, Mosc. Math. J. **8** (2008), no. 3, 547–614.
- [21] M. Shishikura, *Bifurcation of parabolic fixed points*, in: *The Mandelbrot Set, Theme and Variations*, London Math. Soc. Lecture Note Ser. **274**, Cambridge Univ. Press, 2000, 325–363.
- [22] H. Trappmann and D. Kouznetsov, *Computation of the two regular super-exponentials to base  $\exp(1/e)$* , Math. Comp. **81** (2012), 2207–2227.
- [23] H. Trappmann and D. Kouznetsov, *Uniqueness of holomorphic Abel functions at a complex fixed point pair*, Aequationes Math. **81** (2011), 65–76.
- [24] S. M. Voronin, *Analytic classification of germs of conformal mappings  $(\mathbb{C}, 0) \rightarrow (\mathbb{C}, 0)$* , Funct. Anal. Appl. **15** (1981), 1–13.

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